

Onyx — Kernel internals

Onyx · multi-process OS on Raspberry Pi 4

This document describes the inner workings of the **Onyx kernel** (the code under `kernel/`). It does **not** describe Circle, which is used as the HAL/driver stack; we only point out the seams with Circle where necessary.

All file paths are relative to the repository root. The values (addresses, constants) come from the code; the layout constants live in `kernel/include/kern/layout.h`.

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1. Scope: ours vs. Circle

Written by Onyx	Reused from Circle
Per-process address spaces (<code>mm/addrspace.cpp</code>)	64 KB page allocator (<code>palloc/pfree</code>)
Scheduler (<code>sched/scheduler.cpp</code> , <code>sched/task.cpp</code>)	<code>taskswitch.S</code> , <code>mutex/semaphore/event</code>
VBAR_EL1 vectors, trap frame (<code>arch/aarch64/</code>)	<code>InterruptHandler</code> (GIC + EOI), <code>FIQStub</code>
ELF loader (<code>proc/elf.cpp</code>)	EMMC + FatFs, USB HID
ABI table + kapi impl. (<code>sys/</code>)	<code>C2DGraphics</code> (framebuffer), <code>CTimer</code>
GUI: rendering, compositor + window manager, shared surfaces (<code>gui/</code>)	<code>CCharGenerator</code> (bitmap font)

The kernel is linked against Circle's static libraries. **Two Circle files are replaced**

(`lib/sched/scheduler.cpp` and `task.cpp`): because our `.o` files appear in `OBJS` *before* the `--start-group` archives, the linker resolves `CScheduler/CTask` from our versions and never pulls in those from `libsched.a`.

A **shadow header** (`kernel/compat/circle/sched/scheduler.h`) is placed **before** `circle/include` on the include path, so that `<circle/sched/scheduler.h>` resolves to ours (which adds `OnTimerTick()`, the notion of a time slice, etc., while keeping Circle's public + friend API).

2. Boot sequence

The entry point is `main()` (`kernel/main.cpp`), called by Circle's `sysinit` **after** it has already: switched the CPU to EL1, enabled the MMU (64 KB granule), initialized the GIC, run the static C++ constructors.

All the logic lives in the `CKernel` class (`kernel/kernel.cpp`):

`CKernel::Initialize()` — in order

1. **Text console:** `m_Screen` (HDMI) then `m_Serial` (115200). The boot log goes to the HDMI screen so it is visible **without a serial cable**. A `CLogSwitch` routes the logger's output (see §13).
2. **Interrupts + timer:** `m_Interrupt.Initialize()`, `m_Timer.Initialize()`.
3. **Exception vectors:** `install_vectors()` installs our `VBAR_EL1` (`KVectorTable`) over Circle's.
4. **Scheduler:**
 - `m_Timer.RegisterPeriodicHandler(PeriodicTick)` — wires the tick to `OnTimerTick()`.
 - `AddrSpaceInit()` — captures the kernel's TTBR0 (base of the shared L2 table).
 - `RegisterTaskSwitchHandler(AddressSpaceTaskSwitch)` — on each task switch, activates the new task's address space.
 - `RegisterTaskTerminationHandler(AddressSpaceTaskTerminate)` — frees the address space when a task ends.
5. **ABI table:** `KApiTableInit()` fills the `kapi` pointer table (see §8), then `EL0Init()` builds from it the EL0 table and code page every process maps (§6, *Protected mode*); core 0's EL0 controls (`EL0CoreInit`). Then the **cores 1–3** (`COnyxCores`): core 1 the sound (§12), cores 2–3 the app cores (§14), core 3 the network instead with `netcore=1` (§11); `dispdma=` / `gpudirect=` read here.
6. **Graphics + SD:** `m_2DGraphics.Initialize()` (32 bpp framebuffer); `m_EMMC + f_mount()` (FatFs) of SD: (partition 1, the boot FAT32 one), then of **SD1: ... SD3:** (partitions 2–4) when they hold a FAT or **exFAT** file system (see *Volumes* in §8). The theme from `SD:/etc/theme.txt`; the mouse cursor is built in (`BuiltinCursor`, the 12×19 arrow that was `SD:skins/mousecur.bin` — the card has no `skins/` folder any more: the window chrome is drawn user-side).
7. **RAM:** `RamFsInit()` — the RAM volume, its size from `system.ini`'s `ramfs=` (§16).
8. **USB:** `m_USB.Initialize()` (mouse + HID keyboard, hot-plug).

`CKernel::Run()` — the service loop

1. Logs machine info.
2. Launches the **init program** (`StartAutostart`): the ELF named by the `cmdline.txt` option `init=` (e.g. `init=SD:/bin/init`), defaulting to `SD:bin/init` when the option is absent. The kernel does not parse any launch list itself — `init` (PID-1 style) reads `SD:/etc/autostart` and starts everything

from there (by default `voronoy`, which paints the wallpaper then exits, and `panel`, the shell).

Pointing `init=` at a different ELF swaps the whole userland launcher (e.g. a recovery shell) without rebuilding the kernel.

3. Launches the **kernel service tasks**:

- **CCompositorTask** — presents the screen at up to ~60 Hz, **only what changed** (dirty rectangles): every visual change says where — `ScreenDirtyRect` for a part (`kapi_present/SYS_present` and chrome: the caller's window; window add / remove / raise / move / resize / alpha: its old and new places; the cursor: its old and new 16×16 spots, with the drag badge) or `ScreenDirty` for all of it (wallpaper, full screen end, drag & drop start / end). Rectangles that touch are merged (at most 16; more, or over 2/3 of the screen, = the whole screen) and `g_nScreenGen` is bumped. The task takes the damage (`ScreenTakeDamage`) and for each rectangle composites with the screen image **clipped** to it (`GImage::SetClip`: every blit and fill stays inside) and sends just that rectangle to the display (`C2DGraphics::UpdateDisplay (x, y, w, h)`), our Circle patches 6 / 12: the frame buffer's 2D DMA reads the rectangle in place, source and destination strides; `CCompositorTask::Present` starts it with `UpdateDisplayAsync` and **yields** to the other tasks until its interrupt says it is done, instead of spinning); the whole screen only when told, plus a safety refresh every 2 s. A window animating (a game, an emulator) costs its own area instead of the whole screen, a mouse move two cursor-sized spots. An idle desktop thus costs almost no CPU (the heartbeat's fps shows the real composites). It is created **before** `init` (no boot-log pause any more) and the main task hands it the CPU at once with `CScheduler::YieldTo()` (scan starts at that task instead of the round-robin next), so it presents before the first app's window. Starting it late — after a 6 s pause, with a single GUI app already running — used to hang the boot.
- **CReaperTask** — reaps terminated tasks every ~50 ms (frees `CTask` + stack + address space outside the scheduler core). It also blinks the green ACT LED as a headless sign of life: 1 s period while the network is down, 0.2 s once `NetIsUp()`; a frozen LED means the scheduler stopped.
- **CInputTask** (if USB is present) — pumps keyboard/mouse events to the window manager; every 100 ms it also looks for new gamepads (`upad1..4`) and USB MIDI devices (`umidi1..4`, v68: §8 *USB MIDI*), also after a hot plug. It takes **every** USB keyboard (`ukbd1..4`, each with the layout snapshot), not only `ukbd1`: a wireless mouse's receiver (Logitech's nano receiver, e.g. the M187's) has a boot-keyboard interface of its own, which may enumerate first and take `ukbd1` — the real keyboard, `ukbd2`, was then ignored. Each keyboard's raw report is kept and the window manager gets them merged (modifiers ORed, held keys joined), so one keyboard's empty report does not release the keys held on another. The mouse is still `mouse1` only. **Print Screen** (USB usage 0x46) is caught there, on its press (`s_nPrintScreen`: 1, or 2 with Alt), and handled by the main task's loop (it now wakes every 50 ms: `PrintScreenPoll`, the network's notices every 250 ms as before): the **Screenshot** app is told through its service `screenshot` (message type 1, "now" or "window <id>" — with Alt, the window that has the keyboard, found with `Snapshot / HasKeyFocus`), else started (`ExecPath ("SD:apps/screenshot.app/main", "--now" | "--window <id>", "screenshot")`). No key event reaches the app in front (the keymap gives Print Screen no character).
- **CGuiWatchdogTask** (skipped with `watchdog=0` in `cmdline.txt`) — once a second, checks the GUI and writes to the kernel log (`kmsg`): `compositor STALLED` when `CWindowManager::FrameCount()` has not moved for 2 s (with every task's `name:state`), app

'<title>' NOT PUMPING events when a window has queued events and its owner has not called `PopEvent` for 2 s (plus the matching "recovered" lines), and a **heartbeat** every heartbeat= seconds from `cmdline.txt` (default 5, 0 = off): uptime, fps, mouse/key deltas, tasks by state, windows with queued/dropped events. Counters live in `CWindowManager` (frames, `OnMouse/OnKey` calls, `Snapshot()`) and `CWindow` (`QueuedEvents`, `DroppedEvents`, `LastPumpTicks`).

4. The main task runs an idle loop (`MsSleep`).

3. Address-space map

One `TTBR0` per process. The kernel (identity) must stay at the **low VAs** (because identity requires `VA == PA`), so **kernel and user coexist in `TTBR0`**: the kernel L2 entries are **shared**, the user L2/L3 are **per process**.

VA	Content	Attributes
0x0_0000_0000	KERNEL IDENTITY (RAM, MMIO, PCIe) mapped by Circle	EL1 RW, global (nG=0), shared in every table (4 GB = <code>KERNEL_IDENTITY_END</code>)
0x1_0000_0000 (4 GB)	unmapped hole (guard)	
0x2_0000_0000 (8 GB)	.text / .data / .bss (app) heap (sbrk), lazy (v75)	<code>USER_VA_BASE</code> (8 GB) EL0 (AP=*_ALL), ASID-tagged (nG=1)
0x2_8000_0000	window canvas (+ chrome copies)	<code>USER_HEAP_BASE</code> (10 GB)
0x3_0000_0000	wallpaper buffer	<code>USER_WINDOW_CANVAS</code> (12 GB)
0x3_4000_0000	shared surfaces	<code>USER_WALLPAPER_CANVAS</code> (13 GB)
0x3_6000_0000	EL0 kapi table (read-only)	<code>USER_SURFACE_BASE</code> (13.5 GB)
0x3_8000_0000	EL0 code page: stubs + blob (RX)	<code>KAPI_TABLE_VA</code> (14 GB)
0x3_8001_0000	generated code (a JIT, RWX EL0)	<code>KAPI_STUBS_VA</code> (14 GB + 64 KB)
0x3_9000_0000	full-screen buffer / the screen	<code>USER_CODE_BASE</code> (14.25 GB, v58)
0x3_C000_0000		<code>USER_FULLSCREEN_CANVAS</code> (15 GB)
0x4_0000_0000 (16 GB)	user stack (grows down), lazy (unused up to 32 GB)	<code>USER_STACK_TOP</code> (16 GB)
...	threads' user stacks, 32 MB slots	8 MB (app.txt stack, ≤ 64 MB)
0x8_0000_0000	mmap arena (vm_map only), lazy	<code>USER_THREAD_STACKS</code> (32 GB), lazy
0x8_8000_0000		<code>USER_MMAP_BASE</code> (34 GB, v75)
...		
0xF_0000_0000		<code>USER_VA_END</code> = <code>USER_MMAP_END</code> (60 GB)
0x10_0000_0000		T0SZ ceiling (64 GB) on RPi 4

- **64 KB granule, EL1&0 stage-1 only**, `TTBR1` disabled.
- 2 levels: one **L2 entry = 512 MB** (points to an L3 table), one **L3 page = 64 KB**.
- **8-bit ASID** (256 contexts), taken from `TTBR0_EL1[63:48]`.
- `USER_LOAD_BASE` = `USER_VA_BASE`: apps are linked at 8 GB (see `user.ld`).
- **Lazy** (v75, §4 *Demand paging*): the heap, the stacks and the mmap arena are regions whose pages are filled on first touch; the image, the canvases, surfaces, the kapi pages, the code arena, the full-screen buffers and the sound ring are mapped at once (**FIXED / IMAGE**).

4. Memory management: `CAddressSpace`

Source: `kernel/mm/addrspace.cpp`, `kernel/include/kern/addrspace.h`.

A `CAddressSpace` object = a process. It owns: the L2 table, the ASID, a PID, the window pointer, the stdin/stdout streams, the process handle, the exit code, and the argv/cwd string.

Where the app pages come from, by board RAM. An app's own frames (ELF segments, heap, stack, `vm_map`, `code_alloc`) are taken by `MapNewPage` with `palloc_high()`: the **high zone** (1–3 GB, plus the RAM reclaimed above 3 GB) on a 2 / 4 / 8 GB Pi 4. A **1 GB** Pi 4 has no high zone: `palloc_high()` falls back to the low pager (`PAGE_RESERVE`, 256 MB), which the apps then share with every process's page tables. The kernel heap (window canvases, the wallpaper, the full-screen buffer, the GPU's memory, ~680 MB) is the same on every board. So Onyx runs on 1 GB — the desktop and the small apps take a few MB each — but the apps all together have ~250 MB instead of ~2 GB or more: the big ones (Jet Browser, the N64 / GameCube emulators) can run short (`sbrk` / `new` return 0). the Task Manager's Memory tab then shows an app pool of 0 (`ram_detail`: the high zone only).

Construction

1. **Allocates a fresh L2 table** (`palloc`, one 64 KB page).
2. **Copies the kernel L2 descriptors** from the template captured at boot (`memcpy` of the whole L2 page). The kernel entries point to the **same shared L3 tables**; the L2 slots covering the user area stay invalid (zero).
3. **Allocates an ASID** (1..255; 0 is reserved for the kernel/global).
4. **Maps the EL0 kapi table** read-only at `KAPI_TABLE_VA` (14 GB) with `KPAGE_ATTR_APP_RODATA`, and the **EL0 code page** (the system-call stubs and the user-side routines, §6 *Protected mode*) at `KAPI_STUBS_VA` with `KPAGE_ATTR_APP_CODE`. Both physical pages are kernel globals shared by every process (not "owned" by it → never freed at destruction).

`MapPage(VA, PA, attrs, bOwned)`

Maps a 64 KB page (VA and PA aligned to 64 KB):

- `GetOrCreateL3(L2_INDEX(VA))`: if the L2 entry is invalid, allocates a fresh L3 table (zeroed, `DSB ISHST`) and writes the L2 table descriptor.
- Fills the L3 page descriptor: `AttrIdx`, `AP`, `SH`, `AF=1`, `nG`, output = `PA`, `PXN`, `UXN`. The software bit `PAGE_SW_OWNED` (the *Ignored* field, bit 55) marks the frames to free at destruction. Since v75 both descriptors are built aside and **stored as one 64-bit word**: an app core running a job of this space may walk the table at that very moment.

Page attribute matrix (`layout.h` presets)

Use	AttrIdx	AP	nG	PXN	UXN
<code>KPAGE_ATTR_APP_CODE</code> (app code, the EL0 code page)	NORMAL	RO_ALL	1	1	0 (exec. EL0)
<code>KPAGE_ATTR_APP_DATA</code> (data/heap/stacks/canvases/surfaces)	NORMAL	RW_ALL	1	1	1
<code>KPAGE_ATTR_APP_RODATA</code>	NORMAL	RO_ALL	1	1	1

Use	AttrIndx	AP	nG	PXN	UXN
A (the ELO kapi table; v77: a program's read-only segment without PF_X)					
KPAGE_ATTR_APP_RWX (generated code, v58 code_alloc)	NORMAL	RW_ALL	1	1	0 (exec. ELO)
KPAGE_ATTR_APP_SCREEN (fullscreen_direct, the displayed framebuffer)	COHERENT (uncached)	RW_ALL	1	1	1

App pages are **accessible at ELO** (AP=*_ALL; the kernel works in them in place: the A72 has no PAN), executable **at ELO only** for code (UXN=0, PXN=1 everywhere: the kernel never runs app code), and nG=1 (ASID-tagged) → isolation between applications. The identity region keeps Circle's AP=RW_EL1: out of the apps' reach (§6, *Protected mode*).

Activation and context switch

```
// CAddressSpace::Activate()
u64 ttbr0 = MAKE_TTBR0(phys(L2), asid); // = phys | (asid << 48)
asm volatile ("msr ttbr0_el1, %0; isb" :: "r"(ttbr0) : "memory");
```

No TLB flush on switch (TLB tagged by ASID). The AddressSpaceTaskSwitch hook calls `Activate()` for app tasks, or `ActivateKernelAddressSpace()` for kernel tasks (which have no address space).

Demand paging (v75)

Source: `kernel/sys/vm.cpp`, `kern/vm.h`; the plan: `docs/POSIX-PLAN.md` §3.1.

- **Regions.** Each space has a `TVmSpace`: a sorted array of `TVma { start, end, prot, kind }` (binary-searched, at most 4096). Lazy kinds: `ANON` (`vm_map`, only in the mmap arena [34 GB, 60 GB)), `HEAP` ([`USER_HEAP_BASE`, `break` rounded up)), `STACK` (the main stack below 16 GB, 8 MB by default; each thread's at the top of its 32 MB slot). Eager kinds, noted for `vm_query` only: `IMAGE` (the ELF loader), `FIXED` (`MapContig`: canvases, surfaces, the sound ring, the kapi pages, the full-screen buffers; the code arena). Nothing is mapped for a lazy region until it is touched. (v76) `SHM`: a shared memory object mapped by `shm_map` (in the arena, lazy, its frames the object's and not owned by the space: §8 v76: *IPC*).

- **The fill** (`VmFaultIn`): the region must be lazy and allow the access; a zeroed frame from `palloc_high`, a `DSB ISHST`, the PTE (`AP` from the region: `RW_ALL`, `RO_ALL`, or `RO_EL1` for `PROT_NONE` — the frame kept, the app shut out), `DSB`. A software walk: it never yields and works for any space. **No TLBI** (an invalid entry is never cached). Who fills: an **EL0 translation fault** (`sys/el0.cpp`: the access retried by the `eret`; a permission fault the region now allows — a stale TLB entry — gets a local `TLBI VAE1`); a **kernel probe** (`UserReadable / UserWritable`, §6); a **failed fault-safe copy** (retried in C after the fill, never in the exception); the **app-core pager** (§14); `vm_map` (`POPULATE`) and `MADV_WILLNEED` (a yield every 64 pages).
- **Out of memory.** A fill is refused when the app pool (the high zone + the low pager, whose free lists `ram_detail` reads) is under **16 MB** (`VM_RESERVE`). At EL0 the process is killed: `kmsg vm: <name> (pid N) killed: out of memory (page fault at ..., ... KB resident)`, the notice "ran out of memory", term reason `KAPI_PROC_OOM`, status `-9`. In a kapi the call fails with its error value; on an app core the job ends `CORE_FAULT`. **Heuristic overcommit:** one `sbrk` growth or one writable `vm_map` (without `NORESERVE`) larger than the free pool – 16 MB fails (`sbrk` → `-1`, `vm_map` → `-ENOMEM`); nothing else is counted. (Pi round 1: Circle's `CPageAllocator::Allocate` advanced its bump pointer before it found its region full and never stepped it back, so each failed attempt — `palloc_high` tries the full segment 0 before every page it takes elsewhere — made `GetFreeSpace` and `meminfo` one page short for ever: an OOM kill seemed to lose ~1.1 GB and the OOM check fired with half the memory free. Fixed in the Circle fork, docs/05 §24.)
- **Frames leaving** (`vm_unmap`, `MADV_DONTNEED/FREE`, an `sbrk` shrink, an ended thread's stack) and **protections lowered** (`vm_protect`): the PTE written, `DSB ISHST`, `TLBI VAE1IS` (`va >> 12 | ASID << 48`) per page (`TLBI ASIDE1IS` above 64 pages), `DSB ISH`, `ISB` — inner shareable, so cores 2–3 are told — and only then the frames freed (`pfree`, after waking any `wait_word` sleeper on them). Protection changes need no break-before-make (the output address never changes).
- **Pins and the deferred zap.** A probe pins its range for the calling task until its system call returns (`Syscall` → `VmUnpinTask`; one union per task, 40 slots). A page under **another** task's pin is not dropped (nor write-protected) at once: its PTE gets a software bit (`ZAP / SYNC`, bits 56–57), the range is queued, and the last unpin over it drops it (or gives it its region's protection). The region change itself is immediate. So a kapi working in place (a file read yielding between 64 KB chunks, a blocked pipe read) never sees its buffer vanish, nobody waits, and the kernel never touches a freed frame. A range mapped again before that adopts the pages left, zeroed.
- **The EL1 safety net:** `SyncHandlerEL1`, after the `uaccess` fixup, fills a lazy page that kernel code touched outside the helpers (core 0, the current app, a translation fault at a user VA), logs `vm: kernel touched an unpopulated user page at pc ...` once, and retries — instead of a panic.
- **Costs.** A touched page costs 64 KB (a thread's first stack page, a lone `malloc` page); one L3 table (64 KB, low pager) per 512 MB slot touched; reservations cost nothing. Zeroing a page takes about 10 μs. `vm_stats` reports a process's resident bytes (page tables included), its lazy and writable VA, its fills and its page tables.

Destruction (and an important pitfall)

When destroying the address space:

1. Signals EOF on stdout, releases the streams, marks the process handle as terminated.
2. **Invalidates the TLB for this ASID** *before* freeing anything: `tlbi aside1is, Xt ; dsb ish ; isb`. (Reusing a page-table frame while the walk is still cached would break translations.)
3. Removes and destroys the window.

4. (v75) `VmTeardown`: the regions and the pending zaps freed (after the app cores and the GPU let go of the space).
5. Walks the **user** L2/L3 slots and frees the frames marked `PAGE_SW_OWNED` (pages whose zap was deferred included).
6. (v77) Drops its reference on its program's image (§7): the image's shared frames are mapped **not owned**, so step 5 leaves them; the last process of a program that is not preloaded frees them here, after the tables that named them. (When an app core did not stop, the space is leaked as before, and so is this reference.)

⚠ **Pitfall: never free an L3 table shared with the kernel.** Because each address space **copies** the kernel L2 descriptors, some L2 entries point to the **same** L3 tables as the kernel. Before freeing an L3 from the user area, the destructor **compares** the table address against the reference kernel L2's and **skips** any shared table. Freeing a shared kernel table would corrupt the translations of the *entire system*. (See also the design note "per-AS L2 copies the kernel L2".)

5. Scheduling

Source: `kernel/sched/scheduler.cpp`, `kernel/sched/task.cpp`,
`kernel/compat/circle/sched/scheduler.h`.

Preemptive scheduling (track A)

The kernel **preempts application code**. A 100 Hz timer tick drives `OnTimerTick`, which counts down the running task's time slice (`SCHED_SLICE_TICKS` = 20 ms) and sets a reschedule flag when it expires. Every app runs at **EL0** (§6, *Protected mode*): an IRQ taken from EL0 enters through `EL0IrqEntry`, which builds the app's full frame (all registers, FP/SIMD) **on the task's own kernel stack** and runs Circle's interrupt handler there; `EL0IrqExit` then, when the slice is over, calls `OnPreempt` + `Yield` right on that stack — the task is parked **exactly like a voluntary yielder** (one resume path), and is resumed by `EL0Return`'s `eret` to the interrupted instruction. That is the **only** preemption point. (Before kapi v74 apps ran at EL1t on their task's stack, and a `PreemptTrampoline` redirected the IRQ's return to switch there; it is gone.)

An IRQ taken at EL1 — kernel code, including an app's kapi call — never switches tasks: the **kernel is non-preemptive** (the classic Unix model). A long, non-yielding *kernel* loop would still block other tasks — drop a `if (IsReschedPending()) Yield();` cooperative point into any such loop if one appears. Tasks also still switch **voluntarily** (`Yield`, `MsSleep`, `present`, `wait`, ...), unchanged.

Preemption points in file I/O. The long kernel operations an app can ask for are cut into pieces with a `Yield` between them: `kapi_read` / `kapi_save_file` move at most 64 KB per `f_read` / `f_write` (`ChunkedRead` / `ChunkedWrite` in `sys/kapi.cpp`), the ELF loader 128 KB. Without them one read of a 28 MB file (a Doom WAD, a GBA ROM, a newlib app's `fopen`, which reads the whole file) stopped every other task — the compositor, the cursor, the sound feeders — for seconds. Between two pieces the FatFs volume lock is free, so other tasks' file calls get through; the caller's buffer stays valid (its address space is active again when it resumes).

File-system speed. The FatFs disk layer keeps a 1 MB **sector cache** of the single-sector reads (the FAT, the folders), write-through (`docs/05 §8`; `sdcache=0` turns it off): opening files and walking folders stopped asking the card for every sector. And `f_read` / `f_write` move a file's contiguous clusters in **one** SD command (`docs/05 §10`), instead of one per cluster (with 512-byte clusters, one per 512 bytes: 2 MB/s). A

read of 1 MB or more logs its time split (fs: read ... SD: N commands, waiting, data port, other tasks). The card runs in High Speed (50 MHz, docs/05 §9; sdhs=0 = 25 MHz). The boot log says what is in use (SD card mounted (SD:): ...); /bin/fsbench measures.

Yielding SD waits. The SD driver also yields while the card keeps it waiting: after a 2 ms spin (a normal command ends within that: yielding for it cost a whole time slice per command, 2 MB/s), CEMMCDevice::TimeoutWait calls OnyxDriverWait () (a weak hook of our Circle fork, docs/05 §7), which Yields if IRQs are on. The FatFs volume lock is held during that wait, so it is a **sleeping** lock (OnyxFsLockTake / OnyxFsLockGive, sys/fslock.cpp, re-entrant, its waiters Yield): with Circle's spin lock a second task entering FatFs would spin forever against a holder it never lets run. The holder is in a **no-kill section** (CScheduler::EnterNoKill / LeaveNoKill): TerminateTask on it only marks it, and it ends in LeaveNoKill, once the lock is free — a dead owner would lock the card forever. (Before: 100–200 ms freezes each time the menu bar read the app.txt files, or an app opened a file in a big directory.) Many **short** waits in a row also yield (2026-10-01): a long multi-block write (one SD command for a file's contiguous clusters, the card taking a few hundred µs per block, each wait well under 2 ms) kept core 0 for 200 ms (stall: jet:cache ran 213 ms without yielding, PC in TimeoutWait). Each turn of a shorter wait calls OnyxDriverPoll () (a second weak hook, docs/05 §7), which Yields once the task has run **10 ms** (DRIVER_SLICE_US) since its last Yield (CScheduler::GetLastYield), under the same conditions (a scheduler on this core, IRQs on, not core 1's crash dump). Between two blocks of a transfer is a safe point: the volume lock is held, and the SD host holds a PIO transfer until it is served. Reads (40 MB/s) yield about every 400 KB, writes every 50–100 KB. Nothing outside FatFs drives the SD card while tasks run: the boot's CEMMCDevice::Initialize comes before the mount, and core 1's crash dump writes raw sectors only with core 0 stopped (it never yields).

Stall watchdog. To find the kernel code that still keeps the CPU too long, the scheduler notes the time of every Yield(). When the running task (not idle) has not passed through Yield() for more than SCHED_STALL_US (100 ms), the IRQ exit path (KernelIRQExit → CScheduler::StallSample) samples the interrupted PC and LR every SCHED_STALL_SAMPLE_US (50 ms, up to 8 samples). The task's next Yield() closes the report (task name, duration) into a small ring, and the reaper task writes it to the kernel log:

```
stall: doom ran 850 ms without yielding; pc/lr: 1a2b40/1a2a10 1a2b48/1a2a10 ...
```

Kernel addresses resolve with kernel/kernel18-rpi4.map (or aarch64-none-elf-addr2line -f -e kernel/kernel18-rpi4.elf <pc>). No sample at all ("IRQs masked") means the task ran with interrupts off the whole time. (Its first catch: 12dbe4/138044 = LeaveCritical from CTerminalDevice::Write — the boot-console log writes, see §11 MuteNormal.)

CPU hogs vs yielders (dynamic priority). Circle's network and Wi-Fi code waits in Yield() loops (qlock/sleep/tsleep in addon/wlan/p9proc.cpp, the CNetTask Process loop, socket send/recv — also inside an app's kapi_* call, e.g. vncd). In plain round-robin, each of those yields hands the CPU to a CPU-bound app for a full slice, so the loops crawl. Driver timeouts then expire and the Wi-Fi link drops (the spin test app took the network down, and VNC/telnet with it). So the scheduler lowers the priority of the tasks the timer has to preempt:

- The preemption (El0IrqExit, sys/el0.cpp) calls CScheduler::OnPreempt(), which bumps the task's **preemption streak** (m_nPreemptStreak[slot]). Any voluntary Yield (a kapi wait, msleep, present, a kernel task's loop) resets it.
- A task with a streak ≥ SCHED_HOG_STREAK (2) is a **CPU hog**. An app that computes more than a slice now and then but yields in between (vncd encoding a frame, then sending it) is not one.

- A hog's preemption opens a **burst** of `SCHED_BURST_US` (60 ms). During the burst `GetNextTask` **skips the hogs** (and idle): kernel tasks and apps that yield are served first. The burst ends early once none of them is ready.
- A hog also runs with a **one-tick slice** (≤ 10 ms instead of `SCHED_SLICE_TICKS`), so the kernel's yield loops never wait more than ~ 10 ms behind it.

A hog thus keeps roughly 15 % of the CPU while others need it, and all of it otherwise. Both are tunable at boot for A/B testing: `cmdline.txt slice=` (ticks) and `hogsched=0` (plain round-robin) → `CScheduler::Configure`.

CScheduler (shadow) and CTask

- Circle's public + friend API is preserved (`Yield`, `Sleep/MsSleep`, `GetCurrentTask`, `AddTask`, `BlockTask`, `WakeTasks`...) → Circle drivers that block (USB, etc.) work without modification.
- **The task list has no limit.** Circle's table of `MAX_TASKS` (40) pointers — and its `LogPanic` ("System limit of tasks exceeded") when full — is gone: the tasks are a **circular linked list** of `TSchedNodes` (`scheduler.h`), each with the scheduler's own state about its task (the no-kill depth, a pending kill, the preemption streak, killed from outside). `CTask` itself cannot grow (its layout is Circle's, shared with the prebuilt libraries), hence the separate node; `AddTask` allocates it, the reaper frees it. The round robin scans the circle from the node after `m_pScan` (the last one chosen; `YieldTo` points it just before its task); `m_pCurNode` is the running task's node, so the current task's state is reached without a search. `EnumerateTasks`'s callback must not yield.
- **A process's tasks are one group** — the tasks whose `TASK_USER_DATA_USER` is the same address space (the app's threads, §7). `TerminateTask` on an app's task terminates the whole group (`TerminateGroup`: all but the caller); `CountGroup` counts its live ones. A task **killed** from outside may still be on a wait list of its process (a mutex, an event, a socket): the reaper frees it only once no task of its group is alive, and runs the termination handlers of a batch **before** deleting any of its tasks — the last handler frees the process (its sockets closed, its objects' wait lists emptied) while the tasks still exist.
- `WakeTasks` **tolerates a task that is no longer blocked**: a `BlockTask` with a timeout that expired is made Ready by `GetNextTask` but stays on the event's list until it runs again and unlinks itself; an event set in that window (the V3D's frame-done interrupt right after one of the drawing task's 2 ms waits ended) found it Ready, and Circle's assertion halted all four cores — the random freeze of `n64emu` with the GPU (found with the crash record, `sched/scheduler.cpp(742)`). Such a task is now only unlinked.
- **"Real time" tasks (v68, `SetPriority`, `kapi_thread_priority`).** A node's `nPrio` (`SCHED_PRIO_HIGH`) makes `GetNextTask` try `ScanPrio` first — only while such tasks exist (`m_nPrioTasks`): the first ready real-time task after `m_pScan` wins, before the burst and the round robin. The current task is left out of that scan, so a real-time thread that yields while still ready (a spin on `kapi_lock` held by an ordinary thread) lets the others run. A tick (`OnTimerTick`, after `WordWaitTick`) that finds one ready while an ordinary task runs sets the resched flag (`m_bPrioPreempt`): the IRQ exit preempts the app at once (in its own code only, as always), and `OnPreempt` does not count that towards the app's hog streak. The priority holds only while the task yields by itself: preempted at the end of its slice (a busy loop), it is marked `bPrioSpent` — an ordinary task — until its next voluntary `Yield` (sleep, wait). So a real-time thread cannot starve the system; an audio pump that renders and sleeps keeps it.
- `Yield()` performs the context switch **with the IRQ disabled** for atomicity, then **restores the full DAIF** of the resumed task (each task keeps its own IRQ enable state).

- The **only** change in `task.cpp: TaskEntry()` re-enables the IRQ (`msr daifclr, #2`) so that a **fresh task** does not start with the IRQ masked inherited from the switcher.
- The idle task uses `wfi` (on core 0; on the network core it only pauses: no timer tick there).
- **One scheduler per core that runs tasks.** `CScheduler::Get ()` returns the scheduler of the calling core (`s_pThis[core]`, the core read from `MPIDR_EL1`): core 0's, and with `netcore=1` core 3's (§11). A `CTask` joins the scheduler of the core that creates it, so Circle's code — tasks, `CSynchronizationEvent`, `MsSleep`, `Yield` — works unchanged on either core. The wait lists of the events share one spin lock (an event may be set from another core). Cores 1-2 have none (`IsActive ()` is `FALSE` there; `Get ()` falls back to core 0's, as before). Circle's own libraries, built with Circle's `scheduler.h`, read `s_pThis` as a single pointer in their inline `IsActive ()`: element 0, core 0's.

Per-task data

`CTask`'s `TASK_USER_DATA_USER` slot holds the task's `CAddressSpace*` pointer (0 for kernel tasks) — the same one for all the tasks of a process (its threads). The PID, the state (R/S/B/N), the name and the ASID are accessible via the address space and the enumeration API (see §8). The address space counts its tasks (`AddTask / DropTask`; the first is its **main task**, `GetMainTask`); the reaping (`reaper`) calls the termination handler — which drops the task, and frees the address space with the last one — then destroys the `CTask`.

6. Exceptions and vectors

Source: `kernel/arch/aarch64/vectors.S`, `kernel/arch/aarch64/exception.cpp`, `kernel/include/kern/trapframe.h`.

KVectorTable (OUR VBAR_EL1)

16 entries (4 groups × 4 types). A crucial point from hardware bring-up: the **EL1t group (Current EL, SP0)** is the one actually taken by the main thread and the tasks (which run in EL1t). It must route to the real handlers (`SyncEL1Entry / IrqEntry / FIQStub`) and **not** to `BadModeEntry` — otherwise the first tick crashes.

Group	Sync	IRQ	FIQ	SError
Current EL SP0 (EL1t)	SyncEL1	Irq	FIQStub	BadMode
Current EL SPx (EL1h)	SyncEL1	Irq	FIQStub	BadMode
Lower EL AArch64 (ELO)	EIOSync	EIOIrq	FIQStub	BadMode
Lower EL AArch32	BadMode ×4			

IRQ path

`IrqEntry`: `SAVE_TRAP` → **unmasks the FIQ** (`DAIFClr, #1`, required by Circle's `EnterCritical`) → calls Circle's `InterruptHandler` (GIC dispatch + periodic tick + EOI) → `KernelIRQExit` (the crash-log and stall samples;

an app core's stop; **no task switch**: an IRQ at EL1 interrupts kernel code — the apps' IRQs come through `EL0IrqEntry`, §5) → re-masks the FIQ → `RESTORE_TRAP + ERET`.

On cores 2 and 3 (app cores, §14) the same `KVectorTable` is installed: their only IRQ is the stop IPI, and `KernelIRQExit` hands it to `AppCoreOnIRQExit`; a synchronous fault there goes to `AppCoreOnFault` instead of the kernel panic.

Bring-up note: leaving the FIQ masked during `EnterCritical` caused a Circle assertion to fail (`synchronize64.cpp`). Hence the `DAIFClr, #1/DAIFSet, #1` around the call, as Circle's `IRQStub` does.

Trap frame

`TTrapFrame` (800 bytes): `x[0..30]`, `sp_el0`, `elr_el1`, `spsr_el1`, then the full FP/SIMD state — `v[0..31]` (the 128-bit `q` registers), `fpsr`, `fpcr`. Saved by the assembler macro `SAVE_TRAP` onto the stack on exception entry, restored by `RESTORE_TRAP_ERET`.

The FP/SIMD block is saved on **every** trap so floating-point user code keeps its registers across a context switch. The cooperative `TaskSwitch` (Circle) only preserves the callee-saved `d8-d15` mandated by the AArch64 PCS — enough for a voluntary `Yield()`, but **not** enough when the timer preempts a thread mid-computation. Saving the whole register file here (and in `EL0IrqEntry`'s frame, see § "Preemptive scheduling") keeps hardware float correct under both voluntary switches and preemption from the IRQ-exit path. FP/SIMD is enabled at EL0/EL1 at boot (`CPACR_EL1`, `circle/lib/startup64.S`), so the `stp/ldp q-` register forms never trap.

Faults

Every app runs at EL0 (§ *Protected mode* below): a fault in an app arrives through the EL0 vectors and kills that process only — unless it is a translation fault in a lazy region (`v75`, §4 *Demand paging*): the page is filled and the access retried. The kill's `kmsg` says why when the regions tell: (`stack overflow`) within 1 MB below a stack, (`PROT_NONE access`). `SyncHandlerEL1` sees kernel faults only: `UAccessFixup` first (a fault inside a fault-safe copy routine, below: the kapi returns its error, or fills the page and retries), then the `v75` safety net (a lazy page kernel code touched: filled), otherwise it is a kernel bug → the post-mortem console (`DumpAndHalt`) and the crash record.

Fault-safe access to app memory (`kern/uaccess.h`, step 2)

Every kapi checks the pointers it gets from an app **at its entry point** (`kapi_*`; the helpers below them take kernel memory): `sys/uaccess.cpp`, `arch/aarch64/uaccess.S`.

- **Who may point where.** A kernel caller (no scheduler yet, or a task without an address space) is trusted. An app's pointer must lie in the user VA range [`USER_VA_BASE`, `USER_VA_END`), full stop (its `CTask` stack is its kernel stack, never accepted).
- **UserRange / UserRangeAvail:** [`p`, `p+n`) inside what the caller may use, computed as "bytes left from `p`", never `p+n` (no wrap). **UserReadable / UserWritable:** the range, then each 64 KB page translated with the **app's** permissions, AT `S1E0R/W` (`v75`; `S1E1R/W` before) — used where the kernel works directly in the app's buffer (file reads, pipes, sockets, GPU pixels). Since `v75` a page of a lazy region not there yet is **filled** first, and the range is **pinned** until the system call returns: another thread's `vm_unmap` / `DONTNEED` / `mprotect` over it is deferred (§4 *Demand paging*). A probe never yields.

- **Copies:** `UserCopyIn`, `UserCopyOut`, `UserGet`, `UserPut`, `UserStrOut` and `CUserStr` (a kernel copy of an app string, 256 bytes inline, the heap beyond; paths over 511 characters are refused): leaf assembly routines, word-sized when both pointers are 8-aligned (byte-wise otherwise: the display is Device memory). Plain `LDR/STR` after the range check (the A72 has neither PAN nor UAO). (v75) A copy that faulted on an unfilled lazy page (the fixup records `FAR / ESR`) has it filled in C, then runs again — one more page each time, so it ends.
- **The fixup table:** a hand-made `.rodata` list of `{start, end, recovery}`, one per routine; `SyncHandlerEL1` looks the faulting ELR up first and resumes at the recovery label (the routine returns `-1`). It works for kapis reached by `svc` from EL0 too (they run at EL1t).
- Every pointer-taking kapi fails with its usual error value on a bad pointer; the queues (`klog_read`, `mailbox_recv`, `pop_event`, `pop_post`, `readdir`, `vfs_next`, `tcp_accept`, `thread_join`) check before they consume anything; the GPU structs are copied and their inner pointers (`kapi_gpu_frame.pixels`, `kapi_gpu_program.vs/cs/fs`) checked on the copy. Code pointers (handlers, `post`, `thread_create`'s fn) are not data and are not checked. **A new kapi that takes a pointer must do the same.**

The NULL page: Circle maps every identity page, page 0 included, `AP = RW_EL1`, `UXN = 1`: an app's NULL load, store or call faults at EL0 (the app is killed), and a NULL pointer handed to a kapi is refused by the range check.

Per-process handles (`kern/handle.h`, step 1)

`open`, `opendir`, `pipe`, `file_in`, `file_out`, `spawn`, `stdin_stream`, `stdout_stream` return a **per-process handle**, not a kernel pointer: `(generation << 16) | (index + 1)` (never 0, at most `0xFFFFF`, through the same `void *`). The `CHandleTable` is a member of `CAddressSpace` (threads share it; kernel tasks share one kernel table); entries `{object, type FILE/DIR/STREAM/PROCESS, kind FATFS/RAMFS/VFS, generation 1..255, closing, pins}`; 16 entries growing to 4096. A lookup needs index, generation and type to match an open entry; anything else (a forged or stale handle, a raw pointer) gets the kapi's failure value (`read -1`, `fsize 0`, `stream_write -1`, `wait -1`, `proc_done 1`, the closes do nothing). A kapi that may yield **pins** the entry (`CHandleUse`): a close by another thread meanwhile only marks it, the last unpin closes. **The handles still open are closed when the process ends** (`~CAddressSpace → CloseAll`; a stream's last release, which may write to the card, is deferred to the reaper task: `HandlesRunDeferred`). Streams and `CProcess` records are **reference-counted** (a pipe: the creator's handle and each child; a `CProcess`: the spawner's handle and the child), so either side can end first. **Sockets** are owner-checked (the request's pid, also on the network core); a descendant of the owner adopts one on first use (`ftpd` hands its sessions their socket). (v75) `OFILE` entries (`file_open`); (v76) `LSOCK` (a local socket: its handle's value is its socket number) and `SHM` (a shared memory object, the kind is the access) — and every passable object (open file, stream, local socket, shm) can be **carried into another process's table** by `sock_sendmsg` or `spawn_ex2` (§8 v76: *IPC*). `vfs_req_data` / `vfs_reply` only take requests addressed to the calling provider.

Protected mode: apps at EL0 (steps 3–5)

Source: `kernel/arch/aarch64/el0.S`, `el0blob.S`, `kernel/sys/el0.cpp`, `kern/el0.h`. Design and history: [EL0-PROTECTED-MODE.md](#).

Every process runs at EL0 (kapi v74; v73 had it opt-in): the apps, the `/bin` tools, Koton's plugins, the BASIC runner — anything loaded from an ELF. The kernel and its own tasks run at EL1 and are not preempted. (`cmdline.txt appmode= / protected= / appfault= / nullguard=` and `app.txt mode = of v73` are gone; old cards that still set them get no effect.)

- **Mappings** (the `KPAGE_ATTR_APP_*` presets of `layout.h`): data `RW_ALL`, read-only data `RO_ALL`, code (and `code_alloc`, and `v78 PROT_EXEC` regions) `UXN = 0`, `PXN = 1`, everything else `PXN = UXN = 1`; the screen (`fullscreen_direct`) is Device memory: the user-side `memcpy/memset` align their stores. The identity region 0–4 GB keeps Circle's `AP = RW_EL1`, `UXN = 1`: EL0 can neither read, write nor execute it. There is no PAN on the A72: the kernel reads and writes EL0 pages directly.
- **Stacks**: the `CTask` stack (256 KB, `EL0_KSTACK_SIZE`) is the task's **kernel** stack; the user stack (app.txt's `stack`, 1–64 MB) is mapped in the address space; a thread's at `USER_THREAD_STACKS` (32 GB) + (index + 1) × 32 MB with a 16 MB+ guard below.
- **Entry from EL0** (`EL0_ENTRY`, sync and IRQ): `TPIDR_EL1` holds the top of the running task's kernel stack (written by `EL0Return` just before each `eret`, so it follows the scheduler with no switch hook); the full `TTrapFrame` (`x0–x30`, `SP_EL0`, `ELR`, `SPSR`, `q0–q31`, `FPSR`, `FPCR`) is built at `TPIDR_EL1 - 800`, `SP_EL0` pointed at it and `SPSel` set to 0: the handler runs at **EL1t on the task's kernel stack**, so the kapis that yield (a file read, a wait) switch tasks as any kernel code does. **Exit** (`EL0Return`): `DAIF` masked, `SPSel = 1`, `TPIDR_EL1` set, the frame restored (`x0` last), `eret; dsb nsh; isb`. First entry: `EL0Enter(entry, user_sp, arg, 1r)` (a zeroed frame, `SPSR 0x300 = EL0t` with `IRQ/FIQ` on).
- **System calls** (`EL0SyncHandler`): `svc #0` with the table slot in `x8`, arguments in `x0–x7` (no kapi takes more, none on the stack or in FP registers), result in `x0`. Refused (`x0 = 0`): slot 0, out of range, a null kernel slot, and the user-side slots (`memcpy`, `memset`, `memmove`, `pump_events`, `wait_for_exit`, `pump_wait` — the kernel's pump would run user handlers at EL1).
- **IRQ from EL0** (`EL0IrqEntry` → `EL0IrqExit`): Circle's `InterruptHandler` on the kernel stack; on core 0 the crash-log and stall samples, and at the end of the slice `OnPreempt + Yield` right there — **the only preemption point**: an IRQ taken at EL1 (kernel code) never switches tasks; on cores 2–3 `AppCoreOnIRQExit`.
- **A fault from EL0** (any synchronous exception but `svc`) kills the process: a kmsg line `e10: <name> (pid N) killed: ...`, then (2026-10-03) `e10: <name> backtrace: <1r> <1r> ...` — up to 16 return addresses read along the frame-pointer chain (`x29` → {next `x29`, `1r`}, `UAccessCopy`, the chain must climb), the callers to give to `addr2line` (a program built with frame pointers: the POSIX ports and WebKit are) —, a desktop notice (`IpcNotify`), `kapi_exit(-11)`. `SError` still goes to `BadModeEntry`.
- **The per-process kapi table**: at boot `EL0Init` fills two pages shared by every protected process: the **EL0 table**, mapped read-only at `KAPI_TABLE_VA` (14 GB), and the **EL0 code page** at `KAPI_STUBS_VA` (14 GB + 64 KB). Slot `n` points at the stub `movz x8,#n; svc #0; ret; nop` at `+n × 16`; the user-side entries point into the blob (`e10blob.S`, position-independent, at `+8` KB): `memcpy/memmove/memset` in user code, `pump_events` (posts via `pop_post`, then events via `pop_event`, a key handler bracketed by `event_mods` so `get_modifiers` is right), `wait_for_exit`, `pump_wait` (`pump_sleep` then the pump), and the return paths `EL0ThreadReturn` (→ `thread_exit`), `EL0MainReturn` (→ `exit(0)`), `EL0CoreReturn`. Existing binaries run unmodified.
- **Per core** (`EL0CoreInit`): `CNTKCTL_EL1.EL0PCTEN/EL0VCTEN` (the counters at EL0); `SCTLR_EL1` UCI (`dc cvau/ic ivau` for the JITS), `nTWE/nTWI` (`wfe/wfi`), `UCT` (`ctr_el0`), `DZE` (`dc zva`); **`TPIDRRO_EL0 = the core number`** (`kapi_core` in `user/kapi.h` reads it from `v73` — apps built earlier read `mpidr_el1` and are killed at EL0: rebuild them); `PMUSERENR_EL0.EN` only with `cmdline.txt` `e10pmu=1`. `TPIDR_EL0` (the thread pointer, writable at EL0) is the app's: the kernel never writes it but for a thread's start (`v75: thread_create_ex's t1s`, 0 before) and an app-core job's (its caller's value), and Circle's `TaskSwitch` saves and restores it per task — an EL0 preemption goes through it too.
- **Threads**: `thread_create` in a protected process `erets` to `fn` on its user stack, returning through `EL0ThreadReturn`. **App cores**: `core_run` for a protected owner enters `fn` at EL0 on the app's stack; the job's end is an `svc` from `EL0CoreReturn` (→ `AppCoreEL0Done`); any other `svc` or fault → `KAPI_CORE_FAULT`.

- **ID register reads** (v74): on the A72 an `mrs` of an EL1 ID register at EL0 is an undefined instruction; `EL0SyncHandler` emulates it (`EmulateMrs`: the instruction decoded, the value written into its register, `ELR + 4`), as Linux does: `MIDR_EL1`, `REVIDR_EL1` (real), `MPIDR_EL1` (the current core), `ID_AA64PFR0_EL1` (masked to EL0/EL1, FP, AdvSIMD, SVE, DIT, CSV2/3), `ID_AA64DFR0_EL1` (DebugVer only), `DFR1/AFR0/AFR1` and the AArch32 ID registers 0, the other `ID_AA64*` (ISAR, MMFR, PFR1, ZFR0) the hardware's (a snapshot of core 0's at boot). Any other system register or an `msr` kills the app. Counted in `proc_stats`' `emulated`.
- **System-call statistics** (v74): each process counts its `svcs` (a total, a per-second rate over a window of `CNTPCT`, a saturating count per table slot: 820 bytes); `proc_stats(pid, out)` returns them (§8). `ps` (SYSC/s), the Task Manager and `/bin/sysstat` show them.
- The kernel's `TKApiTable` is only the system-call dispatch table (never mapped into an app); its `pump_events`, `wait_for_exit`, `pump_wait`, `memset`, `memcpy`, `memmove` slots are 0 (the EL0 table points them at the user-side code).
- **The tool** `tools/el0scan.sh [file|dir...]` disassembles ELF files and archives and reports any instruction an app cannot run at EL0 (privileged system registers, cache/TLB maintenance other than `dc cvau/cvac/civac/zva` and `ic ivau, eret, hvc, smc...`): run it on the card after a build (`./tools/el0scan.sh sdcard/apps sdcard/bin sdcard/koton`).

Not closed yet: the powerful `kapis` (`reboot`, `kill_pid`, `inject_*`, `screen_grab`, `win_read...`) stay open to every app, and the GPU (an app's V3D shaders can reach physical memory).

7. ELF loader and process model

Source: `kernel/proc/elf.cpp`, `kernel/include/kern/elf.h`, `kernel/proc/image.cpp`, `kernel/include/kern/image.h`, task model in `kernel.cpp`.

Program images (v77): loaded once, shared, kept

*The study and the stages: `docs/ELF-LOADER-PLAN.md`. Until v77 a start read the whole file into the kernel heap, then copied its segments into frames of the process: each byte moved twice, an 80 MB program (the WebKit port's) took 80 MB of kernel heap for a moment and 80 MB of frames **per process**, and every start read the card again.*

The headers first (ElfReadPlan). The ELF header and the program headers are read on their own and checked: the ELF64 magic, `ELFCLASS64`, `EM_AARCH64 = 183`, type `ET_EXEC / ET_DYN`; the program headers inside the file; for each `PT_LOAD` segment with memory, its bytes inside the file and its addresses inside the user area (no wrap); at most 16 such segments; **no two segments in one 64 KB page** (before v77 the second one silently replaced the first one's page — our programs are linked with 64 KB pages, so none did). A file size above the memory size is cut to it, as before. A refusal comes with a reason, which the start's error line shows (`cannot load SD:/bin/x: segment past end of image`).

The image object (TImage). One per program in memory:

- for each **read-only** segment (no `PF_W`: R+X code, R data — 99.8 % of a large program): its 64 KB frames, allocated from the app pool, zeroed, and filled **straight from the file** — no whole-file buffer, no second copy, and the file's other sections (symbols, debug information) are never read. Reads are gathered up to 128 KB where the frames follow each other in memory (a read per page where they do not), with a yield every 128 KB, as the old loop did;

- for each **writable** segment: a copy of its file bytes in the kernel heap (156 KB at most in today's programs), from which every process gets its own pages. A process therefore never needs the file again, which is what lets `pkg` replace a program's file while it runs.

A process's mapping (ImageMap). The read-only segments' frames are mapped **not owned** (`MapPage (... , bOwned = FALSE)`) with `KPAGE_ATTR_APP_CODE (PF_X)` or `KPAGE_ATTR_APP_RODATA`; the writable ones get fresh owned pages (`MapNewPage, APP_DATA, or APP_CODE` if `PF_X`, as before), the copy's bytes, and a zero bss. The regions are noted as before (`KAPI_VMK_IMAGE`). The address space holds one reference on the image (`CAddressSpace::SetImage`), dropped at the very end of its destructor. The caller then runs `SyncDataAndInstructionCache()` once, as `LoadELF` always did.

Nothing writes a shared frame after the load. The loader writes it through the kernel's identity mapping before any page table names it; in every process its pages are `AP = RO_ALL` — read-only for EL0 **and** for EL1 — so a kapi's copy-out into a program's code or constants fails as it always did (`UserWritable` uses `AT_S1E0W`), no vm call can change them (`vm_protect, vm_unmap` and `vm_advise` only work in the mmap arena and the lazy regions), the kernel has no debugger stub that patches code, and the teardown frees only the frames flagged `PAGE_SW_OWNED`. One difference from before: a read-only segment **without** `PF_X` was mapped read-write; it is now really read-only (none of today's programs has one: their constants are in the `R+X` segment).

The key is the program's full path, in its canonical form — `ImageCanonPath`, the one function that the run, the preload, the unload, the query and the file layer's hook all use:

Rule	Example
lower case, the volume included — ASCII letters only	<code>SD:/Apps/MonApp.App/main → sd:/apps/monapp.app/main</code>
the volume always first: the path's own, else the caller's (<code>/x</code> : the root of the working directory's volume; anything else: under the working directory, as <code>ResolvePath</code>); no working directory (the kernel's own launches): <code>SD:/</code> , which is what FatFs makes of a path without a volume	<code>ls</code> in <code>SD:/bin → sd:/bin/ls</code> ; <code>SD:apps/x.app/main → sd:/apps/x.app/main</code>
the spellings of one volume collapse: its name in any case, FatFs's numeric form (<code>0:</code> is the first name of <code>FF_VOLUME_STRS</code> , <code>1:</code> the second...) and the kernel's <code>SD0:</code>	<code>0:/bin/ls, SD0:/bin/ls, sd:/bin/ls → sd:/bin/ls</code>
two volumes stay two keys: <code>SD:</code> (the card's first partition) and <code>SD1:</code> ... <code>SD3:</code> (its others), <code>USB:</code> ...	<code>SD1:/apps/x.app/main → sd1:/apps/x.app/main</code>
<code>/</code> between the names (<code>\</code> too, as FatFs takes it), none doubled, none at the end, no <code>.</code> / <code>..</code> , a name's trailing dots and spaces dropped (FatFs drops them)	<code>SD:\apps//x.app/./main. → sd:/apps/x.app/main</code>
at most 255 characters, 64 names deep; beyond: no key (the program is loaded,	

Rule	Example
never shared)	

Bytes that are not ASCII letters are kept as they are. FAT also folds accented letters, so two spellings of one accented name (É... / é...) are two keys: the file may be in memory twice, and a change made through the other spelling is not seen by the hook below — the wrong *file* is never run for a path, but an old *version* could be, until `unload`. The same holds for a file reached by its 8.3 short name. Onyx's programs have plain ASCII names.

A start (CUserProcessTask::Run, ProgramImage). `ImageOpen (path)`: an image with that key → a reference, **the card is not touched** — no open, no directory lookup (the launchers' existence check asks the image list first; the user stack's size, `app.txt's stack` =, is kept in the image after the first start). None → the file is opened and streamed into a new image. An image **being loaded** by another task → the start waits for it (it looks again every 5 ms; the loader yields), then shares it; if that load fails, the waiters get its error and nothing is left. A task that is loading cannot be killed (it has no address space yet, so no pid).

Lifetime. An image lives while a process maps it or while it is **pinned** (a preload). Dropping the last reference of an image that is not pinned frees its frames (their `wait_word` waiters woken first).

`image_unload` — and the file layer's hook — take the **name** and the pin away at once: no new process maps it, the processes running it keep it, and its frames go with the last of them. Images do not survive a restart.

Where the file changes, the image loses its name (ImageFileChanged). Since a run of a path that has an image never looks at the file, the file kapis tell the image code:

Operation	Hook
<code>remove, path_unlink</code>	the path (a folder: every program under it)
<code>rename, path_rename</code>	both names; a renamed folder: every program under either name
<code>file_open</code> for writing, with <code>KAPI_O_CREAT</code> or <code>KAPI_O_TRUNC</code> ; <code>file_out</code> (truncate or append); <code>save_file</code>	the path, at the open and at the close of the written file (an image made from the half-written file in between is dropped too)
a write to <code><folder>/app.txt</code>	the image of <code><folder>/main</code> stays; it forgets its stack size (read again at its next start)

Not covered: the accented and short-name spellings above; a card changed on a PC (images do not survive a restart, so there is nothing to cover); the kernel's own writes (`lastcrash.txt`, the clock file: never programs); `RAM:` and the providers' volumes (`FTP:`): programs are only run from FatFs volumes. The hook costs one comparison when no image exists, a path canonicalisation and a walk of the image list otherwise; it does no I/O.

Preload (image_preload, /bin/preload). A kernel task (`CPreloadTask`) loads the program's image and pins it; the call returns at once. A start during that load waits for it. A preload is refused when the image would take the app pool's 16 MB reserve (`VmCommitOK`); a process's own start takes what there is, as

before. `pkg` asks `image_list (path)` before it replaces a file: a kept program is unloaded, replaced, and preloaded again (`user/pkg/pkglib.h` move).

Accounting. A process's owned pages (`ps' PAGES`, `vm_stats.resident`, the crash record) no longer include its program's read-only segments: they are counted once, in `meminfo`'s app figure (`g_nUserPages + ImagePagesTotal ()`), and listed by `preload`.

The log. One line per start, in the kernel log (`kmsg, docs/04 §8`): `image SD:/bin/wctest: loaded in 5123 ms, mapped in 3 ms (81856 KB shared, 192 KB private) — loaded (read from the card now), shared (in memory: no read) or shared after a wait; a preload logs image sd:/bin/wctest: loaded in 5123 ms, kept (81856 KB shared).`

Tests: on the PC `sh tools/tests/run_image_test.sh` (the real `image.cpp` and `elf.cpp`, the kernel around them stubbed, ASan: the canonical path, the header checks against crafted files, the load, two processes on the same frames, a start waiting for another task's load, a failed load, the references and the pin, unload while in use, the hook).

`LoadELF(image, size, AS, &entry)`

The old entry point, over a whole file in memory. Nothing in the kernel calls it any more; it goes through the same code (an image of its own, without a path: never shared), then `SyncDataAndInstructionCache()`, and returns `e_entry`.

`CUserProcessTask` — one application = one task

`CUserProcessTask` (subclass of `CTask`; its `CTask` stack is the process's 256 KB kernel stack; the app's **user** stack is 8 MB (v75: lazy, so it costs only what is touched), or more if the app's folder's `app.txt` asks for it, `stack = 16M` — `AppUserStack`, rounded up to 64 KB, 8–64 MB):

1. Creates a fresh `CAddressSpace`.
2. Installs `stdin/stdout`, the process handle, `argv`, `cwd`.
3. Its program's image (v77, *Program images* above): found in memory by its path, else streamed from the file; mapped into the address space (`ImageMap`).
4. `SetUserData(AS, TASK_USER_DATA_USER) + AS->AddTask(this)` (its main task) + `Activate()` (switches `TTBR0/ASID`).
5. Makes the **user stack** below `USER_STACK_TOP` a lazy `STACK` region (8 MB, or `app.txt`'s `stack`; nothing below it: an overflow is a "stack overflow" kill) and **enters the app at EL0**: `EL0Enter (entry, USER_STACK_TOP, 0, EL0MainReturnVA ())` — no return; the task's own stack (`EL0_KSTACK_SIZE`, 256 KB) is from then on only the process's **kernel** stack (its traps, the `kapi` it calls).
6. `main`'s return goes through the blob's `EL0MainReturn` ($\rightarrow$ `exit (0)`); `kapi_exit`, a fault or a kill end the process in the kernel: `ThreadsEndProcess()` (its other threads end with it) and `Terminate()`; the reaper reaps the tasks and frees the address space with the last one.

Threads (v67)

Source: `kernel/sys/thread.cpp`, `kernel/include/kern/thread.h`.

A thread is a `CUserThreadTask`: one more `CTask` whose `TASK_USER_DATA_USER` is the app's `CAddressSpace` (set in its constructor, before it is first scheduled: no `Yield` in between), so the task switch activates the app's page table and every `kapi` sees the same process — window, heap, files, sockets, `cwd`. It enters `fn`

(arg) at **EL0** (`EL0Enter`) on its own user stack, a lazy `STACK` region (v75) at the top of its record's slot, `USER_THREAD_STACKS` (32 GB) + (index + 1) × 32 MB, the rest of the slot unmapped (the guard); 256 KB by default for `thread_create` (16 KB .. 16 MB), 8 MB for `thread_create_ex`. A slot reused at the same size keeps its region; an ended thread's stack pages are dropped (`thread_exit`). Its `TPIDR_EL0` starts at 0, or at `thread_create_ex`'s `tls` (set before `EL0Enter`; Circle's `TaskSwitch` saves and restores it per task from then on); its `CTask` stack is its kernel stack. The timer preempts it in its own code like any app (§5). `fn`'s return, or `kapi_thread_exit`, records its exit code and ends the task; the reaper frees it while the process goes on. A process runs at most `THREADS_MAX` (32) threads besides its main one.

- **The process ends with its main task** (return from `main`, `kapi_exit`), or when any thread calls `kapi_exit`: `ThreadsEndProcess` → `TerminateGroup` (all but the caller). A kill (task manager, `kill_pid`, an orphan) ends the group the same way (§5). A fault in any of its threads kills the whole process (§6, *Protected mode*), never the machine.
- **The per-process state** — `CProcThreads`, made on first use, freed by `~CAddressSpace` (`ThreadsFree`, first): the thread records (`tid` → task, done, exit code; 64: the ended ones are kept until joined, the oldest reused first), the synchronisation objects (256 handles), the posted-call ring (256) and two events: `DoneEv` (a thread ended: its joiners) and `WakeEv` (a post, a window event, the close box: `pump_wait`).
- **The objects** — mutex (recursive, an owner; released when its owner thread ends), event (manual or auto reset), barrier (a count, a round number). A `kapi` runs on core 0 and is not preempted, so an object's check-then-wait is atomic: the objects need no lock. A waiter blocks on the object's `CSynchronizationEvent`, only ever set-then-cleared (Circle's `Pulse` is private) — a condition variable's broadcast; it counts itself in `nWaiters` across the wait and checks its condition again when woken. An object closed while waited on (`sync_close`) is freed by its last waiter (they return -2); those left at the process's end are pulsed — unlinking the killed tasks still on their wait list, which `~CSynchronizationEvent` asserts is empty — and deleted.
- **Posts**: `post (fn, ctx, value)` appends to the ring and pulses `WakeEv`; `pump_events` runs the pending posts (at most one ring's worth) before the window's events, on the calling thread. `pump_wait` points the window's wake-up (`CWindow::SetWake`: `PushEvent` and `RequestExit` pulse it) at `WakeEv`, sleeps if nothing is pending, then pumps; `wait_for_exit` sleeps on it too (16 ms at most, as before).
- **Word waits (v68, a futex)** — `wait_word (addr, expected, ms) / wake_word (addr)`. The word's **physical** address is the key (`WordPhys`: since v75 `UserReadable` — a lazy page filled, pinned for the call, so its frame stays while the caller sleeps — then `AT S1E0R` on the caller's `TTBR0` and `PAR_EL1`; below 4 GB, the kernel's identity map): the same word of a shared surface, mapped at another address in each process, wakes across processes. A waiter (`TWordWaiter`: the address, the value, its process, a `CSynchronizationEvent`) lives on its task's stack, linked in one kernel list under an IRQ spin lock; it is linked, then the word read again, then it waits — with the IRQ masked until it is on the event's list (Circle's `Wait` tests then blocks: a tick's `Set` in between would be lost; `Yield` keeps each task's own `DAIF`). `wake_word` sets the events of the waiters on that address. **Code on an app core changes words without any kapi**: `WordWaitTick`, called by `PeriodicTick` in the 100 Hz timer interrupt, reads every sleeping word through the identity map and wakes those whose value moved (nothing to do, and nothing read, while nobody sleeps). Spurious wakes are allowed (callers loop). A dying process unlinks its waiters first (`ThreadsFree` → `WordWaitsFree`: a killed task's record is on its stack, freed after the batch's handlers). A frame leaving its space (v75: an unmap, `DONTNEED`) wakes its waiters first (`WordWaitsZap`). Test: `/bin/futextest`, `/bin/memtest`.
- **Priority (v68)**: `thread_priority (tid, 1)` makes a thread "real time" (§5): picked first whenever it is ready, as long as it sleeps before its slice ends — for an audio pump.

- **The lists show a process once:** `list_windows`, `list_tasks` and `list_procs` skip the tasks that are not its main task; a thread's name is `<app>:<name>` (or `<app>:<tid>`), which the crash reports' task lists show. `kill` / `kill_pid` refuse the caller's own process.
- **User side:** `umm_malloc` / `umm_free` take a lock (`kapi_lock`: an exclusive swap, a `yield` while it is held — a spin on an app core, which makes no `kapi` call), and `user/libc/onyx_syscalls.c` defines newlib's retargetable locks (the 8 static ones and the 10 `__retarget_lock_*` functions, which keeps newlib's `lock.o` out); a recursive lock's owner is the tid on core 0, the core on an app core. `errno` stays shared.

Launch entry points

- **LaunchApp(name):** builds `SD:apps/<name>.app/main`, reads it into RAM, creates a `CUserProcessTask` (without `stdio`). Exposed via `kapi_launch`.
 - **SpawnProcess(path, args, stdin, stdout, cwd):** creates a `CProcess` handle, increments the streams' refs, creates the task. Exposed via `kapi_spawn` (the terminal for pipes/redirections). `kapi_wait/kapi_proc_done` query the handle.
 - **ExecPath(path, args):** "fire-and-forget" (without `stdio` or handle; the task name is derived from the path). Exposed via `kapi_exec` (the file manager to open a document in an app, or launch a program).
 - **Only ELF programs:** the kernel no longer knows BASIC (the former `BasicRedirect`). The formats a program runs (`.bas` / `.bax` → `SD:/bin/basic`) are chosen in user space: `SD:/etc/runners.ini` + `user/launch.h`, which starts the runner with `kapi_exec_as (path, args, name)` (v49, `ExecPath` with a name), so the process keeps the app's name (`windows`, `raise_app`, `list_windows`).
-

8. The kapi ABI table

Source: `kernel/include/kern/kapi_abi.h`, `kernel/sys/kapitable.cpp`, `kernel/sys/kapi.cpp`. App side: `user/kapi.h`.

The mechanism

Rather than linking the apps against the **kernel symbol addresses** (which move on each rebuild), the kernel publishes a **function-pointer table** (`struct TKApiTable`) at a **fixed virtual address**:

- `KAPI_TABLE_VA = 14 GB` — stable "forever" (between the canvas at 12 GB and the stack at 16 GB).
- The kernel's table is a static variable aligned to 64 KB (`s_Table` in `kapitable.cpp`). `KApiTableInit()` fills all the pointers + the `version` field. It is the **system-call dispatch table** only: it is never mapped into an app.
- What an app sees at `KAPI_TABLE_VA` is the **ELO table** (`ELOInit`, `sys/el0.cpp`), one page shared by every process and mapped **read-only** in each `CAddressSpace` (cf. §4): the same layout and `version`, but slot `n` points at a stub `mov x8, #n; svc #0; ret` in the ELO code page (`KAPI_STUBS_VA`), and the user-side slots (`memcpy`, `memset`, `memmove`, `pump_events`, `wait_for_exit`, `pump_wait`) at routines that run in the app itself (`arch/aarch64/el0blob.S`; §6, *Protected mode*).
- App side, `kapi.h` defines `#define KT ((const struct TKApiTable *) KAPI_TABLE_VA)` and one inline function per entry (`kapi_create_window`, `kapi_open`, ...) that does nothing but call through the table — a plain indirect call into the stub, which makes the system call. `user/kapi_names.h` (generated by `tools/gen_kapi_names.py` from `kapi_abi.h`) names the slots for `/bin/sysstat`.

Consequence: an application binary **embeds no kernel address** and keeps working against any kernel that exposes the same ABI → no rebuild of the apps when the kernel changes.

The *append-only* contract

KAPI_ABI_VERSION = 77. The TKApiTable struct is **strictly append-only**: you never remove or reorder a field; you add new ones **at the end** and you increment the version. An old app only touches the prefix it knows → it stays compatible. The history of additions is annotated in the file (v1 = app_dir, v2 = set_click_handler, v3 = opendir/readdir, v4 = streams/spawn, ... v16 = set_window_theme, v17 = chdir/getcwd, v18 = stdin_stream/stdout_stream, v19 = klog_read, v20 = set_verbose/get_verbose, v21 = the TCP socket calls, v22 = set_pointer_handler, v23 = meminfo, v24 = sbrk, v25 = reboot, v26 = kbd_ready, v27 = set_keymap_data, v28 = get_chrome/draw_text_buf (user-side window chrome), v29 = compat break (kernel-drawn widget API removed, table consolidated), v30 = random (hardware RNG), v33 = ram_detail, v34 = set_wheel_speed/get_wheel_speed, v35 = shared surfaces (surface_*) + shell IPC (register_shell, shell_request, mailbox_send/mailbox_recv), v36 = memset/memcpy/memmove, v37 = tcp_listen/tcp_accept, v38 = screen_grab/inject_pointer/inject_key, v39 = set_menu/get_menu/menu_command, v40 = ipc_register/ipc_lookup, clipboard_set/clipboard_get, set_window_alpha, shutdown, v41 = fullscreen_begin/present_fb/fullscreen_end, v42 = drag_begin/drag_data, get_modifiers/inject_modifiers, v43 = net_ping/net_resolve/net_info, v44 = vfs_register/vfs_next/vfs_req_data/vfs_reply, v45 = wlan_scan, v46 = sound_acquire/sound_release/sound_start/sound_stop/sound_write/sound_status, v47 = sound_instrument, v48 = key_held/inject_key_held, v49 = exec_as, v50 = pad_state, v51 = core_acquire/core_run/core_state/core_release, v52 = gpu_info/gpu_draw, v53 = gpu_texture/gpu_render, v54 = kapi_gpu_vertex3 second (added) colour r2 g2 b2 a2 in place of reserved, KAPI_GPU_B_ALPHATEST(t), v55 = fullscreen_direct, v56 = win_list/win_read/win_raise/win_close, v57 = seek, v58 = code_alloc, v59 = fsize64, v60 = sound_volume/wlan_reconnect, v61 = gpu_program/gpu_render2, v62 = gpu_render3, v63 = gpu_vbuf, v64 = win_minimise/win_geometry/resize_window2 — the modernised CDE desktop's windows: the frame's metrics and title buttons shared with the apps (KAPI_FRAME_*), GUI_EVENT_WINCTL, KAPI_WIN_MINIMISED, the frames' rounded corners, WIN_FLAG_ALPHA), v65 = desk/win_desk — the workspaces (virtual desktops: KAPI_WIN_OFFDESK, KAPI_WIN_DESK, KAPI_DESK_MAX), v66 = screen_set — the resolution changed while running, GUI_EVENT_DISPLAY_RESIZE; a window as big as the screen (was 1024 × 768 at most), v67 = thread_create/thread_exit/thread_join/thread_self, mutex_*/event_*/barrier_*/sync_close, post/pump_wait — threads (§7), their synchronisation objects, calls posted to the event pump; the scheduler's task list has no limit, v68 = sound_config/sound_map — low-latency sound and the mapped PCM ring (§12), wait_word/wake_word — a futex (§7), thread_priority — "real time" threads (§5), midi_read/midi_devices — USB MIDI input (struct kapi_midi_event), v69 = screen_native — the monitor's own resolution (its EDID), set_timezone — the time zone while running, and the window flag WIN_FLAG_FIXED (§10.2) — for Setup, the first-run wizard, v70 = gpu_texture_rect — a rectangle of a GPU texture replaced (no whole re-upload), the frame flag KAPI_GPU_F_ALPHA (a target that keeps its alpha), the GPU's handles shared fairly by the programs using it at once (KAPI_GPU_MAX_TEXTURES 1024 in all, KAPI_GPU_MAX_TEXTURES_AS 512 a program; gpu_vbuf 32 in all, 8 a program) — for the GPU compositing service (user/gpucomp, §15), v71 = vol_info — a volume's size, used and free bytes, type (struct kapi_vol_info, KAPI_VOL_RAM); and the **RAM: volume** (§16): the file calls (open ... rename, save_file, file_in / file_out, chdir, seek, fsize64) reach a file system in memory on RAM: paths — no new call for that, v72 = gpu_render's **compositing blend presets** KAPI_GPU_BLEND_MULCOL ... DSTOUT (5–12, §15: the layer blend modes of user/gpucomp — multiply, screen, plus, subtract, lighten, mask, cut out); an older kernel draws 5–15 as ALPHA (the service checks the version) — no new call, v73 = **protected**

mode (§6): `pop_event`, `event_mods`, `pop_post`, `pump_sleep` — the pieces of the event pump an EL0 app runs on its own side (an EL1 app, as apps were then by default, never needed them); the core number in `TPIDRRO_EL0` (`kapi__core`); the file / stream / process handles per process and every pointer checked (no change for a well-behaved app), v74 = **every process at EL0** (the EL1 mode removed), `proc_stats` (a process's system calls: `struct kapi_syscall_stats`), the ID register reads emulated; the kernel table's six user-side slots 0, v75 = **the POSIX layer's kernel half** (`docs/POSIX-PLAN.md`), 43 entries in three blocks after `proc_stats`: memory (slots 199–206), files and processes (207–228), sockets and poll (229–241), with their structures and the `KAPI_E*` error values (newlib's `errno` numbers: a v75 call returns ≥ 0 , or `-KAPI_Exxx`). **The skeleton**: the entries exist (the slot numbers are `static_asserted` in `kapi_abi.h`) and return `-KAPI_ENOSYS` until their work package lands; the subsections below say what is implemented. `user/kapi.h`'s wrappers also return `-KAPI_ENOSYS` on an older kernel or where a host table (the PC simulator, the Windows / macOS builds) leaves the slot 0, v76 = **IPC between processes** (`docs/POSIX-PLAN.md` §14, WP-IPC: what WebKit2 needs), 11 entries after `poll` (slots 242–252): `sock_pair` (local sockets, `STREAM` / `SEQPACKET` / `DGRAM`), `sock_sendmsg` / `sock_recvmsg` (handles carried between processes), `shm_create` / `shm_open` / `shm_unlink` / `shm_ctl` / `shm_map` (shared memory), `handle_close`, `spawn_ex2` / `get_handles` (handles given to a child); `KAPI_SO_RCVBUF` / `SNDBUF` / `PEERPID` / `DOMAIN`, `KAPI_VMK_SHM` (v76: *IPC* below), v77 = **program images** (§7 *Program images*): a program is streamed from its file once and its read-only segments are shared by its processes — no call changes for that —, 3 entries after `get_handles` (slots 253–255): `image_preload`, `image_unload`, `image_list`; `struct kapi_image_info`, `KAPI_IMG_*` (v77: *program images* below), v78 = **executable memory for a JIT** (WebKit roadmap step 3, `docs/08`): `vm_map` and `vm_protect` accept `KAPI_PROT_EXEC` for anonymous regions; no new entry (v78: *PROT_EXEC* below), v79 = **what the kernel is**: `kernel_info` (slot 256), for `/bin/uname` (v79: *kernel_info* below), v80 = **the cores' load and the network's bytes by process**: `cpu_stats`, `net_stats` (slots 257, 258), for the Task Manager's Processor and Network tabs (v80: *cpu_stats*, *net_stats* below).

The callbacks' value (`gui_handler`: sender, event, value) is the type `gui_value`: `long` on Onyx (64 bits: a pointer event packs its wheel, buttons and position there), `long long` where `long` has 32 bits -- the Windows build of the apps (`pc/Koton`, `docs/03 Koton for Windows`), whose table is filled by a Win32 layer. On Onyx it is the same type as before: no ABI change, no new version.

Categories of exposed functions

Category	Examples
Windowing	<code>create_window(_ex)</code> (the canvas; 0 when the client area is bigger than the screen — <code>g_nScreenWidth/Height</code> ; before v66, 1024×768 — or memory is short — an app must check it: drawing into a null canvas faults, and the app is killed), <code>resize_window</code> (the client size shown, $\leq$ the canvas made at creation; the frame — <code>OuterW/H</code> , the chrome copies' size — follows it, and the app redraws its chrome: <code>wk_decorate_window</code>), <code>move_window</code> , <code>present</code> , <code>exit</code> . Window flags: <code>WIN_FLAG_BORDERLESS</code> , <code>WIN_FLAG_BACKMOST</code> (desktop, bottom

Category	Examples
	band), <code>WIN_FLAG_TOPMOST</code> (the menu bar: top band, never the active app nor the key target; at <code>y=0</code> it reserves its smallest logical height — <code>CWindowManager::TopInset()</code> — so auto-placement and title-bar drags stay below it), <code>WIN_FLAG_TRANSPARENT</code> (client blitted with the magenta key), <code>WIN_FLAG_SYSTEM</code> (a shell component — menu bar, notifications, panel, app list: skipped by <code>list_windows</code>, so never in the taskbar; a plain flag bit, no ABI change). The z-order is three bands: backmost < normal < topmost (<code>Add/RaiseLocked</code> keep them). The key target is the frontmost non-topmost window; the active app (menus, chrome highlight uses the key target) is the frontmost window that is neither topmost, backmost nor borderless.
Menu bar (v39)	<code>set_menu(spec, handler)</code> stores the app's menu spec (≤ 2 KB; lines <code>M<title>, I<id>\t<label>\t<shortcut>, -)</code> + a <code>GUI_EVENT_MENU</code> (14) handler on its <code>CWindow</code>; <code>get_menu(buf, cap, title, tcap)</code> returns the active app's spec + title and a serial that changes with the active window or its menu (0 = none); <code>menu_command(id)</code> queues <code>GUI_EVENT_MENU(id)</code> to the active window (<code>MENU_QUIT</code> = -1 $\rightarrow$ <code>RequestExit</code>, like the close box). Used by <code>menubar</code> + <code>wtk::Menu</code>.
Launch/management	<code>launch</code>, <code>toggle_app</code>, <code>raise_app</code>, <code>exec</code>, <code>kill</code>, <code>kill_pid</code>
Threads (v67)	<code>thread_create(fn, arg, stack_size, name)</code> $\rightarrow$ <code>tid</code> ≥ 2 (main: 1), -1 no memory, -2 too many (32); <code>thread_exit(code)</code> (the main thread: the process); <code>thread_join(tid, timeout_ms, &code)</code> $\rightarrow$ 0, -1 timeout, -2 none / joined already, -3 itself; <code>thread_self</code>. <code>mutex_create/mutex_lock(h, timeout)/mutex_unlock (recursive), event_create(manual,</code>

Category	Examples
	<code>initial)/event_set/event_reset/event_wait(h, timeout),</code> <code>barrier_create(count)/barrier_wait</code> (1 for the last one in), <code>sync_close</code> — handles, 256 per process; timeouts in ms, 0 = only try, <code>KAPI_WAIT_FOREVER</code>. <code>post(fn, ctx, value)</code> → queued for the pump (−1 full: 256); <code>pump_wait(timeout)</code> sleeps until an event / a post / the close box, pumps → what was pending. See §7.
Word waits, priority (v68)	<code>wait_word(addr, expected, timeout_ms)</code> sleeps while the 4-byte word <code>*addr == expected</code> → 0 (woken, or the value differs), 1 timeout (0 ms: only check), −1 bad address (unaligned, unmapped); <code>wake_word(addr)</code> → the sleepers woken. Keyed by the physical address (a word of a shared surface wakes across processes); the 100 Hz tick also reads every sleeping word and wakes those that changed — an app core's write needs no <code>wake_word</code> (≤ 10 ms). <code>thread_priority(tid 0 self / 1 main / ≥ 2, prio 0 / 1 / −1 ask)</code> → the previous one, −1 bad prio, −2 no such thread: a "real time" task is picked first when ready and a tick preempts an app for it, while it yields by itself (§5). See §7.
Enumeration	<code>list_apps, list_windows, list_tasks, list_procs, get_datetime</code>
Widgets	none since v29 (the kernel-drawn <code>add_button...</code> removed: §10.3; apps use wtk)
Events	<code>pump_events, wait_for_exit, should_exit, set_key_handler, set_click_handler, set_pointer_handler</code> (full pointer stream, v22 — incl. <code>GUI_EVENT_PTR_WHEEL</code>, a signed scroll-notch delta in the <code>lvalue</code> wheel field via <code>GUI_PTR_WHEEL</code>)
App-drawn text	<code>draw_text, font_width, font_height</code>
System-call statistics (v74)	<code>proc_stats(pid, out)</code> (pid 0: the caller)

Category	Examples
	→ 0 and <code>struct kapi_syscall_stats { syscalls, emulated; rate, slots; top_slot[8], top_count[8]; reserved[4] }</code> (104 bytes): the <code>svcs</code> since the process started, the ID register reads <code>emulated</code>, the calls per second (the last full window ≥ 1 s), the table's slot count, the 8 slots most called (a slot = the field's index in <code>TKApiTable</code> in 8-byte words, <code>version = 0</code>; <code>user/kapi_names.h</code>, generated by <code>tools/gen_kapi_names.py</code>, names them) → -1 no such process / a kernel task, -2 a bad pointer.
Protected mode (v73)	<code>pop_event(struct kapi_event *ev)</code> → 1 and the next window event <code>{ handler, sender, value, event, mods }</code> (the handler not called), 0 none / no window; <code>event_mods(mods)</code> sets what <code>get_modifiers</code> reports while a key handler runs → the previous value (<code>0xFFFFFFFF = live</code>); <code>pop_post(struct kapi_posted *p)</code> → 1 and the next posted call <code>{ fn, ctx, value }</code> (not run), 0 none; <code>pump_sleep(timeout_ms) = pump_wait</code> without the pump → what is pending, -1 not a process. The user-side <code>pump_events / wait_for_exit / pump_wait</code> of a protected app (§6) are built on them. <code>user/kapi.h</code>: the wrappers (version ≥ 73) and <code>kapi_is_protected()</code>.
Files	<code>open/read/fsize/close, save_file, opendir/readdir/closedir, mkdir/remove/rename, chdir/getcwd</code> (current working directory, inherited by children). <code>fsize</code> (and <code>readdir</code>'s size) is clamped to 4 GB - 1; <code>fsize64(h)</code> (v59) gives an exFAT file's real 64-bit size. <code>rename</code> across two volumes fails (-1): the caller copies then deletes (FatFs' <code>f_rename</code> would otherwise rename inside the source volume). All of these (and the streams, <code>seek</code>, <code>fsize64</code>, <code>chdir</code>) work on RAM: paths too (v71, §16): the path is resolved first (<code>ResolvePath</code>:

Category	Examples
	relative to a current folder on RAM: as well), then a RAM: path goes to <code>sys/ramfs.cpp</code>, a provider's (FTP:) to <code>sys/vfs.cpp</code>, the rest to FatFs; since v73 a handle is a per-process handle (§6, <i>Per-process handles</i>) whose entry records the kind (FatFs, RAM, provider). v75: POSIX open files beside them: <code>file_open/read/write/seek/truncate/sync/stat/close</code> (64-bit offsets, <code>pread / pwrite, O_CREAT/EXCL/TRUNC/APPEND</code>), <code>path_stat/unlink/mkdir/rename/utime</code>, <code>dir_read</code> (255-character names), <code>unlink / rename</code> of open files, <code>-KAPI_Exxx</code> errors (v75: <i>files and processes</i> below).
Volumes (v59)	FatFs volume strings (<code>FF_STR_VOLUME_ID</code>, docs/05 §13): SD: = the SD card's first FAT volume (partition 1, the boot FAT32 one, found as before; SD0: is an alias), SD1: SD2: SD3: = MBR partitions 2–4 (<code>FF_MULTI_PARTITION</code>) (FAT12/16/32 or exFAT, mounted at boot when present), USB:..., FD:, NVME: (declared, not mounted yet). <code>ResolvePath</code>: a volume prefix is upper-cased (<code>sd1:</code> → SD1:, SD0: → SD:), a path starting with <code>/</code> is relative to the current directory's volume root, anything else to the current directory. RAM: (v71) is the RAM volume (§16), not a FatFs one. The four SD volumes share one FatFs lock slot (<code>LockSlot</code>, <code>sys/fslock.cpp</code>): they are one card, one command at a time.
Streams/processes	<code>pipe</code>, <code>file_in/out</code>, <code>stream_read(_nb)/write/close/eof</code>, <code>stdin_read</code>, <code>stdout_write</code>, <code>spawn</code>, <code>wait</code>, <code>proc_done</code>, <code>get_args</code>
Modal dialogs	none (v9/v10 <code>message_box</code>, <code>file_open</code>, <code>file_save</code> removed: §10.5)
Desktop	<code>screen_size</code>, <code>wallpaper_generate</code>, <code>wallpaper_buffer</code>, <code>wallpaper_commit</code>, <code>cursor_pos</code>
Appearance/keyboard	(<code>set_window_theme</code>, v16: removed — the chrome is drawn user-side), <code>set_keymap</code>

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	(load a country map <i>by name</i> — deprecated: the kernel compiles in no maps, so it always returns 0; use <code>set_keymap_data</code>, <code>get_keymap</code>, <code>kbd_ready</code> (USB keyboard attached? v26 — informational; <code>keyb</code> no longer needs to poll it), <code>set_keymap_data</code> (load a layout from a <code>.kmap</code> blob, v27 — records it in a persistent snapshot and installs it on the keyboard whenever it attaches, so it needs no keyboard to be present; returns 1 once the blob is accepted, 0 only on a malformed blob. This removed the old boot race where <code>keyb</code> could time out waiting for USB enumeration and leave the keyboard map-less. Ctrl with -, = / +, 0 (2026-10-01, <code>CInputTask::SetKeyMapData</code>): Circle looks a key held with Ctrl up in the layout's Ctrl column unless it is a letter, and the <code>.kmap</code> files leave that column empty there -- nothing came; the snapshot's empty Ctrl entry of the key whose own character is -, = or +, or 0 (its own, or its Shift one on AZERTY) now gets the keypad's key of that character (<code>KeyKP_Subtract</code> / <code>KeyKP_Add</code> / <code>KeyKP_0</code>), whose string Circle sends with or without Ctrl: the app receives -, +, 0 and <code>get_modifiers()</code> says <code>MOD_CTRL</code> (Jet Browser's zoom keys; letters and the keypad untouched; Ctrl+Shift+= still nothing -- Shift's column wins)), <code>app_dir</code>
Logging / memory	<code>klog_read</code>, <code>set_verbose</code>, <code>get_verbose</code>, <code>meminfo</code> (total/free/app KB + page size, v23), <code>sbrk</code> (per-process heap, v24)
Networking (v21, v37)	<code>net_status</code> (live: 0 as soon as the Wi-Fi association drops — <code>CWPASupplicant::IsConnected ()</code> — not only before the first DHCP bind), <code>tcp_connect</code>, <code>tcp_send</code>, <code>tcp_recv</code>, <code>tcp_close</code>; server side (v37): <code>tcp_listen(port)</code> → listening handle (Circle <code>CSocket::Bind+Listen</code>; -6 = port in use), <code>tcp_accept(h, ip, cap)</code> → blocks until a peer connects, returns a

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	connected handle + the peer IP (Circle Accept). All handles share the 16-slot table in <code>sys/net.cpp</code> and are reclaimed when the owner dies. Used by <code>/bin/telnetd</code> .
Power (v25)	<code>reboot</code> (restart the machine — applies settings read only at boot, e.g. the WLAN config rewritten by <code>wpaconf</code>)
Remote screen (v38)	<code>screen_grab(dst, w, h)</code> — runs <code>CWindowManager::Composite</code> (windows, wallpaper, cursor; <code>bCountFrame = FALSE</code> so the watchdog's fps stays the real compositor's) straight into the caller's <code>w*h 0x00RRGGBB</code> buffer (<code>w/h</code> must be the screen size), or returns 2 without touching it when <code>g_nScreenGen</code> has not changed since the previous grab into the same buffer (<code>vncd</code> then skips the diff / encode); <code>inject_pointer(x, y, buttons, wheel) → OnMouse (+ OnMouseWheel)</code> , <code>inject_key(keys) → OnKey</code> — the same paths as the USB mouse/keyboard. Used by <code>/bin/vncd</code> .
Windows as objects (v56)	For the window-level remote desktop (<code>/bin/rdpd</code>): <code>win_list(out, max)</code> fills <code>struct kapi_win_info</code> bottom to top (<code>CWindowManager::Snapshot</code>): <code>id</code> (<code>CWindow::Id</code> , a serial never reused), owner pid, the client area on the screen (<code>x y w h</code> ; the full-screen window: the whole screen, <code>KAPI_WIN_FULLSCREEN</code>), <code>WIN_FLAG_*</code> , <code>alpha</code> , <code>gen</code> (<code>CWindow::Gen</code> , bumped by every <code>Damage</code> — drawn, moved, resized, raised — and by <code>present_fb</code>), <code>KAPI_WIN_KEYS</code> (the key target), the title, the frame (<code>ow oh = OuterW/H, il it = the insets</code> , 0 without a frame) and <code>chromeGen</code> (bumped by <code>get_chrome</code> , i.e. when the app redraws its frame, by a resize, and once more by the first present after that: the frame read again whole). The state also says <code>KAPI_WIN_MINIMISED</code> / <code>KAPI_WIN_OFFDESK</code> (not shown: <code>rdpd</code> sends a hidden window's pixels only when it is shown again) and the

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	window's desk (v65). <code>win_read(id, part, x, y, w, h, dst, stride)</code>: a rectangle of the canvas (part 0; the full-screen window: its back buffer, or the screen itself when direct) or of a chrome copy (1 active, 2 inactive), clipped. <code>win_raise(id) = Raise, win_close(id) = RequestExit</code> (as the close box). The desktop is listed first as a pseudo-window, <code>KAPI_WIN_DESKTOP</code> (0xFFFFFFFF, backmost, screen-sized, <code>gen</code> = the wallpaper's counter + the backmost windows'): <code>win_read</code> of it (whole only) runs <code>CWindowManager::CompositeDesktop</code> (the wallpaper + the backmost windows) straight into the caller's buffer. A <code>CWindow</code> is never freed (see <code>Composite</code>), so an id found in the snapshot is safe to read.
Generated code (v58)	<code>code_alloc(size)</code> — zeroed memory the app may write and execute (a JIT's code: <code>gcemu</code>'s PowerPC → AArch64 translation), rounded up to 64 KB pages: <code>CAddressSpace::CodeAlloc</code> maps fresh owned frames (<code>MapNewPage</code>, freed at teardown) with <code>KPAGE_ATTR_APP_RWX</code> (EL1 RW, PXN = 0) in a per-process bump arena <code>USER_CODE_BASE</code> (14.25 GB) ... <code>USER_CODE_END</code> (15 GB, 768 MB); 0 when full. Circle clears <code>SCTLR_EL1.WXN</code>, so a writable page may be executable. After writing code the app cleans the D-cache and invalidates the I-cache over it (DC CVAU, DSB ISH, IC IVAU, DSB ISH, ISB).
File seek (v57)	<code>seek(h, pos)</code> — the read position of a file opened with <code>open (f_1seek)</code>; 0, or -1 (a VFS provider's file: not seekable). A file bigger than 4 MB gets its cluster map at its first seek (FatFs fast seek, <code>CREATE_LINKMAP</code>, docs/05 §11), freed by <code>close</code>: any position without walking the FAT. For <code>gcemu</code>'s disc images (1.4 GB, read on demand).
Shared surfaces (v35)	<code>surface_create(w, h)</code> → an id > 0: a

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	page-aligned, physically contiguous 0x00RRGGBB buffer (<code>CSurface</code>, <code>kern/gui/surface.h</code>) owned by the caller; <code>surface_map(id)</code> maps the same frames into the caller (the id travels to another process over IPC) → their address; <code>surface_size</code>, <code>surface_present</code> (yields toward the process composing it), <code>surface_destroy</code>. A process mapping a surface it does not own becomes one of its users (v65, ≤ 4): the frames are freed once the owner let it go (<code>surface_destroy</code>, or its end: <code>DestroyByOwner</code>) and every user has ended too — no process keeps a mapping of freed frames, whichever ends first. The Control Panel's applets draw into their host's surface this way (<code>user/applet_proto.h</code>, the developer guide).
IPC services (v40)	<code>ipc_register(name)</code> makes the caller the service <code>name</code> (≤ 16 services; a name held by a live process is refused; freed when the owner dies — <code>IpcOnProcessGone</code>); <code>ipc_lookup(name)</code> → pid or 0. Messages use the per-process mailboxes (<code>mailbox_send/mailbox_recv</code>), now up to 512 bytes (<code>MAILBOX_MSG_MAX</code>). The kernel itself can post: <code>IpcPost(service, type, data, len)</code> (from pid 0, to a running service: FALSE when it is not registered) and over it <code>IpcNotify(title, text)</code> → the notify service (e.g. "Network ... connected"); Print Screen → the <code>screenshot</code> service (§2, <code>CInputTask</code>). A blocking <code>mailbox_recv</code> sleeps until a message lands (<code>IoWait</code> on the I/O generation; <code>CMailbox::Push</code> calls <code>IoWake</code>) — until 2026-10-03 it was a loop of <code>Yield</code>, the receiver ready for ever: <code>clipd</code>, the one service that waits this way, took most of core 0 (ps: state R, no system call).
Clipboard (v40)	<code>clipboard_set(type, data, len)</code> / <code>clipboard_get(&type, buf, cap,</code>

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	<code>&serial</code>): one typed blob (≤ 64 KB) held by the kernel so it outlives the app that copied (<code>CLIP_TEXT</code> 1, <code>CLIP_FILES</code> 2, <code>CLIP_FILES_CUT</code> 3 — paths <code>\n</code>-separated); the serial bumps on every set. Since the shared clipboard (2026-10-01: the service <code>clipd</code>, <code>user/clipboard.h</code>, <code>docs/clipboard/README.md</code>) it is only a fallback: <code>clipboard.h</code> writes the kernel's copy too, and reads it when <code>clipd</code> cannot be reached; the history (10 items, any format) is <code>clipd</code>'s, in its own memory.
Window opacity / session (v40)	<code>set_window_alpha(0..255)</code> on the caller's window: <code>CWindow::DrawTo</code> blends chrome + client over what is below (<code>BlendRect</code>, magenta-keyed if <code>TRANSPARENT</code>; 0 = not drawn) — used for fades. <code>shutdown(mode): f_mount(0)</code> unmounts/flushes the SD card, then <code>reboot()</code> (mode 1) or ACT LED off + <code>halt()</code> (mode 0).
Full-screen apps (v41)	<code>fullscreen_begin(&w, &h)</code> maps a kernel-owned, screen-sized back buffer (<code>CWindowManager::EnsureFullscreenBuffer</code>, 64 KB-aligned) at <code>USER_FULLSCREEN_CANVAS</code> (15 GB) and makes the caller's window (created if missing, moved to 0,0) the full-screen window: the compositor task skips its frames, <code>OnMouse/OnMouseWheel</code> send the whole pointer stream to it in screen coordinates and it is the key target. <code>present_fb()</code> copies the buffer into the displayed <code>C2DGraphics</code> buffer + <code>UpdateDisplay()</code>, then yields. <code>fullscreen_end()</code> — or the window's removal when the app exits — gives the desktop back. The buffer is made at the first <code>fullscreen_begin</code> and made again, bigger, when the screen has grown since (<code>screen_set</code>, v66; the old one is left allocated, as the wallpaper's: it may still be mapped) — before, it kept its first size and a full-screen app after a

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	change to a larger resolution wrote past its end, over the kernel heap. <code>screen_grab</code> (VNC) returns the full-screen buffer meanwhile. v55 <code>fullscreen_direct(&w, &h, &stride)</code> (after <code>fullscreen_begin</code>): maps the framebuffer the display scans out (CBcmFrameBuffer, the C2DGraphics display, 32 bpp) at <code>USER_FULLSCREEN_SCREEN</code> (15.5 GB), normal uncached (<code>KPAGE_ATTR_APP_SCREEN</code>); <code>present_fb</code> then copies nothing (it only yields); <code>screen_grab</code> (VNC) reads the screen through the same uncached mapping in the grabber's address space (the kernel's own map of the framebuffer is Device memory, slow to read). What is drawn there shows at once (no double buffering: tearing is possible); the GPU renders there directly (below 1 GB, contiguous), so a full-screen GPU frame costs no copy at all. Returns 0 when not possible (keep the back buffer).
Drag & drop (v42)	<code>drag_begin(type, data, len, label)</code> — only while the left button is held — copies the payload (≤ 4 KB: 1 text, 2 \n-separated paths) into a kapi-side buffer and calls <code>CWindowManager::DragBegin(src, label)</code>. While the session lasts, <code>OnMouse</code> sends <code>GUI_EVENT_DRAG_OVER</code> (16) to the window under the cursor (<code>DND_F_LEAVE</code> when it leaves), and <code>Composite</code> draws a label badge next to the cursor (a + when Ctrl is held). At the left-button release, <code>DndFinishLocked</code> sends <code>GUI_EVENT_DROP</code> (15) to the window under the cursor — (<code>flags << 32</code>) (<code>x << 16</code>) <code>y</code>, client coords, <code>DND_F_COPY</code> if Ctrl — and <code>GUI_EVENT_DRAG_DONE</code> (17) to the source: (<code>flags << 32</code>) <code>target pid</code> (<code>CWindow::OwnerPid</code>, set by <code>CreateWindow</code>), flags <code>DND_F_COPY</code> / <code>DND_F_CANCEL</code> (Esc, via <code>OnKey</code>) / <code>DND_F_DESKTOP</code> (no window or a backmost one). All three go to the

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	pointer handler; the normal pointer stream still reaches the source (its capture), so its widgets see the button go up. The target reads the payload with <code>drag_data(&type, buf, cap)</code>. A window removed mid-drag ends it. Modifiers: <code>get_modifiers() = MOD_CTRL 1 / MOD_SHIFT 2 / MOD_ALT 4</code> — from the USB keyboard's raw report (<code>RegisterKeyStatusHandlerRaw</code> in mixed mode, so the cooked key path is unchanged) or <code>inject_modifiers()</code> (<code>vncd</code>, from the RFB Control/Shift/Alt keysyms). Per key event: <code>OnKey</code> stores the modifiers in each <code>GUIEvent</code> (<code>nMods =</code> the global state OR the xterm parameter of <code>ESC[1;<m>X / ESC[n;<m>~</code>, <code>m - 1 =</code> Shift 1 + Alt 2 + Ctrl 4, which Circle's keymap and <code>vncd</code> send for navigation keys); <code>kapi_pump_events</code> sets <code>CWindow::m_nKeyEventMods</code> around the app's key handler, and <code>get_modifiers()</code> returns it while the handler runs — so Shift+arrow selects even when the live state is late or clobbered (VNC, a second keyboard).
Network tools (v43)	<code>net_ping(host, seq, timeout_ms, ip, cap)</code> resolves the host (<code>CDNSClient</code> unless a dotted quad), builds an ICMP echo request (id <code>0x4F4E</code>, 32 data bytes, <code>CChecksumCalculator</code>) and sends it with <code>CNetworkLayer::Send(..., IPPROTO_ICMP)</code>; the reply is read from Circle's secondary ICMP queue (<code>EnableReceiveICMP</code>, enabled only while a ping is in flight, <code>ReceiveICMP</code>), matching type 0 + id + seq + sender, yielding while it waits → RTT in <code>µs</code> or -1 down / -3 unresolved / -4 timeout / -5 send failed. <code>net_resolve =</code> DNS only. <code>net_info =</code> a text dump for <code>netstat</code>: up, hostname, ip, mask, gateway, dns, dhcp lines (<code>CNetConfig</code>), then one tcp <code><handle> listen conn <local port> <remote ip> <pid></code> per socket slot (<code>TSocketSlot.bListen</code>); since v75 also <code>`udp bound <default peer</code>

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User-space file systems (v44)	<code>sys/vfs.cpp</code>, <code>kern/vfs.h</code> — FUSE-like. A provider app calls <code>vfs_register(prefix)</code> (<code>/bin/ftpfs:FTP:</code>, <code>FTP:</code>, <code>FTPS:</code>). <code>kapi_open/read/fsize/close</code>, <code>save_file</code>, <code>opendir/readdir/closedir</code>, <code>mkdir/remove/rename</code> on a path with a registered prefix (<code>VfsHandles</code>) become requests (<code>VfsCall</code>): the calling task fills a slot (<code>op</code>, <code>path</code>, <code>path2</code>, <code>a0–a2</code>, a kernel copy of the payload), sets the provider's <code>CSynchronizationEvent</code> and waits on the slot's own event (200 ms re-checks: provider alive via <code>IpCPidAlive</code>, 120 s timeout). The provider takes it with <code>vfs_next</code> (blocking = up to 0.5 s, so it can also poll its mailbox), reads the payload with <code>vfs_req_data</code>, answers with <code>vfs_reply(id, status, data, len)</code> (copied into a kernel buffer, then into the caller's). Ops: <code>OPEN</code> (→ <code>fid</code> + <code>size</code>), <code>READ</code> (<code>fid</code>, <code>offset</code>, ≤ 64 KB), <code>CLOSE</code>, <code>LIST</code> (packed <code>u32 size</code>, <code>u8 is_dir</code>, <code>name\0</code> entries), <code>SAVE</code>, <code>MKDIR</code>, <code>REMOVE</code>, <code>RENAME</code>. Provider-backed handles live in static tables (<code>s_File</code>, <code>s_Dir</code>), so the file kapis tell them from FatFs <code>FIL/DIR</code> by address. <code>FTP:/FTPS:</code> are auto-started: the first use execs <code>SD:/bin/ftpfs</code> and waits up to 5 s for it to register. A dying provider is dropped and its pending requests fail (<code>VfsOnProcessGone</code>, from <code>IpCOnProcessGone</code>). Streams (<code>kapi_file_in</code>: <code>cat</code>, redirections) go to FatFs (and to the RAM volume on <code>RAM:</code>, §16), not to a provider.
Volumes' room, RAM: (v71)	<code>vol_info(path, out)</code> → 0 and <code>struct kapi_vol_info { total, free, used; files, dirs; flags; type[12] }</code> for the volume of <code>path</code> ("<code>SD:</code>", "<code>SD1:/roms</code>", "<code>RAM:</code>", a relative path: the current folder's), –1 no such volume. A FatFs volume: <code>f_getfree</code> (its free clusters — FSINFO on FAT32, else counted once by FatFs: the first call on a big card can take a moment), <code>type</code>

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	FAT12/FAT16/FAT32/exFAT. RAM: (§16): its size, the pages its files take, what can still be written (also bounded by the free page memory less the reserve), its files and folders, <code>flags KAPI_VOL_RAM</code> (lost at a restart), <code>type RAM</code> . <code>/bin/df</code> prints it; <code>/bin/ramtest</code> uses it to check that the memory comes back.
Wi-Fi scan (v45)	<code>wlan_scan(out, max) → struct kapi_wlan_ap</code> (ssid, bssid, security <code>WLAN_SEC_OPEN/WEP/WPA/WPA2</code> , channel, freq, level dBm, connected), strongest first, one per BSSID. <code>NetWlanScan</code> in <code>sys/net.cpp</code> — no Circle patch : it drives the BCM4343 firmware's <i>escan</i> through <code>CBcm4343Device::Control</code> ("escan 5"), collects <code>ReceiveScanResult</code> messages for ~3.5 s (the firmware's <code>brcmf_escan_result_le</code> layout, as in hostap's <code>driver_circle.cpp</code>), then <code>escan 0</code> . Security from the capability privacy bit + the RSN (48) / WPA vendor (221) IEs; <code>connected = the BSSID GetBSSID()</code> reports while <code>CWPASupplicant::IsConnected()</code> . <code>wpa_supplicant</code> reads the same result queue for its own scans: while it is still looking for its network, a scan here may take its results (it scans again). Used by <code>/bin/wifiscan</code> and <code>wpaconf</code> .
Master volume (v60)	<code>sound_volume(volume 0..10, mute 0/1)</code> (−1 keeps a value) → the volume `
Wi-Fi join (v60)	<code>wlan_reconnect()</code> — <code>NetWlanReconnect</code> (a core-3 request with <code>netcore=1</code>): <code>wpa_supplicant</code> 's <code>SIGHUP</code> handler, caught at link time (— <code>wrap=eloop_register_signal_reconfig</code> , <code>kernel/Makefile</code> ; docs/05 §14) and run from its own event loop (a 0 s <code>eloop</code> timeout) — deauthenticate, read <code>SD:/etc/wpa_supplicant.conf</code> again, rescan, join the highest priority network in range — then <code>CDHCPClient::Restart()</code> (a new lease: another network). 0 asked, −1 no Wi-Fi running. Used by the

Category	Examples
	Wi-Fi menu (wifimenu).
Low-latency sound (v68)	For the sound owner: <code>sound_config(chunk_frames 64..1024 (0: 1024), ahead 1..4 (0: 4))</code> → the latency now in frames, $(ahead + 1) \times chunk$ (−1 not the owner); <code>sound_map()</code> → the mapped PCM ring (struct <code>kapi_sound_ring</code>: one 64 KB page, 8192 frames, <code>wr</code> moved by the app after its frames, <code>rd</code> by the kernel, <code>dry</code> underruns, the chunk / ahead now), mixed with the voices and the stream until the owner releases the output — plain memory, so an app core fills it (<code>kapi_sound_ring_write</code>). Both are back to the defaults / off at <code>sound_release</code> or the owner's death. See §12. Test: <code>/bin/ringtest</code>.
Sound (v46)	<code>sound_acquire</code> (1 ok / 0 busy / −1 no audio: the caller becomes the owner; the output starts on first use), <code>sound_release</code>, <code>sound_start(voice 0..15, milliHz, wave SOUND_SQUARE/SINE/TRIANGLE/SAW/NOISE, volume 0..255)</code> (plays until stopped), <code>sound_stop(voice or -1)</code>, <code>sound_write(s16 stereo frames, n)</code> → frames taken (PCM ring, non-blocking), <code>sound_status(&rate, &free, &owner)</code>. Non-owners get −1; the owner's exit silences it. See §12.
Run as (v49)	<code>exec_as(path, args, name)</code> → <code>ExecPath</code> with the process named <code>name</code> instead of after the path (1 = started). User space runs a format's program this way (<code>launch.h</code>: <code>SD:/bin/basic</code> for an app's <code>main.bax</code> is named after the app).
Monitor, time zone (v69)	<code>screen_native(&w, &h)</code> → 1 and the monitor's own resolution, 0 unknown: the EDID's first block (the firmware's <code>PROPTAG_GET_EDID_BLOCK</code>, Circle's <code>CBcmPropertyTags</code>), its header checked, the first detailed timing descriptor (bytes 54..71, the preferred timing) read — the active pixels, 8 low bits and 4 high ones of a shared byte, horizontally

Category	Examples
	(bytes 2, 4) then vertically (5, 7); 0 when it is a display descriptor or too small (no monitor, an analog adapter). <code>set_timezone(minutes)</code> → 1: the local time's offset from UTC now (−720 .. 840, <code>CTimer::SetTimeZone: get_datetime</code>, the menu bar's clock), 0 out of range — <code>system.ini</code>'s <code>timezone=</code> sets it at boot. Setup (the first-run wizard) marks the monitor's size "Best" and changes the zone as it is picked.
USB MIDI (v68)	<code>midi_read(ev, max)</code> → up to <code>max</code> <code>struct kapi_midi_event</code> (<code>time_us</code> — the kernel's μs clock, <code>CTimer::GetClockTicks</code>, which <code>kapi_clock_us</code> reads user side, on an app core too —, <code>cable</code>, <code>status</code>, <code>data1</code>, <code>data2</code>, <code>device</code> (its <code>umidiN</code>), <code>length</code> 1..3), oldest first, never waits; −1 bad arguments. One queue for the system (256 events, the newest dropped when full). <code>midi_devices()</code> → attached. Circle's USB device factory makes a <code>CUSBMIDIHostDevice</code> for every class-compliant MIDI interface (<code>int1-3-0</code>) and its <code>CUSBMIDIDevice</code> names itself <code>umidiN</code>; the input task finds <code>umidi1..4</code> every 100 ms (a hot plug too), registers its packet handler (<code>MidiPacket</code>: at USB-completion time, stamps the packet and queues it under an IRQ spin lock) and a removed handler that frees the slot. No Circle change. Test: <code>/bin/miditest</code>.
Gamepads (v50)	<code>pad_state(index, out)</code> → 1 and <code>struct kapi_pad</code> filled for USB gamepad 0..3 (<code>KAPI_PAD_MAX</code>), else 0: <code>vid/pid</code>, <code>props</code> (Circle's <code>TGamePadProperty</code>, bit 0 = a known mapping), <code>focus</code> (the caller's window has the keyboard), <code>seq</code> (reports received), <code>nbuttons/buttons</code>, <code>naxes/axes[16]</code> (value, min, max), <code>nhats/hats[6]</code> (0..7 = N..NW). Raw state: for pads Circle knows (Xbox 360 / One, PS3 / PS4, Switch Pro) <code>buttons</code> are its <code>TGamePadButton</code> bits, for other HID pads the report's own. The input task finds

Category	Examples
	<code>upad1..4</code> (Circle's names) every 100 ms, registers a status handler that copies each report into a slot under a sequence count (odd while writing: the handler runs at USB-completion time), and a removed handler that frees the slot. The mapping to one button set is user space (<code>user/gamepad.h</code> , <code>SD:/etc/gamepad.ini</code>).
App cores (v51)	<code>core_acquire()</code> → 2 or 3 (a free app core, now the caller's) or -1; <code>core_run(core, fn, arg, stack_top)</code> → 0, or -1 (not yours / still running / fn or the stack not a user address): the core calls <code>fn (arg)</code> in the caller's address space on the given stack (16-byte aligned, the caller's memory); <code>core_state(core)</code> → <code>KAPI_CORE_IDLE</code> (0: fn returned), <code>KAPI_CORE_RUNNING</code> (1), <code>KAPI_CORE_FAULT</code> (-2: fn faulted and was stopped, logged to <code>kmsg</code>) or <code>KAPI_CORE_NOTYOURS</code> (-1); <code>core_release(core)</code> stops <code>fn</code> if it runs and frees the core (done at the app's exit anyway). <code>fn</code> makes no kapi call and no allocation. See §14.
GPU (v52)	<code>gpu_info(buf, cap)</code> → 1 (the V3D is up; <code>buf = "V3D 4.2 (1 core)"</code>) or 0 (<code>buf</code> says why); the first call brings the GPU up. <code>gpu_draw(v, n, clear, pixels, w, h, stride)</code>: <code>n</code> vertices struct <code>kapi_gpu_vertex { float x, y, z; u8 r, g, b, a; }</code> (a triangle list; normalized device coordinates, <code>y</code> up, <code>z</code> -1 near ... 1 far; depth test <i>less</i>, both faces; colours interpolated) rendered by the GPU into <code>pixels</code> (0x00RRGGBB, <code>w × h ≤ 2048</code>, <code>stride</code> pixels a row) after clearing it to <code>clear</code> (0xRRGGBB) → 0, -1 no GPU, -2 bad arguments / too many vertices (<code>KAPI_GPU_MAX_VERTS = 196608</code>), -3 the GPU did not finish (it is then left off). See §15.
GPU (v53)	<code>gpu_texture(handle, pixels, w, h, stride)</code>: a texture of <code>w × h (≤ 2048)</code> pixels 0xAARRGGBB; <code>handle < 0</code> makes one (<code>≤ 256</code> in all; v70: <code>≤ 1024</code> in all, <code>≤</code>

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	512 a program), ≥ 0 replaces its pixels, <code>pixels = 0</code> frees it → the handle, -1 no GPU, -2 bad arguments, -4 no memory / no free handle; a program's textures are freed when it ends. <code>gpu_render(f, v, nv, b, nb)</code>: one frame into struct <code>kapi_gpu_frame { pixels, w, h, stride, clear, flags }</code> (<code>KAPI_GPU_F_KEEP</code>: drawn over the pixels instead of clearing them) of <code>nv</code> vertices struct <code>kapi_gpu_vertex3 { float x, y, z, w, s, t; u8 r, g, b, a; u8 r2, g2, b2, a2; }</code> (v54: the second colour is added after the texel × colour product and clamped to 1 — 0 for the v53 behaviour) in <code>nb</code> (≤ 4096) batches struct <code>kapi_gpu_batch { first, count, texture, flags, float matrix[16]; }</code>: each batch draws its triangles with its own matrix (row by row, clip = $M \cdot (x \ y \ z \ w)$, transformed by the GPU, which divides by <code>w</code> and clips; <code>KAPI_GPU_B_NOMATRIX</code> = identity), texture (-1: colour only, else texel × colour; <code>B_LINEAR</code>, <code>B_WRAP_S/T(REPEAT, CLAMP, MIRROR)</code>), depth test (<code>B_ZFUNC</code>: less by default, ... always) and writes (<code>B_NOZWRITE</code>), culling (<code>B_CULL_BACK/FRONT</code>, front = counter-clockwise, y up), alpha test (v54: <code>B_ALPHATEST(t)</code>, fragments whose $\alpha < t / 255$ are discarded) and blending (<code>B_BLEND(ALPHA, ADD, MUL, PREMUL)</code>; v72: also the premultiplied compositing presets <code>MULCOL</code> (colour $s \cdot d + d(1 - sa)$, alpha kept), <code>UNDER</code> ($s(1 - da) + d$, alpha over), <code>SCREEN</code> ($s + d(1 - s)$), <code>PLUS</code> ($s + d$), <code>RSUB</code> ($d - s$, alpha kept), <code>LIGHTEN</code> (max), <code>DSTIN</code> ($d \cdot sa$, colour and alpha: a mask), <code>DSTOUT</code> ($d(1 - sa)$: cut out) — each a pair of V3D blend equations, colour and alpha apart) → 0, -1, -2, -3 as <code>gpu_draw</code>. The kernel gives the V3D only well-formed triangles: each one is transformed by its batch's matrix on the CPU and clipped against the near plane ($z \geq -w$, $w > 0$) and a guard band $4 \times$ the screen (<code>kern/v3d_clip.h</code>, the

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	batches then NOMATRIX; test <code>tools/tests/run_v3d_clip_test.sh</code>) — Ocarina of Time's triangles crossing the eye plane, projected ~50 000 screens away, wedged the GPU and froze the whole Pi. See §15.
GPU (v61)	<code>gpu_program(handle, p)</code>: the app's own shaders — <code>struct kapi_gpu_program { vs, cs, fs; nvs, ncs, nfs; inputs, csInputs, csOutputs, varyings, flags }</code>, three V3D 4.2 QPU programs (≤ 4096 instructions each; built at run time by <code>user/v3d/qpu.h</code>) copied into GPU memory; a vertex is <code>inputs</code> floats (4..64, the clip-space $x\ y\ z\ w$ first; the coordinate shader reads the first <code>csInputs</code>, writes <code>csOutputs</code> ≥ 6: $X_c\ Y_c\ Z_c\ W_c\ X_s\ Y_s$), the vertex shader writes $X_s\ Y_s\ Z_s\ 1/W_c$ then <code>varyings</code> (≤ 64); flags <code>KAPI_GPU_P_FS_4WAY</code> (4 threads, else 2), <code>P_FS_FINAL</code> (no thread switch — unreliable on wide targets, see §15), <code>P_FS_ZWRITE</code>; <code>handle < 0</code> makes one (≤ 256), ≥ 0 replaces it, <code>p = 0</code> frees it $\rightarrow$ the <code>handle</code>, -1, -2, -4 as <code>gpu_texture</code>; freed when the program ends. <code>gpu_render2(f, v, nv, stride, b, nb, uni, nuni)</code>: one frame (as <code>gpu_render</code>) of <code>nv</code> vertices of <code>stride</code> floats in <code>nb</code> batches <code>struct kapi_gpu_batch2 { first, count, program, flags, blend, wmask, scissor[4], vsUni, vsNUni, csUni, csNUni, fsUni, fsNUni, tex[8], texFlags[8], texUni[8] }</code>: each with its program, its uniforms (three ranges of <code>uni[nuni]</code>, $\leq 2^{20}$ words), up to 8 textures — the kernel writes each one's TMU words <code>p0</code> (texture state, 16-bit float RG / BA) and <code>p1</code> (a sampler from <code>texFlags</code>: <code>B_LINEAR</code>, <code>B_WRAP_S/T</code>) at <code>fsUni + texUni[i]</code> —, depth / cull flags (<code>B_ZFUNC</code>, <code>B_NOZWRITE</code>, <code>B_CULL_*</code>), blending <code>KAPI_GPU_BLEND2(cSrc, cDst, aSrc, aDst, cEq, aEq)</code> (the V3D factors and equations), a colour write mask (<code>wmask</code>: the channels not written) and a scissor ($x\ y\ w\ h$ in the target, top-

Category	Examples
	left origin; $w \leq 0$: none) $\rightarrow 0, -1, -2, -3$. The triangles are clipped on the CPU as for <code>gpu_render (V3DClipTriangleN</code>: every float of the vertex interpolated). See §15.
GPU (v62)	<code>gpu_render3(f, v, nfloats, b, nb, uni, nuni, view)</code>: <code>gpu_render2</code> with each batch's vertices where the app keeps them — <code>struct kapi_gpu_batch3 { struct kapi_gpu_batch2 b; unsigned off, stride; }; b.count</code> vertices of <code>stride</code> floats ($\geq$ the program's inputs, ≤ 64) from float <code>off</code> of <code>v[nfloats]</code> (<code>b.first</code> unused) — and their x / y framed by the kernel on the way, $x' = \text{view}[0] \ x + \text{view}[1] \ w, y' = \text{view}[2] \ y + \text{view}[3] \ w$ (view 0: as they are): no common array for the app to make first $\rightarrow$ as <code>gpu_render2</code>. gcemu's TEV renderer gives it its recorder's frame as it is. See §15.
GPU (v63)	<code>gpu_vbuf(bytes)</code>: memory the GPU reads too — low, physically contiguous, mapped into the program (the surface arena); up to 8 a program (v70: 32 in all; 8 in all before), 64 MB each, freed when it ends $\rightarrow$ its address, 0 none. <code>gpu_render3</code> with <code>v[nfloats]</code> inside one draws the vertices where they are: their x / y framed in place (the program draws the same vertices again with <code>view</code> 0), the triangles inside every plane drawn as runs of them, the others clipped into the buffer's end past <code>nfloats</code> (keep room there) — nothing copied.
GPU (v70)	<code>gpu_texture_rect(handle, x, y, w, h, pixels, stride)</code>: the rectangle <code>x, y, w $\times$ h</code> of texture <code>handle</code> (the caller's) replaced by <code>pixels</code> (0xAARRGGBB, <code>stride</code> pixels a row), the rest kept $\rightarrow 0, -1$ no GPU, -2 bad arguments (not the caller's handle, the rectangle not inside the texture). The texels are stored where the texture unit reads them for the texture's size (<code>StoreRect</code>,

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	<code>kern/v3d_tiling.h</code>: a utile row — 4 texels, 16 contiguous bytes in every layout — per offset computed) and only the span of bytes written is cleaned from the CPU caches; it waits for a frame in flight, as <code>gpu_texture</code>. <code>KAPI_GPU_F_ALPHA</code> (a frame's flag, <code>gpu_render / 2 / 3</code>): the target is 0xAARRGGBB premultiplied — its alpha loaded with <code>KAPI_GPU_F_KEEP</code> (not forced to 1), blended, written (the colour write mask no longer spares it in the direct mode), stored, and cleared to <code>clear</code>'s top byte; without it nothing changes (0x00RRGGBB: the direct mode leaves the top byte, the copy mode writes 0). Handles: <code>gpu_texture</code> gives 1024 in all (256 before), 512 at most a program (−4 past that), so an emulator's texture cache and the browser's layers fit side by side; <code>gpu_vbuf</code> 32 blocks in all, 8 a program (8 in all before). See §15 <i>Sharing the GPU</i> and <i>The compositing service</i>.
Windows (v64)	The modernised CDE desktop's windows. <code>win_minimise(id)</code>: window <code>id</code> (0: the caller's) minimised — not drawn, not hit, never active nor the keys' target — until <code>win_raise / raise_app</code> brings it back (<code>KAPI_WIN_MINIMISED</code> in <code>win_list</code>'s state) → 0 / −1. <code>win_geometry(out)</code>: <code>struct kapi_win_geom { x, y, w, h; cw, ch; ax, ay, aw, ah; state }</code> — the caller's whole window (frame included), its client size, the work area (the screen less the menu bar at the top and the topmost windows standing on the bottom edge: the dock) → 0 / −1. <code>resize_window2(w, h, &stride)</code>: as <code>resize_window</code>, but the canvas and the frame's copies grow past their first size when needed (new memory at the same addresses; their pixels are lost: redraw, <code>get_chrome</code> again) → the canvas and its stride, 0 (no memory: the size kept). With it: the frame's metrics

Category	Examples
	<code>KAPI_FRAME_TITLE_H</code> 28, <code>KAPI_FRAME_BORDER</code> 4, <code>KAPI_FRAME_RADIUS</code> 8 (the chrome copies' top byte: a transparency, heeded in the corner squares), the title buttons' places <code>KAPI_FRAME_BTN_W/H/Y/EDGE/STEP</code> (the window menu at the left; close, maximise, minimise from the right: <code>KAPI_FRAME_MENU/CLOSE/MAXIMISE/MINIMISE</code>); <code>GUI_EVENT_WINCTL</code> (18: the window menu, maximise — for the app); <code>WIN_FLAG_ALPHA</code> (32: a borderless window's pixels carry their transparency). See §10.2.
Screen size (v66)	<code>screen_set(w, h)</code>: the screen's resolution now (640 × 480 .. 2560 × 1600, <code>w</code> even; <code>ScreenResizeRequest</code>, <code>kernel.cpp</code>). The compositor does it between two frames (the display DMA is idle — each present waits for its end): <code>C2DGraphics::Resize</code> (Circle's: the frame buffer asked of the firmware again at that size, the drawing buffer made again), the old size back if the firmware refuses; then <code>CWindowManager::OnScreenResized</code>: the cursor and every window kept on the screen (moved in when past the right / bottom edge; a window parked at a negative place is left there), the app-written wallpaper dropped (the next <code>wallpaper_buffer</code> is a new buffer of the new size — the old one is left allocated: an app may still have it mapped), and <code>GUI_EVENT_DISPLAY_RESIZE</code> (19, <code>lValue = w << 16 h</code>, <code>GUI_DISPLAY_W/H</code>) to every window's pointer handler. The caller waits for it → 0; −1 out of bounds; −2 not now (a full-screen app owns the display — its mapping of the frame buffer would be stale —, the debug console, another change under way); −3 the firmware refused it (the old size kept). <code>screen_grab</code> refuses the old size (<code>vncd</code> / <code>rdpd</code> see it and take the new

Category	Examples
	one). Not kept across a reboot: <code>cmdline.txt</code> 's <code>width=</code> / <code>height=</code> are (the Display applet writes both). See §10.2.
Workspaces (v65)	The virtual desktops. <code>desk(set, count)</code> : <code>set ≥ 0</code> shows desk <code>set</code> , <code>count > 0</code> sets how many there are (1 .. <code>KAPI_DESK_MAX</code> = 8; the dock keeps 1–6; the windows of the desks dropped go onto the last one); <code>−1 / 0</code> keep them → the current desk`
Held keys (v48)	<code>key_held(key)</code> → 1 while the key is held and the caller's window has the keyboard (<code>KeyTargetLocked</code>), else 0 — for games, since key events only report presses. Keys: <code>KEY_UP/DOWN/LEFT/RIGHT</code> , <code>KEY_ENTER</code> , 27, ' ', 'a' .. 'z' (the US position of the key), '0' .. '9'. The WM keeps two bitsets over the logical codes: the USB one, rebuilt from every raw report (<code>KeyRawStub</code> → <code>SetUsbHeld</code> , HID usage → key), and the injected one (<code>inject_key_held(key, down)</code> , from <code>vncd</code> 's RFB key down / up); <code>KeyHeld</code> ORs them.
FM (v47)	<code>sound_instrument(voice, const struct kapi_fm_instrument *)</code> — a 2-operator FM instrument (OPL2 style, <code>struct kapi_fm_op op[2] =</code> modulator / carrier: <code>mult, level, ksl, attack, decay, sustain, release, wave, flags</code> <code>FM_SUSTAINED / FM_TREMOLO / FM_VIBRATO / FM_KSR</code> ; <code>feedback, connection</code>) for that voice; then <code>sound_start(voice, milliHz, SOUND_FM, volume)</code> . Owner only. See §12.
Memory primitives (v36)	<code>memset</code> , <code>memcpy</code> , <code>memmove</code> — since v74 user-side routines of the ELO code page (<code>el0blob.S</code> , no system call; general registers only; they align the destination for the Device-memory framebuffer), the kernel table's slots 0 (before: Circle's kernel implementations). <code>user/kapi.h</code> wraps them as weak <code>kapi_memset/kapi_memcpy/kapi_memmove</code> symbols, and the freestanding app

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	Makefiles alias the C names onto them (<code>-Wl,--defsym,memset=kapi_memset, ...</code>): GCC may emit these calls on its own (array/struct initialization, copies) even with <code>-ffreestanding</code> , and freestanding apps have no libc. Newlib programs keep newlib's own.
Crypto (v30)	<code>random</code> (fill a buffer with random bytes — a software PRNG (splitmix64) seeded from the high-resolution timer: the BCM2711's hardware RNG stalls the bus in this setup, <code>sys/kapi.cpp</code> ; not cryptographically strong; for seeding — the TLS entropy source in <code>user/tls/onyx_tls.hpp</code> feeds mbedTLS's CTR_DRBG from it)

All the functions run **in the kernel (EL1), on the calling process's task**: the app's `svc` enters on that task's own kernel stack with the app's page table still active, so the kernel reads and writes the app's memory in place — **after checking every pointer** (`kern/uaccess.h`, §6). A call that waits (a file read in pieces, a socket, `wait`) yields there, on that kernel stack, while the compositor and the other apps keep running.

v75: memory

Work package WP-MEM (`docs/POSIX-PLAN.md` §3.1; `kernel/sys/vm.cpp`, `kern/vm.h`; the threads' two in `sys/thread.cpp`). The mechanism — lazy regions, pins, the deferred zap, the TLB rules, the OOM policy — is §4 *Demand paging*. Every entry returns ≥ 0 or `-KAPI_Exxx`. Sizes: 64 KB pages; addresses and lengths are rounded to them where noted.

Slot	Entry	Does → returns
199	<code>vm_map (addr, len, prot, flags)</code>	A zero-filled <code>ANON</code> region of <code>len</code> (rounded up) in the mmap arena [<code>34 GB, 60 GB</code>), filled on first touch (<code>KAPI_MAP_POPULATE</code> : now, a yield every 64 pages, a failure not reported). <code>addr</code> is a hint (taken if free, else the lowest gap) unless <code>KAPI_MAP_FIXED</code> (aligned, inside the arena; replaces what is there) or <code>KAPI_MAP_FIXED_NO_REPLACE</code> (<code>-EEXIST</code> if anything is there). <code>prot</code> = <code>KAPI_PROT_NONE/READ/WRI</code>

Slot	Entry	Does → returns
		TE, v78 EXEC (executable at EL0: a JIT; before v78: -ENOTSUP). → the address / -EINVAL (len 0, a bad FIXED), -ENOMEM (no room, 4096 regions, or a writable map larger than the free app pool – 16 MB without KAPI_MAP_NORESERVE)
200	vm_unmap (addr, len)	Inside the arena (holes allowed; regions split): the pages dropped (a pinned one when its kapi ends) → 0 / -EINVAL (unaligned, outside the arena) / -ENOMEM (a split at 4096 regions)
201	vm_protect (addr, len, prot)	ANON regions covering the range without a hole: their protection (split / merged), the present pages re-protected + TLBI (a write taken from a page under another thread's kapi: when it ends) → 0 / -EINVAL / -ENOMEM (split cap) / -ENOTSUP (EXEC over an SHM region; before v78: any EXEC)
202	vm_advise (addr, len, advice)	Lazy regions (ANON, HEAP, STACK) covering the range: KAPI_MADV_WILLNEED fills (-ENOMEM); DONTNEED / FREE drop the pages — zeros on the next touch (ANON and HEAP only); NORMAL / RANDOM / SEQUENTIAL nothing → 0 / -EINVAL
203	vm_query (addr, out)	struct kapi_vm_region { start, end, prot, kind (KAPI_VMK_ANON/HEAP/STACK/IMAGE/FIXED), resident (pages

Slot	Entry	Does → returns
		present), flags (KAPI_VMF_LAZY) } of the region holding addr → 0, or of the next one above → 1; -ENOMEM none above, -EFAULT
204	vm_stats (pid, out)	struct kapi_vm_stats { resident (bytes of owned frames, page tables included), lazy (VA of lazy regions), writable (VA of writable regions), faults (pages filled), pt_bytes, limit (0) }, pid 0 = self → 0 / -ESRCH / -EFAULT
205	thread_create_ex (attr)	struct kapi_thread_attr { fn, arg, stack_size (0 = 8 MB; 16 KB .. 16 MB, lazy), tls (its initial TPIDR_EL0), name, flags (KAPI_THREAD_DETACHED: no join, its record freed when it ends), prio (0, 1 = "real time"), reserved[2] = 0 } → tid ≥ 2 / -EAGAIN (32 running) / -ENOMEM / -EINVAL / -EFAULT
206	thread_info (tid, out)	struct kapi_thread_info { stack_lo, stack_hi (its top: the initial SP), tid, state (0 running, 1 ended, joinable), guard (unmapped bytes below stack_lo) }, tid 0 = self, 1 = main → 0 / -ESRCH / -EFAULT

What changed for every app, with no call: the main stack (8 MB by default), the threads' stacks and the heap are **lazy** (a process's resident memory drops by its untouched stack — 1 MB to 8 MB before — and heap); `sbrk` shrinking returns pages; an overflow of a stack is a clean kill with "stack overflow" in `kmsg`; a page fault the app pool cannot serve kills the app (`KAPI_PROC_OOM`, -9), not the system. The kernel's probes see the app's permissions (AT `S1E0*`). Test: `/bin/memtest` (docs/04).

v75: files and processes

Work package WP-FILE/PROC (docs/POSIX-PLAN.md §3.2): `kernel/sys/ofile.cpp` + `kern/ofile.h` (slots 207–221), `kernel/sys/procx.cpp` + `kern/procx.h` (222–228), the RAM: node calls in `sys/ramfs.cpp`, the

pipes in `sys/stream.cpp`. Every call returns ≥ 0 or `-KAPI_Exxx`. Host test `sh tools/tests/run_ofile_test.sh` (the real `ofile.cpp` over the fork's FatFs on a RAM disk — SD: FAT32, SD1: exFAT — and the real `RAM:`; it ends by checking that every cluster came back); `run_ramfs_test.sh` covers the `RAM:` node calls too. Pi tests `/bin/filetest`, `/bin/proctest` (docs/04).

Open files (`HANDLE_OFILE`). `file_open (path, KAPI_O_*, mode)` → a handle (> 0). A FatFs file has **one node** (`TFNode`) whatever the number of its openers: one `FIL`, opened `FA_READ`, re-opened `FA_READ` | `FA_WRITE` when its first writer comes (FatFs keeps no share locks, `FF_FS_LOCK 0`: two `FIL`s writing one file would corrupt it). Its key is the absolute path (`ResolvePath`), compared case-insensitively. A handle is an **open-file description** (`TOFile`): the node, its own 64-bit offset, its access mode, `O_APPEND` (`dup` is the libc's: it counts its references). Each call holds the node's lock across its `f_lseek` + `f_read` / `f_write` (FatFs yields while the card works); the node table has a lock of its own (open, close, unlink, rename; the order is table → node). Both are sleeping locks, and each call is a **no-kill** section from its start to its end.

Call	Semantics
<code>file_open</code>	<code>O_RDONLY</code> / <code>WROONLY</code> / <code>RDWR</code> (3: <code>EINVAL</code>), <code>O_CREAT</code> , <code>O_EXCL</code> (<code>EEXIST</code>), <code>O_TRUNC</code> (writable only), <code>O_APPEND</code> ; a folder or a volume root → <code>EISDIR</code> ; <code>mode</code> ignored (FAT has a read-only bit only); a provider's path (<code>FTP:</code>) → <code>ENOTSUP</code> ; a bad path pointer <code>EFAULT</code> , over 511 characters <code>ENAMETOOLONG</code>
<code>file_read (h, buf, len, off)</code>	<code>off = -1</code> : at the handle's offset (advanced), else a pread; ≤ 1 GB a call; 0 at the end; <code>EBADF</code> on a write-only handle. The buffer is checked for what will be filled (the rest of the file)
<code>file_write</code>	<code>off = -1</code> : at the offset, or at the end with <code>O_APPEND</code> (under the node's lock: two appenders never overlap); a write past the end fills the gap with zeros (FatFs leaves it undefined); <code>ENOSPC</code> when nothing fits, <code>EFBIG</code> past 4 GB – 1 on FAT32
<code>file_seek</code>	<code>SEEK_SET</code> / <code>CUR</code> / <code>END</code> → the new offset (no size change); < 0 → <code>EINVAL</code>
<code>file_truncate</code>	shrink (<code>f_truncate</code>), or grow with zeros (64 KB pieces); <code>EINVAL</code> on a read-only handle
<code>file_sync</code>	<code>f_sync</code> (the entry gets the size and the time)
<code>file_stat</code> / <code>path_stat</code>	<code>struct kapi_stat</code> : <code>size</code> (an open file's from its <code>FIL</code>), <code>mtime</code> (FAT's local time → UTC with <code>set_timezone</code> 's zone; an open file written since its last sync: the time of its last write), <code>ino</code> = FNV-1a 64 of the

Call	Semantics
	upper-cased absolute path (<code>RAM::</code> the node's own number, kept by a rename), mode <code>S_IFREG 0644 / S_IFDIR 0755 (0444 / 0555 with the read-only bit)</code> , dev (1 <code>SD:</code> , 2–4 <code>SD1:–SD3:</code> , 5 <code>USB:...</code> , 64 <code>RAM:</code>), <code>blksize</code> (the cluster; <code>RAM:</code> 64 KB), <code>blocks</code> , <code>attr</code> (FAT's), <code>ctime = mtime</code> ; a volume root is a folder
<code>file_close</code>	the description freed; the node's last close flushes (<code>f_close</code>)
<code>path_unlink (p, flags)</code>	a file (<code>EISDIR</code> on a folder); <code>KAPI_UNLINK_DIR</code> : a folder (<code>ENOTDIR</code> , <code>ENOTEMPTY</code>); a root → <code>EBUSY</code> ; a read-only file → <code>EACCES</code>
<code>path_mkdir</code>	<code>EEXIST</code> , <code>ENOENT</code> (no parent)
<code>path_rename</code>	replaces the target (a file by a file; an empty folder by a folder; <code>EISDIR / ENOTDIR / ENOTEMPTY</code>), <code>EXDEV</code> across volumes, a folder into itself <code>EINVAL</code> , a case change allowed
<code>path_ftime (p, mtime)</code>	the entry's time (UTC → local); an open written file is synced first, so its close does not stamp it again
<code>dir_read (dir, out)</code>	the next entry of an <code>opendir</code> handle as <code>struct kapi_dirent2</code> : a 255-character name, 64-bit size, <code>mtime</code> , <code>mode</code> , <code>attr</code> , <code>ino</code> (= <code>path_stat</code> 's; <code>kapi_opendir</code> records each FatFs DIR's path for it: <code>OFileNoteDir</code>) → 1, 0 at the end; the <code>~onyx-deleted</code> folder is not listed
<code>stream_write_nb (h, buf, len)</code>	a pipe: what fits now, <code>EAGAIN</code> when full; a file stream: a plain write

Unlink and rename of open files. An open file that is unlinked (or that a rename replaces) is moved to `<volume>:/.~onyx-deleted/<n>` (a hidden folder) and deleted at its node's last close (the folder too once empty): it stays readable and writable through its handles, and its name is free at once. A file renamed (or hidden) while open has its `FIL` closed, the entry moved, then re-opened at the new path — FatFs keeps the location of the file's directory entry in its `FIL` (`dir_sect`, exFAT's `c_sc1 / c_ofs`) and would write the size and the time into the old one. A folder renamed while files in it are open: their paths follow (their entries do not move). Leftovers of a crash are removed at boot (`OFileBootCleanup`, from `StartAutostart:SD:–SD3:`), on a volume mounted later the first time a file there is hidden. `RAM:` does the same with its own nodes (`RamFsNodeOpen`, `RamFsPRead / PWrite / Truncate`, `RamFsUnlink`, `RamFsRenameEx...`, §16). A teardown (the interrupts masked) only queues its open descriptions; the reaper closes them (`OFileRunDeferred`, from `HandlesRunDeferred`).

The old file calls (`open`, `read`, `save_file`, `file_in` / `file_out`...) keep their own `FILE`s: mixing them with `file_*` on one file at the same time is not coherent (a `save_file` while a `file_*` handle writes the same file: whichever flushes last wins). FatFs' fast seek (`CREATE_LINKMAP`, the old `seek`'s) is never used on a node: FatFs cannot grow a file in that mode.

Pipes. `CPipeStream::PollMask`: `POLLIN` when not empty, `POLLIN | POLLHUP` once the write end is closed, `POLLOUT` when there is room. A blocking `Read` / `Write` sleeps in `IoWait` (≤ 50 ms per turn) instead of yield-spinning, and every change — written, drained, `CloseWrite` — calls `IoWake`. `PIPE_CAP` stays 8 KB. A write to a pipe nobody reads still waits (no `EPIPE`). `IoWait` sleeps in no-kill slices of ≤ 100 ms: a task killed while it waits leaves the event's list before it ends (the reaper would otherwise free it while still listed).

Processes. Each process has a `TProcInfo` (`CAddressSpace::m_pProcInfo`): an **argv block** and an **environment block** ("`a\0b\0\0`", ≤ 64 KB each, kernel heap), made in the spawner's context (`LaunchApp`, `ExecPath`, `SpawnProcess`) and installed by `CUserProcessTask::Run` once the space exists (the spawn record then learns the pid). The environment: `spawn_ex`'s, else the spawner's initial one (`spawn`, `exec`, `exec_as`); a desktop launch or a process the kernel starts (`init`) gets the **system default**, read at boot from `SD:/etc/environment` (KEY=VALUE lines, # comments; without it `HOME=SD:/home`, `PATH=SD:/bin`, `TMPDIR=RAM:/tmp`, `LANG=C.UTF-8`). `argv[0]` is the program's resolved path, then `get_args`' string split as a shell does (blanks, "quotes"). (An empty argument cannot be passed: it would end the block.)

Call	Semantics
<code>spawn_ex (attr)</code>	<code>struct kapi_spawn_attr</code> : <code>path</code> (resolved against the cwd), <code>argv</code> and <code>envp</code> blocks (0: the path alone / the caller's initial environment), <code>cwd</code> (0: the caller's), <code>in</code> / <code>out</code> stream handles → a process handle; <code>ENOENT</code> (no program), <code>EACCES</code> (a folder), <code>EBADF</code> (a stream), <code>EINVAL</code> (a block over 64 KB), <code>EMFILE</code> . Returns once the child has its pid (its first time slice, ≤ 1 s). The child's <code>get_args</code> string is <code>argv[1..]</code> joined with blanks (an argument with a blank in quotes)
<code>proc_wait (h, flags, out)</code>	sleeps (<code>IoWait</code> ; the child's end wakes it) → 1 and <code>struct kapi_proc_status {code, reason, pid}</code> , the handle closed unless <code>KAPI_WAIT_KEEP</code> ; <code>KAPI_WAIT_NOHANG</code> on a running child → 0 with <code>pid</code> set and <code>reason - 1</code> ; <code>EBADF</code>
<code>get_argv / get_env (buf, cap)</code>	the block, filled up to <code>cap</code> → its whole size (call with 0 to size it)
<code>getpid (which)</code>	0 the pid, 1 the parent's (0: none); else <code>EINVAL</code>
<code>clock_info (out)</code>	<code>struct kapi_clock_info</code> : <code>cnt</code> (CNTPCT) and <code>utc_us</code> sampled together (the tick's time: ± 10 ms), <code>freq</code> , <code>tz_minutes</code> , <code>flags</code> <code>KAPI_CLOCK_REALTIME_VALID</code> (a real

Call	Semantics
	date: NTP / RTC), <code>boot_cnt</code> (CNTPCT at boot): a libc reads CNTPCT at EL0 and needs no system call per <code>clock_gettime</code>
<code>sleep_us (us)</code>	under 1 ms: a <code>Yield</code> loop on the μ s clock; else <code>usSleep</code> (the scheduler's 10 ms tick when idle)

How a process ended (`proc_wait`'s reason; `kapi_wait`'s status is the code): `KAPI_PROC_EXITED` (its exit code), `KAPI_PROC_FAULT` (−11: Fault, `sys/el0.cpp`), `KAPI_PROC_KILLED` (−9: kill, `kill_pid` with force or on a windowless app, a dead parent's cascade — before v75 a killed process reported 0), `KAPI_PROC_OOM` (WP-MEM). `ProcInfoTeardown` copies the reason and the pid into the spawn record (`CProcess::nReason`, `nPid`) before it is marked done, and wakes the waiters.

v75: sockets and poll

Work package WP-NET (`docs/POSIX-PLAN.md` §3.3; `kernel/sys/bsdsock.cpp`: the `kapis` and `poll`; `kernel/sys/net.cpp`: the socket table and its "slot layer"; `kern/net.h`). Slots 229–241 (`struct kapi_sockaddr`: IPv4, the port in host order; `struct kapi_pollfd`). Every call returns ≥ 0 or `−KAPI_Exxx`.

Slot	Entry	What it does
229	<code>sock_open (type, flags)</code>	<code>KAPI SOCK_STREAM</code> (TCP) / <code>KAPI SOCK_DGRAM</code> (UDP), <code>KAPI SOCKF_NONBLOCK</code> → the socket number; <code>EPROTONOSUPPORT</code> , <code>EINVAL</code> (flags), <code>ENETDOWN</code> (no network yet), <code>ENFILE</code> (table full)
230	<code>sock_connect (s, to)</code>	TCP: blocking → 0 or the error (<code>ECONNREFUSED</code> , <code>ETIMEDOUT</code> after Circle's retries, about a minute, <code>EHOSTUNREACH</code> , <code>ENETUNREACH</code>); non-blocking → <code>EINPROGRESS</code> , then <code>poll (POLLOUT)</code> and <code>SO_ERROR</code> ; <code>EALREADY</code> , <code>EISCONN</code> . UDP: sets the default peer (only its datagrams are received)
231	<code>sock_bind (s, addr)</code>	the address 0.0.0.0 or the Pi's own (<code>EADDRNOTAVAIL</code>); port 0: an ephemeral port (TCP 61000–61999, UDP Circle's 60000–60999); <code>EADDRINUSE</code> (a bound /

Slot	Entry	What it does
		listening socket of the same protocol has it), <code>EINVAL</code> (bound already)
232	<code>sock_listen (s, backlog)</code>	backlog clamped 1..32; an unbound socket gets an ephemeral port
233	<code>sock_accept (s, peer, flags)</code>	a connection waiting → a new socket (<code>KAPI_SOCKF_NONBLOCK</code> : non-blocking); none → <code>EAGAIN</code> (a non-blocking listener) or a wait (<code>SO_RCVTIMEO</code>); <code>ECONNABORTED</code> (the peer left before it was accepted), <code>ENFILE</code>
234	<code>sock_send (s, buf, len, flags, to)</code>	→ the bytes queued. TCP: all of them when blocking (a wait while Circle's queue holds 64 KB, <code>SO_SNDTIMEO</code> → a short count / <code>EAGAIN</code>), what fits when non-blocking (<code>MSG_DONTWAIT</code>); <code>EPIPE</code> (reset, <code>SHUT_WR</code>), <code>ENOTCONN</code> . UDP: one datagram to <code>to</code> or the default peer (<code>EDESTADDRREQ</code>), at most 1472 bytes (<code>EMSGSIZE</code>); <code>to</code> is ignored on TCP
235	<code>sock_recv (s, buf, len, flags, from)</code>	→ the bytes, 0 = the peer's orderly end (or <code>SHUT_RD</code>); <code>EAGAIN</code> (non-blocking, <code>MSG_DONTWAIT</code> , <code>SO_RCVTIMEO</code>), <code>ECONNRESET</code> , <code>ETIMEDOUT</code> , <code>ENOTCONN</code> . <code>MSG_PEEK</code> (leaves the bytes), <code>MSG_WAITALL</code> (TCP: until <code>len</code> , the end or an error). UDP: one datagram, cut to <code>len</code> (the rest dropped); <code>from</code> = its sender (TCP: the peer)
236	<code>sock_shutdown (s, how)</code>	<code>SHUT_RD</code> : <code>recv</code> answers 0, <code>poll</code> says <code>POLLIN</code> ; <code>SHUT_WR</code> :

Slot	Entry	What it does
		send answers EPIPE — no FIN is sent (Circle's TCP cannot receive after its own FIN; the connection ends at <code>close</code>); <code>ENOTCONN</code>
237	<code>sock_close (s)</code>	0 / <code>EBADF</code> . A socket still connecting or inside a send is closed by its last user
238	<code>sock_getopt (s, opt, &v)</code>	<code>SO_ERROR</code> (the pending error, a positive <code>errno</code> , cleared), <code>SO_NONBLOCK</code> , <code>SO_RCVTIMEO_MS</code> , <code>SO_SNDTIMEO_MS</code> , <code>SO_BROADCAST</code> , <code>SO_NREAD</code> (bytes in the carry buffer, 1 if more is ready), <code>SO_TYPE</code> , <code>SO_ACCEPTCONN</code> ; else <code>ENOPROTOOPT</code>
239	<code>sock_setopt (s, opt, v)</code>	<code>SO_NONBLOCK</code> , <code>SO_RCVTIMEO_MS</code> / <code>SO_SNDTIMEO_MS</code> (0: none), <code>SO_BROADCAST</code> (UDP); else <code>ENOPROTOOPT</code> (libc accepts and ignores <code>TCP_NODELAY</code> , <code>SO_KEEPALIVE</code> , <code>SO_REUSEADDR</code> , the buffer sizes)
240	<code>sock_name (s, peer, out)</code>	peer 0: the local address (the Pi's IP once connected, else 0.0.0.0) and port; 1: the peer (<code>ENOTCONN</code>)
241	<code>poll (fds, n, timeout_ms)</code>	$n \leq 1024$ (<code>EINVAL</code>), <code>timeout_ms - 1</code> forever, 0 only a look → the number of entries with <code>revents ≠ 0</code> . Kinds: <code>KAPI_PK_SOCKET</code> (a socket number), <code>KAPI_PK_STREAM</code> (a stream handle: <code>CStream::PollMask</code>), <code>KAPI_PK_FILE</code> (a file / directory / open-file handle: always IN OUT);

Slot	Entry	What it does
		a bad one <code>POLLNVAL</code> ; kind 0 or <code>h < 0</code> ignored. <code>POLLERR</code> , <code>POLLHUP</code> , <code>POLLNVAL</code> are reported whatever events asks

One table. The BSD sockets live in the same table as the `tcp_*` handles (`sys/net.cpp`, `MAX_SOCKETS 256` since v75, shared by every process; owner pid, adoption by a descendant, `NetCloseByPid`, as in §11). A slot has a type (TCP / UDP), a state (NEW, BOUND, LISTEN, CONNECTING, CONNECTED, FAILED), the pending error, the options, the local port and the peer, a **carry buffer** (a frame, allocated at the first receive that needs it: a segment bigger than the caller's buffer leaves its rest there — so nothing is lost, and `MSG_PEEK` works; `tcp_recv` goes through it too, which fixes its data loss with small buffers when `netcore=0`) and a readiness snapshot. The `tcp_*` handles are sockets of that table: `poll`, `sock_name`, `sock_recv` work on them; a socket number is not a handle (`HANDLE_*`), and it is the same number space for every process (another process's socket: `EBADF`, `POLLNVAL`).

The slot layer never waits. `net.cpp`'s `Slot*` functions run where the stack runs (core 0, or a worker of core 3 with `netcore=1`, through the request slots of §11: a send or receive moves at most 32 KB per round trip) and answer `EAGAIN` instead of blocking. Every wait is `bsdsock.cpp`'s: a few yields, then `IoWait` on the I/O generation (`kern/iowait.h`), then the call again. TCP data goes into Circle with `MSG_DONTWAIT` only while Circle's send queue is under its 64 KB threshold (`GetStatus ().bTxReady`). Receiving: `CSocket::Receive (MSG_DONTWAIT)` straight into the buffer while a whole frame fits, else through the carry buffer; an empty receive with the connection still up (CLOSE-WAIT: the peer's FIN) is the orderly end (0), else the connection's error.

Connect. A TCP connect is run by a task of its own, so that a blocking connect too only waits on `IoWait` (an app killed meanwhile leaves no task inside Circle): with `netcore=0` a one-shot kernel task (`CNetConnectTask`, `netconn`) on core 0; with `netcore=1` a **detached** request (nobody waits on it; the worker frees it; `NetCloseByPid` leaves it alone). It ends with the slot CONNECTED, or FAILED with its error (`SO_ERROR`); a close meanwhile only marks the slot (`bCancel`) and the connector frees it. The same marking protects a send in progress (Circle's `Send` yields between segments) and a `tcp_accept` waiting in Circle's `Accept`.

Readiness. `POLLIN`: data or carry, the peer's end, an error, `SHUT_RD`, a connection to accept (`CSocket::AcceptReady`, Circle patch); `POLLOUT`: connected and under the send threshold, always for UDP; `POLLERR`: a connect that failed (with `POLLOUT` | `POLLHUP`, as Linux); `POLLHUP`: was connected and no longer is (reset, timeout), or a TCP socket never connected. With `netcore=1` the net core's main loop recomputes every open slot's snapshot at each turn (and a worker after each call on a slot) and bumps a generation word when one changed, then sends core 0 an **inter-core interrupt** (`IPI_NET_READY`, one at a time: a flag core 0 clears before it reads the generation); core 0's handler (`NetReadyIPI` → `NetPollTick`) turns that into `IoWake` at once. The 100 Hz tick hook (`IoWaitAddTickHook`) remains as the fallback; a waiter's sleep is capped at 100 ms anyway. (Before the IPI the tick alone did it: every blocking `recv` / `send` / `poll` waited up to 10 ms a turn — an echo's round trip was 10 ms on the LAN.) With `netcore=0` the readiness is evaluated on the spot and nothing announces a change (a connect's end and a close do call `IoWake`), so a wait over sockets looks again at every tick (10 ms). A wait over streams and files only sleeps until `IoWake` (a pipe's `PollMask` and its wakes are WP-FILE/PROC's).

Circle patch (`tools/circle-patches/wp-net.patch`, `docs/05`): a connection's handle stays its socket's until the socket lets it go (`CNetConnection::SetReleased`: before, a reset connection was deleted at the next `Process` and its handle reused by the next connection — which the old socket then read, wrote and closed); `CSocket::AcceptReady` () (a backlog connection is connected: `Accept` will not block; it also replaces backlog connections that died before being accepted, a SYN without its ACK); `CSocket::Accept` lets a failed connection go instead of leaving it listening on its own; `CTransportLayer::IsTerminated` (h).

Test: `/bin/nettest` (`user/bin/nettest.c`; the PC side `tools/tests/nettest_peer.py`): run it with `netcore=0` and with `netcore=1`.

v76: IPC

Work package WP-IPC (`docs/POSIX-PLAN.md` §14: what WebKit2's Unix IPC uses, the spec): what a browser's UI, web and network processes need to talk. `kernel/sys/lsock.cpp` (local sockets, the handles they carry; `kern/lsock.h`), `kernel/sys/shm.cpp` (shared memory objects), `kernel/sys/vm.cpp` (`shm_map`, SHM regions), `kernel/sys/procx.cpp` (`spawn_ex2`, `get_handles`), `bsdsock.cpp`'s dispatch. Slots 242–252 (struct `kapi_iovec`, struct `kapi_handle_xfer` 24 bytes, struct `kapi_msghdr` 48 bytes). Every call returns ≥ 0 or `-KAPI_Exxx`.

Slot	Entry	What it does
242	<code>sock_pair</code> (type, flags, sv)	two connected local sockets: <code>KAPI_SOCKET_STREAM</code> , <code>KAPI_SOCKET_SEQPACKET</code> (5), <code>KAPI_SOCKET_DGRAM</code> ; <code>KAPI_SOCKETF_NONBLOCK</code> → <code>sv[0]</code> , <code>sv[1]</code> (numbers $\geq$ <code>KAPI_SOCKET_LOCAL_BASE</code>); <code>EPROTONOSUPPORT</code> , <code>EMFILE</code> , <code>EFAULT</code>
243	<code>sock_sendmsg</code> (s, m, flags)	the <code>m->iovcnt</code> (≤ 64) iovecs gathered into one message with <code>m->nhandles</code> (≤ 256) handles → the bytes; <code>EAGAIN</code> , <code>EPIPE</code> , <code>EMSGSIZE</code> , <code>ENOBUFS</code> , <code>EBADF</code> (a handle: nothing sent), <code>EINVAL</code> ; an IP socket: <code>EOPNOTSUPP</code>
244	<code>sock_recvmsg</code> (s, m, flags)	into the iovecs; the handles carried added to the caller's table and written to <code>m->handles</code> (<code>m->nhandles</code> in: room, out: count; <code>m->flags</code> : <code>KAPI_MSG_TRUNC</code> , <code>KAPI_MSG_CTRUNC</code>) → the

Slot	Entry	What it does
		bytes, 0 the end
245	<code>shm_create (size, flags)</code>	an anonymous object (<code>KAPI_SHM_ALLOW_SEALING</code> , else <code>KAPI_SEAL_SEAL</code> set) → a read-write handle
246	<code>shm_open (name, oflags, mode)</code>	a named object (<code>/x</code> , ≤ 63 characters, a kernel-wide table): <code>KAPI_O_RDONLY</code> / <code>RDWR</code> , <code>CREAT</code> , <code>EXCL</code> , <code>TRUNC</code> → a handle; <code>ENOENT</code> , <code>EEXIST</code> , <code>EINVAL</code> , <code>ENAMETOOLONG</code>
247	<code>shm_unlink (name)</code>	the name dropped (the object lives while referenced); <code>ENOENT</code>
248	<code>shm_ctl (h, op, arg)</code>	<code>KAPI_SHM_GET_SIZE</code> , <code>SET_SIZE</code> (truncate: <code>EPERM</code> sealed, <code>EBUSY</code> a shrink while mapped, <code>ENOMEM</code>), <code>ADD_SEALS</code> / <code>GET_SEALS</code> (as <code>memfd</code> 's <code>F_SEAL_*</code>), <code>GET_ID</code> , <code>GET_ACCESS</code>
249	<code>shm_map (h, addr, len, prot, flags, off)</code>	the object mapped <code>MAP_SHARED</code> (an SHM region in the mmap arena, placed as <code>vm_map</code>) → the address; <code>EACCESS</code> (write, read-only handle), <code>EPERM</code> (<code>SEAL_WRITE</code>), <code>EBUSY</code> (a fixed place under another task's kapi)
250	<code>handle_close (h)</code>	a shm, local socket, open-file or stream handle closed
251	<code>spawn_ex2 (attr, handles, n)</code>	<code>spawn_ex</code> plus <code>n</code> (≤ 256) handles referenced for the child at the descriptors <code>fd</code> given
252	<code>get_handles (out, cap)</code>	the child's side: those handles, put in its table at the first call → how many

Local sockets. An end (`TLsEnd`) is a core-0 object: its type, a receive queue of messages (`TLsMsg`: the data and the handles carried, one allocation), its peer, the shutdown bits, `bNonBlock`, the timeouts, `SO_RCVBUF`

(the queue's limit, 256 KB by default) and `SO_SNDBUF` (the largest packet, 256 KB), both 4 KB–16 MB. **A local socket's number is a handle** (`HANDLE_LSOCK`) of the caller's table — so it is per process, reference-counted and closed by the teardown, and its value (always $\geq 0x10001$) tells it from an IP socket (0..255, the global table of §11). `bsdsock.cpp` sends every call on such a number here first (`IpLocal*`): `sock_send / recv / shutdown / close / getopt / setopt / name` and `poll` (`connect / bind / listen / accept`: `EOPNOTSUPP`: no named local sockets — WebKit only uses `socketpair`). A send copies into a new message on the **peer's** queue: `STREAM` takes what fits (a blocking send waits for the rest), a packet (`SEQPACKET`, `DGRAM`) is whole or refused (`MSGSIZE` above `SO_SNDBUF`; an empty queue always takes one); a receive copies out of its own queue — `STREAM` across messages but never past one that carried handles (they come with its first byte, as Linux), a packet whole (cut: `MSG_TRUNC`). Every change calls `IoWake`, so a wait (`IoWait`) or a `poll` over local sockets sleeps until it (no 10 ms polling as for IP sockets). The end: a closed peer (its last handle and every message holding it gone) or its `SHUT_WR` $\rightarrow 0$ once the queue is empty; a send to it $\rightarrow$ `EPIPE`. Readiness: `POLLIN` data or the end, `POLLOUT` room in the peer's queue or a send that would fail at once, `POLLHUP` the peer gone. All the queues together hold at most 64 MB (`ENOBUFS`). A call pins its handle (another thread's close waits for the call to end, as for files).

Carried handles (`TIpXfer`). `sendmsg` turns each `{h, kind, tag, flags}` into a reference at once (`IpXferTake`): an open-file description (`HANDLE_OFFILE`: `ofile.cpp`'s `nHolders`, so the offset is shared as by a `dup`), a stream (`CStream::AddRef`), a local socket end, a shm object; an **IP socket** keeps its number and the sender's pid and is **adopted** by the receiver at `recvmsg` (`NetSocketAdopt`: it becomes the receiver's — an IP socket has one owner; one the sender closed meanwhile arrives as `KAPI_HK_NONE`, -1). `recvmsg` adds each to the caller's table (`IpXferGive`); no room in the array or the table: closed, `MSG_CTRUNC`. A message discarded (its end closed, its process gone) closes what it carried (the teardown's way: deferred where a close may wait). Sending the receiving end over its own connection is refused (`EINVAL`: no close could ever free it); longer reference cycles are not collected. The `tag` is the sender's word, given back as it is: `libonyxposix` puts its descriptor type and `O_*` flags there. `KAPI_HXF_WRITER` on a stream (a pipe's write end): the pipe counts one more writer (`CPipeStream::AddWriter`), and its end-of-file comes when every writer has called `stream_eof` or closed (the receiver's entry is `HKIND_STREAM_WRITER`: its close or its process's end counts as its `stream_eof`, §9).

Shared memory. A `TShm` is a size and an array of 64 KB frames, each taken from the app pool (`palloc_high`, refused under `VM_RESERVE`) and zeroed at its first use, all freed — their word waiters woken first — when the last reference goes: the handles (`HANDLE_SHM`, the entry's kind = the access), the messages and spawn records carrying it, the name table, and **each address space mapping it** (one reference per space, `TVmSpace::pShm`). `shm_map` makes a lazy **SHM region** (`KAPI_VMK_SHM`, `TVma::pObj + u10bj0ff`; a split keeps each piece's offset; neighbours merge only over contiguous offsets of the same object). A fault there (`ELO`, a kapi's probe, the safety net, an app core's pager) maps the object's frame with the region's protection **not owned** (`VM_PTE_SW_OWNED` clear): `Release` (`unmap`, `DONTNEED`), the deferred zap and the teardown never free it. A page beyond the object's size faults (the process is killed: "beyond the shared object", as Linux's `SIGBUS`). The space keeps its reference until no region names the object **and** no deferred zap is pending (a PTE marked `ZAP` may still point at a frame), checked after each `unmap / fixed map / settled zap` (`VmShmGc`), dropped by `VmTeardown`. A shrink while any space maps the object is refused (`EBUSY`), so a mapped frame never goes. `vm_protect / vm_advise / vm_query` work on SHM regions (`DONTNEED` drops the mapping, the data stays). The futex (`wait_word / wake_word`) keys on the physical word: two processes meet on a shm word (WebKit's `IPC::Semaphore`).

Spawning with handles. `spawn_ex2` references the handles (as `sendmsg`, but an IP socket is not adopted: the child, a descendant, adopts it at its first use as before) in the child's `TProcInfo`; the child's `get_handles` puts them in its table at the first call (the same list at every call); those never asked for are

closed with the process. CLOEXEC is user space: libonyxposix's `posix_spawn` gives every descriptor without `FD_CLOEXEC` (docs/03 §5.4).

Tests: `/bin/ipctest` (user/bin/ipctest.c, kapi level), `posixtest ipc` (the POSIX calls); on the PC `tools/tests/run_ipc_test.sh` (the real `lsock.cpp`, `shm.cpp`, `handle.cpp` with two handle tables, ASan) and the `posixsim` bench (its fake table implements v76 over Linux socketpairs, `SCM_RIGHTS` and `memfd`).

v77: program images

The loader's image objects (§7 *Program images*: what they are, their key, their lifetime, the hook). `kernel/proc/image.cpp`, the three entries in `kernel/sys/kapi.cpp`, the preload task in `kernel.cpp`. Slots 253–255; `struct kapi_image_info` (280 bytes: `size` — the bytes of memory the image holds, once whatever the processes —, `file_size`, `refs`, `flags`, `path[256]` — its key). A path is a program file's, relative to the caller's working directory; every call returns ≥ 0 or `-KAPI_Exxx`.

Slot	Entry	What it does
253	<code>image_preload (path)</code>	the program loaded ahead and kept : a kernel task reads the file, the call returns at once; from then on a run of that path maps the image without reading the card, and the image stays when no process runs it. Kept already: 0, nothing done $\rightarrow$ 0; <code>ENOENT</code> (no such file), <code>ENAMETOOLONG</code> , <code>ENOMEM</code> , <code>EFAULT</code> . A load that fails later is in the kernel log
254	<code>image_unload (path)</code>	the path's image loses its pin and its name at once: no new process maps it; its memory is freed when the last process running it ends $\rightarrow$ 0; <code>ENOENT</code> (no image), <code>EFAULT</code>
255	<code>image_list (path, out, cap)</code>	<code>path</code> 0: the live images, up to <code>cap</code> written $\rightarrow$ how many there are. <code>path</code> : the image a run of that path would map $\rightarrow$ 1 (<code>out[0]</code> written if <code>cap > 0</code>) / 0. Flags: <code>KAPI_IMG_KEPT</code> (preloaded), <code>KAPI_IMG_LOADING</code> , <code>KAPI_IMG_UNNAMED</code> (unloaded, or its file

Slot	Entry	What it does
		changed: only its processes still use it)

Users: `/bin/preload` (`preload /boot`, the last line of `/etc/autostart`: the list of `SD:/etc/preload.ini`, `user/preloadini.h`, edited by the Control Panel's Preload applet, `user/Apps/preloadconf`), `/bin/unload` (docs/04 §8), `pkg` (`pkglib.h` move: a kept program unloaded before its file is replaced, preloaded again after). `user/kapi.h`'s wrappers return `-KAPI_ENOSYS` on an older kernel. Tests: `sh tools/tests/run_image_test.sh` (§7).

v78: PROT_EXEC

A JIT's memory, as on Unix: `vm_map` (`addr`, `len`, `READ | WRITE | EXEC`, `flags`) gives a **lazy** anonymous region whose pages are mapped `UXN = 0` (executable at ELO; `PXN` stays 1: the kernel never runs an app's code), and `vm_protect` sets or takes away `EXEC` on anonymous regions — so a JIT may keep its code `RWX` (JavaScriptCore's default on Linux) or switch `RW / RX` (`W^X`). A shared object's pages never run: `shm_map` with `EXEC`, or `vm_protect` adding it over an SHM region → `-KAPI_ENOTSUP`. `kernel/sys/vm.cpp`: a region's protection becomes its PTE's AP **and** `UXN` bits (`PteProt`) wherever pages are filled, adopted, settled or reprotected. An instruction abort in a region not filled yet is paged in where the region is executable (`e10.cpp` `PageFault`, `VmProtAt`; an app core asks core 0 as for a data abort, the retried fetch then faults if the region does not allow it). The program writes its code, then makes the caches coherent over it from ELO (`DC CVAU`, `DSB ISH`, `IC IVAU`, `DSB ISH`, `ISB`: `SCTLR_EL1.UCI / UCT`, `e10.cpp`; `libgcc's __builtin__clear_cache` does it). `code_alloc` (v58: an eager region in the code arena) stays for the GameCube emulator. Before v78 `EXEC` was `-KAPI_ENOTSUP` everywhere; `libonyxposix's` `mmap / mprotect` pass the kernel's answer on. Tests: `memtest` (code written, run, made `RX`, rewritten), `posixtest` (`mmap PROT_EXEC`).

v80: cpu_stats, net_stats

Slot	Entry	What it does
257	<code>cpu_stats</code> (out)	<code>struct kapi_cpu_stats</code> (144 bytes): <code>now_us</code> (the clock of the read), <code>cores</code> , then a <code>kapi_cpu_core</code> a core — <code>busy_us</code> (the microseconds it was busy since the boot), <code>role</code> (<code>KAPI_CORE_SYSTEM</code> 0, <code>KAPI_CORE_SOUND</code> 1, <code>KAPI_CORE_APP</code> 2, <code>KAPI_CORE_NETWORK</code> 3), <code>pid</code> (an app core's owner, 0 when free) → 0 / <code>-EFAULT</code> . Two reads make a load: <code>(busy_us' - busy_us) / (now_us' - now_us)</code> .
258	<code>net_stats</code> (<code>pid</code> , out)	<code>struct kapi_net_stats</code> (24 bytes): the payload

Slot	Entry	What it does
		bytes <code>pid</code> 's sockets received and sent (<code>rx_bytes</code> , <code>tx_bytes</code> : TCP and UDP, the old <code>tcp_*</code> calls and the BSD sockets) since it started, its sockets open now; <code>pid</code> 0: every process's since the boot → 0 / <code>-EFAULT</code> . A process that used no socket: zeros.

What "busy" is, core by core. A core with a **scheduler** — core 0, and core 3 with `netcore=1` — counts the time its tasks ran, the idle task apart: `CScheduler::Yield` adds what the leaving task ran (`m_nBusyUs`; `GetBusyUs` adds the running task's time so far, so a read from another core is live). Core 0's idle task sleeps in `wfi`, so its figure is the machine's real load; the interrupt handlers that wake it are counted as idle. The **network core**'s tasks all wait by yielding (it polls the Wi-Fi chip), so while the network works it reads 100 %; once the network has been quiet for 50 ms it sleeps between two questions to the chip (*The network core sleeps*, below) and that sleep is taken off its busy time (`CScheduler::NoteSleptUs`): a few per cent. **Core 1** counts the time it renders sound (`sys/sound.cpp`, around `Render`); an **app core** the time its jobs ran, from the entry at `EL0` to the job's end or drop (`sys/appcore.cpp`; a job that waits in `wfe` still counts).

The network's bytes are counted where the calls return, on core 0 (`NetTcpSend` / `NetTcpRecv`, `NetSockSend` / `NetSockRecv` in `sys/net.cpp`: a process's tasks all run there, so the table needs no lock): one line a process (64 at most; beyond, the totals only), freed when the process ends (`NetCloseByPid`). A byte read with `MSG_PEEK` is counted when it is really read. They are the sockets' payload, not the frames on the air: no header, no retransmission, nothing of the kernel's own traffic (DHCP, DNS, NTP).

v79: kernel_info

Slot	Entry	What it does
256	<code>kernel_info (buf, cap)</code>	what the running kernel is, as <code>key value</code> lines: <code>name</code> (Onyx), <code>abi</code> (<code>KAPI_ABI_VERSION</code>), <code>built</code> (the date and time of the image's link), <code>rev</code> (the source's git revision; <code>+</code> : built from changed sources), <code>machine</code> (aarch64), <code>model</code> (the board's name), <code>ram</code> (MB). Up to <code>cap - 1</code> bytes and a NUL → the text's whole length; <code>EFAULT</code> . Keys may be added: a reader looks

Slot	Entry	What it does
		its keys up.

The date and the revision are `kernel/buildstamp.cpp`'s, an object that depends on every other object and library of the kernel (`kernel/Makefile`): it is compiled again at each link, so the stamp is the image's — `__DATE__` in a file says when that file was last compiled. The boot log's `Built on ...` line prints the same. User: `/bin/uname` (docs/04 §8) — a kernel copied to the card by hand is told from the package's (`uname -v` against `uname -p`). `user/kapi.h`'s wrapper returns `-KAPI_ENOSYS` and an empty text on an older kernel.

Historical note. `ARCHITECTURE.md` §11 describes an earlier approach where the build emitted a `user/kernel_syms.ld` (`kapi_x = 0xADDR;`) and the apps were linked against those addresses. This approach is **no longer used**: the fixed-address table replaced it. The apps no longer depend on the kernel build.

9. Stream / stdio subsystem

Source: `kernel/sys/stream.cpp`, `kernel/include/kern/stream.h`.

This is what makes the terminal's **pipes and redirections** and inter-process communication possible.

Object model

Abstract base class `CStream` with reference counting (`AddRef/Release`; destroyed when the last reference drops). Three implementations:

Class	Role
<code>CPipeStream</code>	in-memory FIFO (ring buffer ~8 KB) between two tasks
<code>CFileStream</code>	FatFs file presented as a stream (read, write/truncate, append)

Virtual methods: `Read` (cooperative blocking: yields until ≥ 1 byte or EOF; 0 = EOF), `ReadNonBlocking` (>0 / 0=EOF / -1=would block), `Write` (may block if full), `CloseWrite` (signals "no more writing" → readers see EOF). (v76) `AddWriter`: a pipe's write end carried to another process (`KAPI_HXF_WRITER`) is one more writer — `CloseWrite` ends the data only when every writer has called it (a received write end's close, or its process's end, counts as its call: `HKIND_STREAM_WRITER`).

- **CPipeStream**: ring buffer `head/tail`. `Read` waits while empty and the write end is not closed; `Write` waits while full. Cooperative → no lock. (v75: the waits sleep in `IoWait` and every change calls `IoWake`; `WriteNonBlocking` / `stream_write_nb`, `PollMask`: §8 v75: *files and processes*.)
- **CFileStream**: wraps a FatFs `FIL`; modes 0=read, 1=write+truncate, 2=append.

Processes and stdio

- `CAddressSpace` owns `m_pStdin/m_pStdout` (set by `SpawnProcess`).
- `kapi_stdin_read/kapi_stdout_write` read/write these streams (or log if absent).
- `kapi_spawn(path, args, in, out)` → `SpawnProcess`: the child task takes a reference on `in/out`. `kapi_wait` blocks (cooperatively) on `CProcess::bDone` then returns the exit code (v75: -9 for a killed child, -11 for a crash; `spawn_ex` / `proc_wait` add `argv` / environment blocks and the reason:

§8). When the child's address space is destroyed, `stdout` gets `CloseWrite` → the reader (the terminal) sees EOF.

The terminal thus chains the `stdout` of one stage to the `stdin` of the next via `CPipeStreams`, and reads the final output non-blocking. App-side details in the [developer guide](#) and the [user guide](#).

10. Graphics subsystem (GUI)

Source: `kernel/gui/{gimage,window,surface}.cpp` + headers (`kern/gui/`). Rendering core ported from the author's FreeBASIC `SimpleOS`.

10.1 `GImage` — software rendering engine

- **Pixel format: 32-bit 0x00RRGGBB** (alpha byte unused).
- **Transparency color: magenta 0xFF00FF** (transparent-blit key).
- A `GImage` owns its buffer (`SetSize`) or **wraps** one (`Wrap`, e.g. the framebuffer's back buffer or a window's canvas).
- Primitives: `Clear`, `SetPixel/GetPixel`, `DrawLine` (Bresenham), `DrawRectangle`, `FillRectangle`, blits `PutOtherRaw/PutOther` (with key) / `PutOtherPart` (sub-rect, used by the 9-slices skin engine).
- **Text**: Circle's bitmap font (`CCharGenerator`). `FontWidth/Height`, `DrawChar` (transparent background), `DrawText`.
- **BMP**: `LoadBMP` (24 bpp uncompressed, BGR→RGB, handles top-down/bottom-up).

10.2 Windows and compositor

- **CWindow**: position, logical size, title, flags (`WIN_FLAG_BORDERLESS`; `BACKMOST`: the desktop's band; `TOPMOST`: the menu bar, the dock — above every window, never active, never the keys; `TRANSPARENT`: the magenta key; `SYSTEM`: not listed as an open app; `ALPHA` (v64): the canvas's top byte is a transparency; `FIXED` (v69): the user cannot move it — no drag, no double-click maximise, `HitTitleButton` finds no button (wtk draws none: `WK_WIN_FIXED`) — and `OnScreenResized` centres it again at a new resolution instead of only moving it in: Setup's), a `GImage` **canvas allocated 64 KB-aligned and physically contiguous** (mapped into the app at `USER_WINDOW_CANVAS` = 12 GB — the app draws directly, with no per-pixel call), the frame's two copies (active, inactive: mapped at `USER_WINDOW_CHROME` / `_INACTIVE`, drawn by the app — kapi v28 `get_chrome`; wtk: `wk_decorate_window`), an event queue (spinlock-protected ring), the app's handlers, its menu.
- **The frame** (v64, the modernised CDE): a 28 px title bar and 4 px borders (`WIN_TITLEBAR_H` / `WIN_BORDER` = `KAPI_FRAME_TITLE_H` / `KAPI_FRAME_BORDER`, shared with the apps in `kapi_abi.h`), rounded corners (radius `KAPI_FRAME_RADIUS` = 8): in the frame's copies a pixel's top byte is its transparency, heeded in the four corner squares only. `DrawTo` blits the frame's **bands** only (the title bar, the bottom border, the two sides beside the client area — never under it) opaque, the corner squares blended by their pixels' transparency (`BlendAlphaRect`), then the client canvas. `CoversOpaque` (`rect`) — may the compositor skip what lies below? — leaves out only the corners' see-through pixels (`CornersIn`: on a corner's row *k* from the edge, its first `CornerSpan` (*k*) pixels — those not wholly inside the arc): a rectangle inside the window still covers. *Borderless* windows have no frame.
- **The present's damage** (`PresentDamage`): an app's present dirties **its client area only** — its whole window when its frame changed since (`get_chrome`, a resize: `ChromeGen`) or it is faded — so a

window refreshing alone (an emulator) keeps the compositor's fast path: its client rectangle is covered, nothing below it is drawn. That first present after a frame's redraw also bumps `ChromeGen` once more: the remote desktop (`rdpd`), which may have read the frame while the app was still drawing it (`get_chrome` bumps it **before** the drawing), reads it again, whole — no frame left without its buttons on the PC. A window's two frame copies are allocated both or none (`AllocChrome`: no half-mapped frame).

- **The title buttons** (v64): the window menu at the left; close, maximise, minimise from the right, at `KAPI_FRAME_BTN_*` (the app draws them, the kernel hit-tests them: `HitTitleButton`). Close and minimise act at their release over them (`RequestExit`, `Minimise`); the window menu (at the press) and maximise (also a double click on the title bar) go to the app as `GUI_EVENT_WINCTL` (18; value `KAPI_FRAME_MENU / KAPI_FRAME_MAXIMIZE`): wtk's `Root` shows its window menu, maximises and restores (the developer guide).
- **Minimised windows** (v64, `SetMinimised`): not drawn, not hit, never the active window nor the keys' target (`ActiveLocked`, `KeyTargetLocked`), their area damaged as they go and come back; `Raise` (`win_raise`, `raise_app`: the dock, the menu bar's Open Windows) brings one back.
- **Workspaces** (v65): a window's desk (`CWindow::Desk`, `-1` = every desk: the topmost, backmost and system windows) is the current one when it is added (`CWindowManager::m_nDesk`, of `m_nDesks`, 4 at boot — the dock sets its number from `SD:/etc/dock.ini`). `SetDeskLocked` flags every window of another desk `OffDesk`; `Hidden ()` (minimised or off-desk) is what the compositor, `DrawTo`, `CoversOpaque`, `HitTest`, `ActiveLocked`, `KeyTargetLocked` and the bottom inset heed, and `SetOffDesk` damages the window's area as it goes and comes back. `ForgetHiddenLocked` drops the pointer's state (hover, capture, a drag, a pressed title button) naming a window that went hidden. `RaiseLocked` of an off-desk window shows its desk first. `OnKey` takes **Ctrl+Alt+Left / Right** (the xterm modifier form `ESC[1;7C / D` too) before the apps: the previous / next desk, with Shift the active window moved along (`MoveToDesk`), not while an app is full screen. `DeskInfo`, `SetDesk`, `MoveToDesk` for kapi desk / `win_desk` (each change bumps `m_nDeskGen`: the dock redraws its pager).
- **See-through windows** (`WIN_FLAG_ALPHA`, borderless): each pixel blended by its top byte (0 opaque .. 255 see-through, times the window's fade); never `CoversOpaque`; a click on a wholly see-through pixel goes to the window below (`HitTest` → `OpaqueAt`). The dock (its rounded corners, the gap under it — almost clear, top byte 254, while a drawer is open, to catch a click elsewhere), the menu bar (its drop-downs), the agenda widget (its text straight on the wallpaper).
- **The work area** (`WorkArea`, v64 `win_geometry`): the screen less the top inset (a topmost window at `y = 0`: the menu bar, its smallest height) and the bottom inset (the topmost windows standing on the bottom edge: the dock, its smallest height). New windows are placed in it; a maximised window fills it.
- **A window that grows** (v64 `resize_window2`): `CWindow::Grow` allocates a bigger canvas and frame copies; the process's mappings are moved to them at the same addresses (`MapContig`, then `CAddressSpace::FlushTLB: tlb_i aside1is`); the old memory is retired and freed by the compositor three frames later (`FreeRetired`: a frame begun before may still be reading it).
- **The screen's size, while running** (v66 `screen_set`, the Control Panel's Display applet): the compositor task makes the frame buffer again (`C2DGraphics::Resize`) between two frames, then `OnScreenResized` keeps the windows on the screen and sends them `GUI_EVENT_DISPLAY_RESIZE`.

The apps placed by the screen's size place themselves again on it: the menu bar (its canvas grown to the screen: `resize_window2`), the dock (`onDisplayResize`: laid out for the new width, along the bottom), the notifications (the top right corner); wtk re-maximises a maximised window and shrinks / moves one past the work area ~0.3 s later (`Root::displayTick`: the dock has moved, the work area is the new one). vncd sends the VNC `DesktopSize` pseudo-rectangle (-223) to a client that takes it (else it closes the session: the client connects again at the new size); rdpd sends `SCREEN` (8: u16 w h) and Onyx Remote resizes its view of the Pi's screen.

- **CWindowManager** (singleton): a Z-ordered list (bands: backmost, normal, topmost; the last = on top), protected by a `CSpinLock`. `Add/Remove/Raise/Minimise`. `Composite(screen)`:
 1. snapshot the list under the lock (the compositor also frees the grown windows' old memory), then blit outside the lock;
 2. find the topmost window that covers the clip rectangle opaquely: nothing below it is drawn;
 3. otherwise clear the desktop and blit the wallpaper;
 4. draw the windows back to front (`DrawTo`, `bActive` for the keys' window);
 5. the drag badge, then the cursor (`mousecur.bin` bitmap or a fallback arrow).
- **Dirty rectangles**: `Composite` honours the screen image's clip rectangle, so the compositor task redraws only the damaged rectangles (see the service tasks above); host tests `sh tools/tests/run_gui_test.sh` (clipped drawing = full drawing inside the clip, nothing touched outside) and `sh tools/tests/desktop_sim/run.sh` (the v64 / v65 windows: `wmtest.cpp` — the corners, `CoversOpaque`, the present's damage, the title buttons, minimise, the see-through windows' clicks, the work area, a canvas growing, the workspaces).

10.3 Events and input (kernel side)

The kernel draws **no widget** any more: the kernel-drawn widget API (buttons, labels, check boxes, text boxes, sliders, scroll bars, icons) was removed from the table by the v29 compat break; every app draws its own controls with **wtk** (`user/wtk`, docs/03 §6). The window manager keeps the window-level work: the hit-test of the frame (title bar, its buttons, the borders), raising and moving windows, focus, and **pushing events** into the window's queue; the app's `pump_events` (user-side since v74: `pop_event` / `pop_post` in a loop, §6) calls its handlers **in the app, at ELO**. Events (`kern/gui/window.h`, mirrored in `user/kapi.h`):

Constant	Meaning
<code>GUI_EVENT_KEY</code> (5)	key pressed (<code>value</code> = char or <code>KEY_*</code> ; the modifiers through <code>get_modifiers</code>)
<code>GUI_EVENT_CANVAS_CLICK</code> / <code>_MOTION</code> (6, 7)	client press / drag: <code>value=(buttons<<32) (x<<16) y</code>
<code>GUI_EVENT_PTR_MOVE</code> ... <code>_WHEEL</code> (8–13)	the pointer stream for app-side toolkits (<code>set_pointer_handler</code> , v22)
<code>GUI_EVENT_MENU</code> (14)	a menu-bar command (<code>set_menu</code> , v39)
<code>GUI_EVENT_DROP</code> , <code>_DRAG_OVER</code> , <code>_DRAG_DONE</code> (15–17)	drag & drop (v42)
<code>GUI_EVENT_WINCTL</code> (18)	a title button handed to the app (v64)
<code>GUI_EVENT_DISPLAY_RESIZE</code> (19)	the screen's size changed (v66)

Mouse handling: hover tracking, press edge (raises the window, hit-test of the frame / canvas), drag (window move / resize), release edge. The keyboard (Circle's "cooked" VT100 strings) is translated into logical keys (`KEY_UP`, `KEY_ENTER`, ...) and delivered to the modal dialog, otherwise to the app's keyboard handler.

10.4 Theme, wallpaper

- **The theme** is the apps' business: the windows' frames, like every control, are drawn by the apps (wtk: `user/wtk/skin.cpp`, `paint.cpp` — by code, no bitmap) from `SD:/etc/theme.txt` (theme = Peach / Steel / Sage / Brick / Slate, or `active`; `inactive`, `face`, `accent`, `outline`, `dock`: read by `wtk/theme.cpp`); the kernel only blits the frames. Of that file the kernel reads only `wheel speed=N` at boot. (The old 9-slice window skin — `wings.bmp` tinted by `CSkin` — and `kapi_set_window_theme` are no longer used by the desktop.)
- **Wallpaper**: `wallpaper_generate` (toroidal Voronoi generated at runtime), or **an app-drawn background** (what the desktop uses: `voronoy` paints the patterns and pictures): `wallpaper_buffer` maps the screen-sized shared buffer at `USER_WALLPAPER_CANVAS` (13 GB), the app draws, `wallpaper_commit` makes it live. The frames are **owned by the kernel** → the background persists after the app exits (the `voronoy` case). The agenda widget reads it to choose its ink.

10.5 Modal dialogs

None in the kernel any more: `message_box`, `file_open`, `file_save` (v9/v10, `CDialog`) were removed; the dialogs are user-side (wtk's `MessageBox`, `FileDialog`...), modal within their app.

11. Network subsystem (WLAN, TCP/IP, NTP)

Sources: `kernel/kernel.cpp` (`CNetBringupTask`), `kernel/sys/net.cpp` (socket backend), `kernel/include/kern/net.h`. Built on Circle's `lib/net` (TCP/IP) + the `addon/wlan` BCM4343 driver + `wpa_supplicant`, all linked into the kernel (see `kernel/Makefile LIBS`).

Two placements, chosen at boot by `cmdline.txt netcore=`:

- `netcore=0` (the default): the whole stack on the primary core, as ordinary cooperative tasks among the others; the socket calls run on the app's task.
- `netcore=1`: **the network core**. Core 3 runs the stack with its own scheduler (§5): at boot it waits in `WFE` (`NetCoreMain`, from `COnyxCores::Run`); once the SD card is mounted, core 0 calls `NetCoreStart`, and core 3 creates its `CScheduler`, the bring-up task (below) and 6 **worker** tasks. Everything the bring-up starts — Circle's `CNetTask`, DHCP, NTP, the WLAN driver's `kprocs`, `wpa_supplicant` — is created on core 3 and runs there, so the stack is still used from one core only, as it was written for. The WLAN firmware and `wpa_supplicant.conf` are read from core 3 through FatFs: the volume lock (`sys/fslock.cpp`) takes its owner with an atomic compare-and-swap. The SDIO interrupt stays on core 0 (its handler only masks the controller's interrupt enable; the driver polls), and so do the TCP / ARP kernel timers (they set flags under spin locks, which the stack reads in `Process`).
 - **Requests**. A socket `kapi` (`tcp_*`, `net_resolve`, `net_ping`, `net_info`, `wlan_scan`) copies its arguments — the app's memory is not mapped on core 3 — into one of 32 request slots

(8 KB of data each: a bigger send goes in pieces), marks it *posted*, and waits: it spins 300 μ s (most answers take microseconds), then yields, then sleeps 1 ms, then 10 ms steps (an *accept* may wait for hours). A worker claims the slot (compare-and-swap), runs the same *Do** function as `netcore=0`, and marks it *done*; a *recv* gathers several segments (up to 8 KB) in one round trip. When every worker waits (accepts, connects), the core adds one (up to 24). `net_status` reads the state directly.

- **BSD sockets (v75).** Their calls are requests too (`NR_S*`), except a connect: a **detached** request (the caller does not wait; the worker frees it; `NetCloseByPid` leaves it alone). The main loop recomputes every open socket's readiness snapshot at each turn and bumps a generation word when one changed, then interrupts core 0 (`IPI_NET_READY`, `COnyxCores::IPIHandler` → `NetReadyIPI`; the 100 Hz tick hook `NetPollTick` as the fallback): `IoWake`, which wakes `poll` and the blocking BSD calls (§8 "v75: sockets and poll").
- **The Wi-Fi driver** is polled on this core instead of waiting for its SDIO interrupt, its scans cover both bands and probe for the networks of `wpa_supplicant.conf` by name, 5 GHz is preferred, a frame is one SDIO command on a 50 MHz bus (`onyx_wlfast = 31`), the frames sent are not aggregated (`onyx_wl_ampdu_tx = 0`: aggregated, half of what the Pi sent during a download was lost — the remote desktop froze) but sent in bursts (`onyx_wl_frameburst = 1`), and TCP scales its window and delays its acknowledgements (`onyx_tcp_ws = 3`, `onyx_tcp_ackn = 8`) (`NetWlanOptions`, `NetWlanNames`; our Circle fork, docs/05 §26 and §27).
- **A trial.** A Pi is often reachable by its Wi-Fi only: a driver change that keeps the Wi-Fi from coming up cannot be taken back from the PC. `SD:/etc/net-trial.txt` holds `name=value` words — `wlfast=` (the driver's fast path bits), `tcpws=`, `tcpwin=`, `ackn=`, `ampdutr=`, `ampdurx=`, `bawsize=`, `rxbawsize=`, `ampdurts=`, `bw5=`, `frameburst=`, `netsleep=`, `netsleepus=`, `netstat=1` (the statistics, the firmware's counters, TCP's timeouts in the log), `secs=` (default 180) —; the bring-up reads it, **deletes it**, applies it for this boot (`NetTrialLoad`), and core 0's main task restarts the Pi after `secs` (`NetTrialPoll`) unless `SD:/etc/net-trial.keep` exists by then: the next boot is without the trial. Before the restart the kernel log's tail (24 KB) is written to `SD:/etc/net-trial.log`: what the driver said during a trial that cut the network is read after it. A new switch is added there, tried, and only then made the default.
- **The network core sleeps when the network is quiet** (2026-10-04). Its tasks wait by yielding, so the core used to turn all the time: 130 000 rounds of its scheduler a second, each asking the chip whether it has a frame. Now `NetCoreMain`, after each round, sleeps (`wfe`) when all of this holds: no frame read from the chip or written to it for 50 ms (`onyx_wl_lastact`, the driver's), no request posted by core 0 for 50 ms, none waiting, and the driver has asked the chip once, whole, since the last sleep and it had nothing (`onyx_wl_polls`: a question is several SDIO commands and each waits a round — sleeping between the rounds made a ping take six sleeps). The sleep ends after 1 ms — the timer's event stream ends a `wfe` every 1.2 ms at most (`CNTKCTL_EL1`, set on that core) — or at once when core 0 posts a request (its `sev`); a `wfe` also ends at every spin lock released on any core (Circle's unlock sends the event), so it is asked again until the time has passed. The first frame or request brings the full pace back. Measured (Pi 4, 150 pings of a quiet Pi): 4.1 ms on average without the sleep, 4.4 ms with it, the core asleep 93 % of the time; the bench, the pings during a download and the echoes are the same with and without. Trial words: `netsleep=<ms>` (0: never sleep), `netsleepus=<us>` (the sleep's length). The chip's SDIO interrupt is not used yet (it would end the sleep at once: the next step).

- **netstat=1** (cmdline.txt): every 5 s the log says the net core's pace (net: core 3: N rounds/s, ... us a round — the stack's tasks all wait by yielding, so a round is the unit of every wait: ~13 μ s) and the driver's (frames a second, a frame's read time, the link's rate).
- **A process that dies** (in a request, or with sockets open): `NetCloseByPid` (its teardown, IRQs masked: nothing waits) drops its posted requests, orphans the running ones (the worker then closes what they opened), and queues its pid in a ring that the net core's main loop reads to close its sockets.
- **Notices.** `IpcNotify` is core 0's: the bring-up leaves its "Connected" notice to `NetCoreNotify`, and the kernel's main task delivers it (`NetCorePoll`, every 250 ms).
- Core 3 is then no longer an app core (§14).
- **Bring-up is non-blocking and non-fatal.** `CNetBringupTask` (a kernel `CTask`, started from `Run()` once the SD card is mounted) runs `CBcm4343Device::Initialize` → `CNetSubSystem::Initialize(FALSE)` → `CWPASupplicant::Initialize`, waits for the DHCP bind, logs net: up, IP ..., then **returns** (the task ends cleanly). A missing firmware / `wpa_supplicant.conf` / access point only logs a warning — the GUI boots regardless.
- **Self-driving.** `CNetSubSystem::Initialize` spawns Circle's own `CNetTask` / `CPHYTask`; they run as ordinary cooperative tasks on our scheduler (§5), so no explicit pumping is needed. (This is also why the net task is CPU-busy — the reason for `netcore=1`.)
- **Globals.** `g_pNet` (the `CNetSubSystem`) and `g_bNetUp` are published in `net.h`; `NetIsUp()` gates the socket calls.
- **Sockets.** `sys/net.cpp` keeps a small table of Circle `CSockets` behind integer handles. `tcp_connect` resolves a dotted-quad or a DNS name and connects (blocking, cooperative); `tcp_send` blocks with a 5 s timeout; `tcp_recv` is **non-blocking** (so a GUI app polls it from its frame loop; with `netcore=1` one call gathers the segments that fit in its 32 KB request buffer, `NET_REQBUF`); `tcp_close` drops it. A connect **takes its slot before it blocks** (`SLOT_CONNECTING`, invisible to the other calls): the net core's workers run other connects during a DNS lookup or a handshake, and two of them once got the same slot (both fetches then read one connection: a browser's style sheets failed). **DNS cache:** 32 names kept 5 minutes (`ResolveName`, used by `tcp_connect`, `net_resolve`, `net_ping`) in front of Circle's `CDNSClient` -- which answers in milliseconds now (it slept 1 s per query: docs/05 §18). Each socket records its **owner pid**; `kapi_exit` calls `NetCloseByPid` as soon as the app's other threads have ended (2026-09-30: it waited for the janitor's reaping, `AddressSpaceTaskTerminate`, which calls it again and finds none left), so a process that exits or dies without closing does not hold its connections or table slots -- a browser relaunched at once found them still held and its pages waited (10 s). The table has **256 slots** since v75 (`MAX_SOCKETS`; 64 before, 16 before that: a browser keeps a dozen open), and it also holds the **BSD sockets** of v75 (TCP and UDP, non-blocking calls, `poll`: §8 "v75: sockets and poll"); `tcp_recv` reads through a slot's carry buffer (a buffer smaller than a segment lost the rest of it with `netcore=0`). `net_info` lists every socket that has a Circle socket: `tcp <n> listen|conn ...` and, since v75, `udp <n> bound <port> <default peer|-> <pid>`.
- **TCP fixes in our Circle fork** (2026-10-01, docs/05 §20–22; host test `tools/tests/run_circletest.sh`): only a **real duplicate ACK** (no data, no SYN / FIN, the window unchanged, data in flight, ACK = SND.UNA: RFC 5681 §2) counts towards a fast retransmit -- a remote desktop client's input messages (data segments that do not advance the ACK) started a spurious fast retransmit + recovery while the mouse moved, the Pi's sending

throttled to ~2 segments a round trip. The **RTO** starts at 1 s (RFC 6298; was 3 s) and its floor stays **1 s** (200 ms was tried and undone: spurious timeouts over Wi-Fi to Windows collapsed the remote desktop's throughput), Karn's algorithm also after a fast retransmit; a connection is given up after 6 timeouts (63 s). **tcp_send's count** is exact: Circle's `CSocket::Send` queues MSS-sized chunks, and when one times out (5 s, its queue full) the chunks before it are counted (it answered the error, those bytes already on their way: an app resending "the rest" put bytes twice in the stream); `NetTcpSend` adds up its 32 KB requests the same way. A short count means the bytes queued, then a timeout: resending the rest is right (rdpd does).

- **Clock.** Once the link is up the bring-up task starts a `CNTPDaemon` (`system.ini ntp=` off or none: no daemon, the clock is not set), which updates `CTimer`'s wall clock; the boot reads `system.ini timezone=` (minutes from UTC) and calls `CTimer::SetTimeZone` so `get_datetime`, the agenda and the log timestamps show local time (`kapi_set_timezone`, v69, changes it while running).
- **Host name.** `system.ini hostname=` (letters, digits, -) is handed to `CNetSubSystem` before the bring-up task starts (`SetHostname`, an Onyx addition to Circle: docs/05) — the name DHCP announces; default Circle's `raspberrypi`. Setup writes it; it takes effect at the next start.
- **Caveats.** The kernel's sockets are plain TCP and UDP (TLS is done in user space: `user/tls`, used by Jet, Mail, `httpsget`, `wget`, the package manager...); the scheduler's task list has no limit any more (§5: the old `MAX_TASKS` of 40, raised for the net workers, is gone); the firmware load uses FatFs and is not locked against concurrent app file I/O (low risk, one-shot at boot) — with `netcore=1` it is (the atomic volume lock).

The apps that use it: Jet Browser, Mail, the IRC client, Lisa (an LLM), the package manager (`pkgman`, `pkgd`, `/bin/pkg`), and the `/bin` network tools (`net`, `ping`, `nslookup`, `netstat`, `wget/httpget/httpsget`, `ftp`, `whois`) and servers (`ftpd`, `telnetd`, `vncd`, `rdpd`) — docs/04 §12 and the `/bin` table.

12. Sound and the second core

Source: `kernel/sys/sound.cpp`, `kern/sound.h`.

- **Multi-core.** Circle is built with `ARM_ALLOW_MULTI_CORE` (fork patch #4, [Circle Changes](#)). `CKernel::Initialize` starts cores 1–3 through a `CMultiCoreSupport` subclass (`COnyxCores`, `kernel.cpp`) right after the kapi table is published. **Everything else stays on core 0** (but the network with `netcore=1`, on core 3: §11): the scheduler, every process, the interrupts (the GIC routes peripherals to core 0; core 1 runs Circle's own `VectorTable`, not our `KVectorTable`). Core 1 runs `SoundCoreMain`; cores 2 and 3 are **app cores** (§14). A failed start is only a warning (no sound producer, no app cores).
- **The device.** `COnyxSoundDevice` derives from Circle's `CPWMSoundBaseDevice` (PWM + DMA, the 3.5 mm jack; 44.1 kHz, 1024-frame chunks) and overrides `GetChunk` — the "producer" — which is called from the DMA completion interrupt. It is created and started on the first `sound_acquire` (not at boot).
- **Core 1 = the producer.** It sleeps in `WFE` until the device starts, then keeps `SND_AHEAD` (4) chunks rendered ahead in a ring; `GetChunk` only copies the next chunk (zeros if core 1 fell behind) and `SEVs` core 1. Without `ARM_ALLOW_MULTI_CORE` the same code renders in `GetChunk` itself (interrupt, core 0).

- **Rendering** (integer only, no FP in the kernel): 16 voices, each a 32-bit phase accumulator (`inc = milliHz · 232 / (1000 · 44100)`), waveforms square / sine (256-entry table, linear interpolation) / triangle / saw / noise (LFSR per period), a linear ~5 ms attack/release envelope (no clicks), mixed ÷4 and clipped; plus the **PCM ring** (0.5 s of s16 stereo frames, `sound_write`). Converted to the PWM range per sample.
- **FM voices (v47)**. `sound_instrument (voice, struct kapi_fm_instrument)` gives a voice a 2-operator FM instrument in the style of the AdLib's OPL2 (op 0 = modulator, op 1 = carrier): multiplier, output level (0.75 dB steps), attack / decay / sustain level / release rates (OPL2 timings: attack 2.8 s at rate 1, decay / release 39 s over 96 dB at rate 1, halving per step), sustained (EG type), tremolo (1 dB, 3.7 Hz) / vibrato (7 cents, 6.1 Hz), waves sine / half / absolute / quarter pulses, feedback 0–7 and connection (FM or additive). Then `sound_start (voice, f, SOUND_FM, volume)` keys it on (both envelopes restart) and `sound_stop` keys it off (the release rate fades it). The envelopes run on an attenuation in 1/256 octave units (4096 = 96 dB), turned into amplitude by a 256-entry $2^{(-x/256)}$ table; the modulator bends the carrier's phase by up to ±4 periods (as OPL2). Tables: `sys/sound_tables.h` (generated). The same file builds on a PC with `SOUND_HOST_TEST` (the Circle parts left out): `tools/tests/run_fms_test.sh` renders instruments with it, and the Windows FM Song player (`tools/fmsplayer`) plays with it.
- **Low latency (v68, sound_config)**. `CPWMSoundBaseDevice`'s chunk size is fixed when it is made — but it is only the size of its two DMA buffers: Circle's `CDMASoundBuffers` programs each transfer with the length `GetChunk` returns. So the device is made once, with the biggest chunk (1024 frames), and a smaller chunk is just a shorter return: nothing re-created, the PWM clock untouched, no Circle change. The owner picks the chunk (`s_nChunkFrames`, 64 .. 1024 frames) and how many chunks core 1 keeps ahead (`s_nAheadCfgr`, 1 .. `SND_AHEAD` 4); each slot of the ahead ring records its own length (a change applies from the next chunk rendered — the ones already rendered play out first). What is heard lags the rendering by about **(ahead + 1) chunks** (the ahead ring + the DMA buffer queued behind the one playing): 1024×4 (the default, what the emulators use) ≈ 116 ms; $256 \times 2 \approx 17$ ms; $128 \times 2 \approx 9$ ms; $64 \times 1 \approx 3$ ms (a DMA interrupt every 1.5 ms; core 1 must never be late: with 1 chunk ahead any hiccup is a click). Add what the app keeps queued in the stream or the ring. `GetChunk` now frees a slot only after copying it (it used to free it first: with 4 slots core 1 could render into the one being copied). `sound_release` / the owner's death restore 1024×4 .
- **The mapped ring (v68, sound_map)**. One 64 KB page (`SoundRing`: a 64 KB-aligned heap block, made on first use and **kept for ever** — core 1 may be reading it when its owner dies) holding `struct kapi_sound_ring`: a header (`wr` the app's, `rd` the kernel's — free-running frame counters —, `dry`, the rate, the chunk / ahead now) and 8192 s16 stereo frames (~186 ms). `kapi_sound_map` maps it into the owner like a shared surface (`CAddressSpace::MapSurface`, the surface arena; the teardown drops the mapping only), once per process. `Render` reads `wr` once per chunk, mixes up to a chunk of frames with the voices and the stream, then moves `rd` (DMB in between both sides); `dry` counts the times the ring ran out while playing. The kernel only ever indexes it masked by its own capacity and treats `wr - rd > 8192` as empty, so nothing an app writes there can make it read elsewhere. Since it is plain memory, **an app core fills it** (`kapi_sound_ring_write`), and a core-0 thread can sleep on `rd` with `wait_word` (the tick sees core 1 move it) — the DAW's engine writes straight into the kernel's ring, no pump thread in the audio path. A former owner keeps the page mapped in its space (the mapping is dropped only with the process), so it could still write into it; the kernel stops mixing it at the release. `ringtest` plays a tone from an app core this way.

- **Ownership.** One pid owns the output (`sound_acquire`); every other call from another pid returns `-1`. `sound_release`, or the owner's exit (`SoundOnProcessGone`, called from `IpcOnProcessGone`), silences the voices, empties the ring and frees the output. The state is shared between core 0 (kapi calls) and core 1 (rendering) under a `CSpinLock`.
- **kapi v46:** `sound_acquire`, `sound_release`, `sound_start` (`voice`, `milliHz`, `wave`, `volume`), `sound_stop` (`voice` | `-1`), `sound_write` (`frames`, `n`), `sound_status`. Users: `/bin/tone`, BASIC `PLAY / SOUND / BEEP / NOTEON / NOTEOFF`.
- **kapi v68:** `sound_config` (`chunk_frames`, `ahead`), `sound_map` () (above).

13. Post-mortem debug console

Source: `kernel/sys/debugcon.cpp`, `kernel/include/kern/debugcon.h`.

Purpose: when an app terminates (or in case of trouble), make the logger's last messages visible **directly on the framebuffer**.

- **CFbConsole:** text console that renders the characters (green on black) on the back buffer of `C2DGraphics`, with line wrapping, scrolling, and `UpdateDisplay()` after each write.
- **CLogSwitch:** routes the logger's output either to the boot console (`m_Screen`), or to the framebuffer console. At boot: `SetNormal(&m_Screen) + DebugConsoleRegister`.
- Once the compositor is started, `MuteNormal()` stops the writes to the boot console: it is no longer shown, and each line drawn there (a text-mode scroll of the whole screen inside `CTerminalDevice::Write`, IRQs masked) froze everything for ~170 ms — an app that `printfs` without a terminal (Doom at start-up) froze the GUI and the network for seconds. `kmsg` still sees every line (CLogger's event ring does not depend on the target).
- When the debug console takes over (`DebugConsoleTakeover`, called on `exit`), the compositor **detects** `DebugConsoleActive()` and **stops presenting** so as not to contend for the framebuffer.

The crash record (`lastcrash.txt`)

Source: `kernel/sys/crashlog.cpp`, `kern/crashlog.h`.

A frozen Pi (the scheduler stopped: the green LED stays lit or dark, no SOS) left nothing to read. Now 64 KB of RAM kept out of the heap (Circle fork, docs/05 §15) hold, for the running session, cleaned to RAM as they are written:

- the **tail of the kernel log** (`CLogSwitch::Write` copies every line, ~62 KB ring);
- the **last 16 places core 0 was interrupted at** (`KernelIRQExit`, each IRQ): PC, LR, SP_ELO, whether IRQs were masked, the task's name — a kernel loop shows itself there;
- **breadcrumbs:** the GPU (`idle` / `clipping` / `binning` / `rendering` / `texture upload`) and the compositor's display copy (`composing` / `display DMA` / `waiting for the display DMA`);
- the reaper's **last pass** (uptime), and a **panic's** registers (`DumpAndHalt`, before the panic screen) with the **return addresses found on the faulting context's stack** (`CrashLogStack`);
- the **free memory**, once a second (`CrashLogMemory`, breadcrumbs 5–7): the kernel heap, the kernel pages (low pager), the app pages (the high zone) — an "Out of memory" shows its approach;
- the **app the GUI watchdog finds frozen** (below): its pid, the last 16 places its task was interrupted at, and its stack's return addresses.

Return addresses. The apps are built `-O2` (no frame pointers to follow): `ScanStack` reads 16 KB up from the context's SP and keeps the words that point just after a `BL / BLR`, in an app's code (`[8 GB, 10 GB)`) or the kernel's (`[0x80000, _etext)`) — 24 at most, newest first. Each page is probed first with `AT S1E1R` (a wild SP, an unmapped page: no fault). Read them with `aarch64-none-elf-addr2line -f -C -e user/<app>.elf <addr>` (or `kernel/kernel8-rpi4.elf`); a few are stale words, the first ones are the calls in progress.

The **BCM watchdog** is armed by the reaper (every second, `cmdline.txt hangreboot=` seconds, default 15, at most ~15; 0 = off): when the scheduler stops, the Pi restarts by itself (a panic too, after its screen and SOS). `kapi_shutdown / kapi_reboot` stop it and mark the session clean. At the next boot, a session that did not end cleanly (its record still in RAM: a power cut loses it) is written to `SD:/etc/lastcrash.txt` and noted in `kmsg` (`crashlog`). Best effort: nothing if the RAM did not keep its contents through the reset.

Core 1 writes the report itself (the RAM does not survive the Pi 4's watchdog reset: the first real freeze left no record). At boot (`CrashLogReport`, SD: mounted) the kernel keeps the previous dump if `SD:/etc/crashdump.txt` holds one (`ONYX CRASH REPORT ... --- end of the report ---` → `SD:/etc/lastcrash.txt`), then fills that 64 KB file with "armed" and notes each of its 128 sectors' LBA (through FatFs: `f_lseek + a one-byte f_read` → `FIL.sect`). Core 1, the sound core (`SoundCoreMain`), is woken at least every ~1.2 ms by the generic timer's event stream (`CNTKCTL_EL1.EVNTEN, EVNTI 15`) and calls `CrashLogCoreCheck`: when the reaper's `CNTPCT` stamp has not moved for **10 s**, it formats the record (no heap, no logger), writes it **raw** through the `emmc1` device into those sectors (runs of consecutive sectors in one write; `g_bCrashDumping` keeps `OnyxDriverWait` from yielding) and restarts the Pi (`CBcmWatchdog::Restart`). No FatFs, heap, logger or scheduler call: core 0 may hold their locks; core 1 takes the sound lock only with `TryAcquire`, so a core 0 frozen while holding it does not stop the watch. Its progress is kept in the RAM record (`nDumpStep`: started, text ready, written, the write failed): when the SD write cannot finish (a wedged bus), the hardware watchdog restarts the Pi and the RAM report (the record of an 8 GB Pi sits at the top of the RAM above 4 GB) says how far core 1 got. **A stuck GUI with the scheduler alive** (the compositor without a frame for 12 s, the GUI watchdog task) asks for the same report (`CrashLogRequest`: core 0 masks its IRQs and waits, core 1 writes it, then the restart). **A Circle panic** (an assertion, the kernel heap's "Out of memory", ...) used to leave nothing: Circle's logger halts **every** core after the panic line (`CMultiCoreSupport::HaltAll`), core 1 too, so the Pi stayed frozen ~15 s and the hardware watchdog restarted it without a report. The logger's panic handler is now `CrashLogPanicHandler`: the log's last line (the panic's) goes into the record and core 1 writes the report at once (`CrashLogDumpNow`), before the halt. An exception (`DumpAndHalt`) asks for it at once too, after its screen and one SOS (it used to wait the 10 s of a stopped reaper).

The app watchdog (`SD:/etc/apphang.txt`). When the GUI watchdog finds an app not taking its window's events (2 s, some queued), it watches that app (`CrashLogWatchPid`: core 0's IRQs note where its task is and, every 8th time, scan its stack) and, every 2 s while it lasts, has the `apphang` task (its own: never the GUI watchdog stuck on the card) rewrite `SD:/etc/apphang.txt`: the app, its tasks' states (running / ready / blocked / sleeping) and pages, then the whole record (the free memory, core 0's samples, the app's samples and stack) and the log's last 12 KB. The compositor without a frame does the same (every 2 s, before the 12 s report). Recovered (the app takes its events again, or is closed / killed): renamed `SD:/etc/lasthang.txt`; a clean shutdown / reboot renames it too. Still there at the next boot (the session never ended cleanly — the freeze grew into a restart): appended to `SD:/etc/lastcrash.txt` (`MergeAppHang`), or written as it when the kernel kept no record of its own. **Power and heat.** Once a second the GUI watchdog task calls `CrashLogPower`: the firmware's `GET_THROTTLED` bits (under-voltage, ARM frequency capped, throttled, soft temperature limit — now, or since boot) and the SoC temperature. A change (or each 5 °C above 60 °C) is logged (`power: SoC 71 C, throttling: under-voltage`).

(`earlier`)), and the last values plus the session's highest temperature go into the crash record (breadcrumbs 2–4), printed in every report — a freeze that only a cold boot brings back smells of the supply or the heat.

The clock across boots. The Pi has no battery-backed clock: until NTP answers (~15 s), the time was 0 and the files written meanwhile (`crashdump.txt`, `lastcrash.txt`) had no date. `SD:/etc/clock` keeps the last time seen (UTC seconds, text; written every 10 minutes by the GUI watchdog task and by `kapi_shutdown` / `kapi_reboot`) and is read back at boot, right after `SD:` is mounted (`CrashLogClockRestore`) — behind by how long the Pi was off, until NTP corrects it.

Headless signs on the green ACT LED (a GPIO set / clear, no lock): **3 s of fast blinks** when core 1 sees the hang, the SD write's own flicker, then **3 s lit** (written) or **3 slow blinks** (the write failed) before the restart; no blinking at all = core 1 stuck too (a wedged bus). `hangreboot=0` turns it off with the watchdog. `kmsg` tells at boot where the record is and whether the dump is armed (`crashlog: ...`). Test: none from an app any more (an app at EL0 can neither mask the IRQs nor stop the scheduler).

The panic screen is now copied into the displayed frame buffer **by the CPU** (not the DMA: the compositor's display DMA may be in flight, and waiting for it hung the panic before its screen and its SOS).

14. App cores (cores 2 and 3)

Source: `kernel/sys/appcore.cpp`, `kern/appcore.h`; user side `user/emucore.h`, test `user/bin/coretest.c`.

Cores 2 and 3 (core 2 only with `netcore=1`: core 3 then runs the network, §11) are a **resource an app acquires**, like the sound output: it gets a whole core and runs one function of its own code there, undisturbed (no scheduler, no timer, no other task on that core). The rest of the system stays on core 0 as before, so none of the kernel, Circle's drivers, FatFs or the network has to be multi-core safe.

- **The core's loop** (`AppCoreMain` → `AppCoreLoop`, from `COnyxCores::Run`): it installs **our** `KVectorTable` on that core (`VBAR_EL1` is per core), notes its kernel stack, enables IRQs (for the stop IPI only: the GIC routes no peripheral there) and waits in `WFE`. A job: load the owner's `TTBR0` (its L2 table + ASID, `CAddressSpace::GetTTBR0`), `tlbi vmal1e1` on that core only (an ASID may have been reused since its last job), then `El0Enter` (`fn`, `stack`, `arg`, `El0CoreReturn`) — at **EL0**, `SP_EL0` = the app's stack, like the app itself on core 0; the job's end is an `svc` from `El0CoreReturn` (→ `AppCoreEl0Done`), any other `svc` or a fault stops the job (`KAPI_CORE_FAULT`). When `fn` returns: back to the kernel address space, state `IDLE`, `SEV`.
- **The kapi side (core 0).** `core_acquire` hands a free, started core to the caller's address space; `core_run` checks the owner, that nothing runs, that `fn` and the stack are user addresses, fills the job, then `bGo` + `DSB` + `SEV`. Every word shared between the cores is plain cacheable memory (inner-shareable, coherent) with `DSB ISH` barriers; only core 0 writes the ownership.
- **Stopping a job** (`core_release`, the app's exit or kill): core 0 raises `bAbort` and sends the core an IPI (`SendIPI`, `IPI_USER`). The IRQ interrupts the job at EL0 and enters `El0IrqEntry` on that core (`IrqEntry` if the core was in its own kernel code); the exit path (`El0IrqExit` / `KernelIRQExit`) sees it is not core 0 (no scheduling there) and calls `AppCoreOnIRQExit`, which **rewrites the trap frame** to return into `AppCoreRestart` on the core's own kernel stack (EL1t, IRQs masked): the job is simply dropped. `AppCoreRestart` goes back to the kernel address space and clears `bAbort` — core 0's signal that the core is out of the app's memory. Core 0 waits for it at most 200 ms.

- **Faults.** A synchronous exception from the job at EL0 (a bad access, an undefined instruction, any svc but the job's end) reaches `EL0SyncHandler`, which hands it to `AppCoreOnEL0Sync` → `AppCoreOnFault` (instead of killing a process): the ESR class, PC and fault address are kept, the state becomes `FAULT` and the frame is rewritten to `AppCoreRestart` as above. Core 0 logs it once (`appcore: core N: fault EC=... at pc ... (address ...)`) when the owner asks the state or releases the core.
- **Memory (v75, §4 Demand paging).** No allocator runs on cores 2–3, so what a job touches is filled beforehand: `core_acquire` fills the owner's heap and makes it **eager** (filled as `sbrk` grows it — the emulators `malloc` on the main thread and touch on the core), `core_run` fills the top 256 KB of the job's stack. Anything else still unfilled (a `vm_map`'ed buffer, deeper in a lazy stack) takes the slow path: the job's translation fault becomes a **page-in request** in its `TAppCore` (`nPageIn = 1`, the address, write or not; `DSB`, `SEV`) and the core waits in `WFE` with its IRQs on (a stop still drops the job). On core 0 an `IoWait` tick hook wakes the `vmpager` task (made at the first `core_acquire`), which fills that page and up to 7 more of the region by a software walk of the owner's tables, answers 2 (retry: the `eret` repeats the access, no TLBI needed) or 3 (no region, or out of memory: `FAULT`) and `SEVs` — up to one 10 ms tick per request. Frames leaving or protections lowered on core 0 reach the job's TLB by the inner-shareable `TLBI`.
- **TLS (v75).** `core_run` reads the caller's `TPIDR_EL0` (the kernel never changes it) and the core writes it before `EL0Enter`: a job shares its caller's thread-local data (errno included).
- **Teardown.** `~CAddressSpace` calls `AppCoreReleaseAS (this)` **before** freeing the window and the frames: a job still running uses them. A core that does not answer the stop (its code masked the interrupts — never do that) is **retired** (`bLost`, never used again) and the dying space's memory is kept (leaked) rather than freed under it.
- **The rules for fn:** no kapi call (the kernel is not called from two cores at once), no allocation (newlib's and umm's `malloc` are not multi-core safe), no masking of the interrupts. It computes, reads the clock directly (`cntpct_el0`) and exchanges data with the app's main thread through memory, with barriers. It may use `WFE`, woken by the main thread's `SEV`.
- **user/emucore.h** packages that for the emulators: the machine runs on the app core (`ec_thread`), the main thread asks for frames (`ec_request`), gets the pictures from a **triple buffer** (`ec_publish` / `ec_take`: the machine never waits for the display and the display always gets the latest complete picture) and the sound from a single-producer ring (`ec_audio_push` / `ec_audio_pop`), and stops the machine between two frames to touch it (`ec_hold` / `ec_resume`: reset, palettes, battery saves). Without a free core, `ec_pump` runs the same frames on the main thread. `gbemu`, `gbaemu`, `nesemu` and `snesemu` use it.
- **Test:** `/bin/coretest` (the same computation on an app core and on core 0, a job stopped by its flag, an endless job stopped by `core_release`, a faulting job, both app cores at once); `coretest exit` leaves a job spinning and exits (the teardown must stop it). `/bin/memtest` (v75): a job writing 4 MB of unfilled memory (the pager, timed), its TLS.
- **Bigger programs on an app core:** newlib's syscalls can run on the main thread when called from an app core (`libc/onyx_syscalls.c`, `onyx_rpc_*`: the caller posts the call and waits in `WFE`, the main thread runs it in `onyx_rpc_serve` and `SEVs`). Doom's engine runs that way (`user/doom`), with its `malloc`, files and saves.

15. The GPU (V3D)

Source: `kernel/sys/v3d.cpp`, the packets `kern/v3d_c1.h`, the shaders `sys/v3d_shaders.qasm`, the texture layouts `kern/v3d_tiling.h`; demos `user/Apps/teapot` (v52), `user/Apps/gpudemo` (v53), `/bin/gpcdemo` (the compositing service over v53 + v70).

The Raspberry Pi 4's VideoCore VI 3D core (**V3D 4.2**) driven from the kernel, bare metal — after Random06457's *rpi4-gpu-bare-metal-examples* (MIT: the packet structs, the three shaders, the control-list recipe) and macoy's *rpi-system* notes (cache cleaning, the binner's memory).

- **Bring-up** (the first `gpu_info` / `gpu_draw`, never at boot): the firmware turns the V3D power domain (10) and clock (5) on and the clock to its maximum (mailbox); the reset is released (`PM_GRAFX V3DRSTN`, with the PM password) and the V3D's async AXI bridges opened (`ASB_V3D_M_CTRL` / `_S_CTRL` at `0xFEC11008/C`, waiting for the ACK bit with a time limit); then `HUB_IDENT1` must say version 4 (kmsg: `HUB_IDENT0..2, CTL_IDENT0`). Any failure leaves the GPU off; `gpu_info` says why. The interrupt is connected there too (see below).
- **Memory**: no V3D MMU — the GPU gets physical addresses, all its buffers come from Circle's low heap (`HEAP_LOW`, < 1 GB, page aligned): the control lists, the shaders and default attributes, the vertices, the render target (raster RGBA8), the tile allocation (the tiles × 64 B + 2 MB), the tile state (256 B a tile) and a 4 MB overflow pool. CPU caches: cleaned after the CPU writes (the lists, the vertices), the target cleaned and invalidated before the GPU writes it and invalidated before the CPU reads it; the GPU's L2T and slice caches are flushed before each job.
- **A frame** (`gpu_draw`): the vertices are copied into the GPU buffer; the **binning list** (`TILE_BINNING_MODE_CFG`, the clip window, `CFG_BITS` with the depth test *less* and depth writes, the viewport scaling and offset, `GL_SHADER_STATE` → a shader record + two attribute records over the same 16-byte vertices — position 3 floats, colour 4 normalized bytes —, `VERTEX_ARRAY_PRIMS` triangles) sorts the triangles into 64 × 64 tile lists; the **rendering list** configures the tile buffer (one RGBA8 target, Z cleared to 1.0), clears, then for each tile runs its list, stores the colours (`STORE_TILE_BUFFER_GENERAL`, raster) and clears again. Jobs: `CT0QMA/QMS/QTS/QBA/QEA` (bin), `CT1QBA/QEA` (render); the end is polled on the frame counters `BFC` / `RFC` (the caller's task yields meanwhile; one frame at a time), 0.5 s at most — past that the state of the control-list executors is logged and the GPU left off. The binner's out-of-memory requests (`INT_STS OUTOMEM`) are answered from the overflow pool (`PTB_BPOA` / `BPOS`). The RGBA8 picture is copied into the caller's pixels as `0x00RRGGBB`.
- **Shaders**: fixed — a vertex shader (position → screen, colour → varyings), its coordinate shader for the binner, a fragment shader writing the interpolated colour: Mesa-compiled QPU code taken from Random06457's example. The geometry, the lighting and the projection are the app's (the CPU); the GPU does the rasterization, the interpolation and the depth test.
- **The full pipeline** (`kapi v53`, `gpu_render`): the same frame recipe with **batches**. Each batch has its own shader record (the attributes: position x y z w — 4 floats —, s t, colour; s t, colour, added colour; 32-byte vertices), its uniforms (the batch's 4 × 4 matrix, then the viewport scales), its `CFG_BITS` (depth function and writes, culling: `clockwise_primitives` set, front = counter-clockwise in y-up coordinates, as GL), its blending (`BLEND_ENABLES` + `BLEND_CFG`: alpha `SRC_ALPHA` / `INV_SRC_ALPHA`, add `SRC_ALPHA` / `ONE`, multiply `DST_COLOR` / `ZERO`, premultiplied `ONE` / `INV_SRC_ALPHA`) and one `VERTEX_ARRAY_PRIMS` over its range of the vertex buffer. `KAPI_GPU_F_KEEP` loads each tile from the target first (`LOAD_TILE_BUFFER_GENERAL`, raster RGBA8, alpha forced to 1) instead of clearing it. The target's load and store packets are built in one place (`V3dLoadTarget` / `V3dStoreTarget`, `kern/v3d_c1.h`). **Fixed (2026-09-30)**: the load packet's packed

bit-fields lacked 3 bits of padding after `r_b_swap`, so its stride landed at bit 25 instead of 28 and the V3D read it divided by 8 — every frame drawn over the target's pixels (the compositing service without a clear; `gpcdemo test`'s "over the target's pixels" failed on the Pi) loaded the wrong rows. `tools/tests/run_v3d_cl_test.sh` now checks each field of the load / store packets against Mesa's `v3d_packet.xml` positions, and the PC's software V3D (`tools/tests/gpucomp/hostkapi.cpp`) decodes the kernel's own packets for the target's stride (it re-implemented the frame and could not see the packets; `gpcdemo test` also drew over a single colour, which hid wrong rows — it draws over a pattern now).

- **Shaders of v53** (hand-written: `sys/v3d_shaders.qasm` → `v3d_shaders.inc`, assembled by `tools/qpu/qpuasm`, see below): `VS_CLIP` — the 10 inputs, the matrix product (16 `ldunif`), $1/w$ on the SFU (`recip`, `r4`), $Xs Ys$ in 24.8 fixed point, Zs , $1/Wc$, then the 10 varyings `s t r g b a r2 g2 b2 a2` (v54); `CS_CLIP` — its binner copy (`clip X Y Z W`, then $Xs Ys$); `FS_COLOR` — the interpolated colour ($varying = ldvary \times W + C$, `C` in `r5` two instructions later); `FS_TEX` — the texture lookup (`T` to `tmult`, the TMU configuration `p0 / p1` by `wrtmuc`, `S` to `tmus` starts it, as Mesa orders them; a thread switch while it runs; `ldtmu` returns `R G` and `B A` as half floats) multiplied by the colour; both then add the second colour and clamp to 1 (v54). `FS_COLOR_AT / FS_TEX_AT` (v54, `B_ALPHATEST`) read the threshold as one more uniform and discard the fragment (`fsub.pushn`, then `setmsf.ifa - , 0`) when alpha is below it. The fragment shaders end with the TLB writes (`vfpack tlb`) after the last-segment thread switch.
- **App shaders** (`kapi v61`, `gpu_program / gpu_render2`): the app brings its QPU code (built at run time with the C++ builder `user/v3d/qpu.h` over Mesa's instruction packer, `tools/qpu/mesa`; tested on the PC by the simulator `tools/qpu/qpusim` — `tools/tests/run_qpu_test.sh`). A program (`TProgram`, ≤ 256 , owned by its process) holds the three shaders in low GPU memory, each at a 256-byte boundary and followed by NOPs (the QPU fetches ahead). Each batch of `gpu_render2` gets its shader record from the program: the VPM segment sizes (`[inputs / 8]`, `[(4 + varyings) / 8]`, the coordinate shader's `[csInputs / 8]` and `[csOutputs / 8]`), the fragment shader's threading (`P_FS_4WAY`, `P_FS_FINAL`) and `Z` writes; one float vec4 attribute per 4 inputs (≤ 16 arrays: the `GL_SHADER_STATE` packet is written by hand, its count has 5 bits), its uniforms copied from the call's array — the samplers first (32-byte aligned), then the vertex, coordinate and fragment ranges, the texture words patched in the latter —, then `CFG_BITS`, `BLEND_CFG` (the batch's own factors and equations), `COLOR_WRITE_MASKS` and `CLIP_WINDOW` (the scissor) when they change. The kernel does not check what the shaders do: a program that never ends is stopped by the frame's time limit (the GPU is then left off), and the TMU's general writes could reach any low memory — apps are trusted, as with `code_alloc`. **P_FS_FINAL is best avoided**: a fragment shader started in its final thread section (no last-segment pair) that reads uniforms stopped the GPU now and then on targets 512 pixels wide or more (5 runs out of 7, mostly the first frame after boot; 64×64 frames always passed; `v3dprog ww`, docs/03). Mesa never uses it for fragment shaders; neither do `user/v3d/gxtev` nor `gcemu` since: they end with the last-segment pair (two `thrs` in a row, the live values in registers of the register file — the accumulators do not survive a thread switch) before the TLB writes.
- **gpu_vbuf / gpu_render3 in place** (v63, `ClipInPlace`): a `vbuf` is a `HEAP_LOW` block (64 KB aligned, below 1 GB: its address is the GPU's) mapped into the program by `MapSurface` (`s_VBuf`, freed by `V3DReleaseAS`). When `gpu_render3`'s array lies in one of the caller's, each batch's vertices are framed where they are and tested; the triangles inside every plane make runs `{first, count}` of the batch's own vertices (`s_pRun`, `s_pRunAt`: `Render2` draws each run with the batch's shader record, its attributes at the batch's address and stride), the others are clipped into the buffer's free end, each fan at a whole number of the batch's strides from it (a run too); then the touched range's cache is cleaned. The copying paths (`ClipFrame2 / 3`) make one run a batch.

- **gpu_render3** (`ClipFrame3`): each batch's triangles are read at its offset and stride in the caller's array, framed (x, y) into the GPU's vertex buffer at that stride — kept as they are when inside every plane, else clipped in their place — so the output is each batch's own packed run (`s_pOutOff`, `s_pOutStride`); `Render2` gives each batch its attribute records at that address and stride, and draws from its vertex 0 (`gpu_render2`'s batches too, at the common stride). The Wind Waker's display no longer copies its ~7 MB of vertices a frame before the call (`gcemu's Out::prepare` 5.9 -> 0.1 ms): the machine's core, sharing the L2 and the RAM, 33.5 -> 29.9 M cycles a field.
- **gpu_render2's clipping** (`ClipFrame2`): a triangle whose three vertices are inside every plane (`V3DInsideN`: the near plane, `w > 0`, the guard band — nearly all of them) is copied once, its program's inputs only, straight into the GPU's vertex buffer (`s_Verts2`, sized for the clipped fans); the others go through `V3DClipTriangleN` (every float interpolated) into the same buffer. Before, each vertex was copied four times (staged, clipped, into a buffer, into the GPU's) — ~30 MB a frame for The Wind Waker's ~100k vertices through the L2 the app cores share: its `gpu_render2` went from ~34 to ~25 ms a frame. Every 256 frames `kmsg` gets `render2`, a frame: `clip ... us, lists ... us, GPU ... us, target ... us; ... vertices` (the clipping and its copy; the lists, the uniforms and the target's cache cleaning; the binning and rendering; the target after) — Wind Waker's game: ~9.8 ms, ~0.9, ~14, ~0.2 for ~80k vertices drawn.
- **Textures** (`gpu_texture`, ≤ 1024 — 512 a program — since v70, owned by the program that made them, freed with its address space): a 32-byte `TEXTURE_SHADER_STATE` (base address, size, RGBA8, swizzle, and for UIF: *level 0 strictly UIF*, its `UB_PAD`, XOR) then the texels, in the layout the texture unit reads for that size (Mesa's `v3d_setup_slices`, `kern/v3d_tiling.h`): **LT** (a line of 4 × 4-texel utiles, a side ≤ 4), **UBLINEAR** 1 / 2 columns (8 × 8 blocks of 2 × 2 utiles, width ≤ 8 / 16), else **UIF** (columns of 4 blocks, padded rows against the page-cache banks, XOR on odd columns when aligned on them). A batch's `SAMPLER_STATE` (filter, wrap S / T) is written per frame; the fragment uniforms are `p0` = the texture state | 3 (two 16-bit words returned) and `p1` = the sampler state. Host test (every texel of every size up to 300 × 300 against Mesa's functions): `sh tools/tests/run_v3d_tiling_test.sh`.
- **The QPU assembler** `tools/qpu/qpuasm` (host, `cd tools/qpu && make`): Mesa's QPU encoder / disassembler (`qpu_instr.c`, `qpu_pack.c`, `qpu_disasm.c`, MIT, vendored in `tools/qpu/mesa/`) behind a parser of the disassembler's own syntax; a line is accepted only if its encoding disassembles back to it, then each shader is checked against the instruction restrictions (Mesa's `qpu_validate.c` and its scheduler's timing rules: `r4 ≥ 3` instructions after an SFU write, `r5 ≥ 2` after `ldvary`, one TMU / SFU / VPM / TLB access a line, no regfile read right after its write, the thread-switch rules, no TLB access before the scoreboard wait). `qpuasm -d HEX...` disassembles. It re-assembles the three v52 shaders bit for bit.
- **Straight into the target** (v52 and v53): the caller's pixel pointer is translated page by page (`AT S1E1R → PAR_EL1`, IRQs off); when the whole span (`(h - 1) × stride + w` pixels) is physically contiguous below 1 GB — a window canvas is — the tile stores (and the `KEEP` loads) go straight there: raster RGBA8 with **R/B swap** (= 0x00RRGGBB in memory), the row pitch = the caller's stride, alpha not written (`COLOR_WRITE_MASKS` 0x8) and cleared to 0; the span is cleaned + invalidated from the CPU caches before and after. Other buffers go through `s_Target` and a copy, as before. `kmsg` says which mode is used (when it changes).
- **Completion by interrupt**: the V3D core interrupt (GIC SPI 74, shared by core and hub in the Pi 4's device tree; the hub's are masked) is connected at bring-up with `FLDONE` (bin done), `FRDONE` (render done) and `OUTOMEM` unmasked. The handler answers the binner's memory requests from the overflow pool (and masks `OUTOMEM` once the pool is used up) and sets the bin / render events;

the drawing task sleeps on them (`WaitWithTimeout`, 2 ms slices, the frame counters `BFC` / `RFC` re-checked each time, 0.5 s at most). Without an interrupt it degrades to that 2 ms polling (logged once) and serves the binner itself.

- **Sharing the GPU between programs** (the GameCube emulator and the browser at once, say). The kernel is not preempted (§5) and every GPU call runs on core 0, so the `while (s_bBusy) Yield (); s_bBusy = TRUE;` that starts each `gpu_texture` / `gpu_texture_rect` / `gpu_program` / `gpu_render*` is atomic: **one GPU job at a time, whole** — a frame's lists, its binning and rendering, its target's copy — and another program's call waits (it yields) until that frame is done. Nothing of a frame survives it: the control lists, the vertex buffers, `s_Target`, the tile memory are rebuilt by each frame (their memory only grows). What a program owns is marked with its address space (textures, programs, vbufs), checked by every call, freed when it ends (`V3DReleaseAS`, after the frame in flight). So two programs can use the GPU together; what they compete for is the pools of handles: until v70 one program could hold all 256 textures (gcemu and n64emu keep up to 256) or all 8 vbufs, and the other got -4 / 0 — **v70**: 1024 textures, 512 a program; 32 vbufs, 8 a program. Still shared: the GPU's time (frames are served in the order the scheduler runs the callers: a 14 ms Wind Waker frame delays a composite by up to that much), and its state — a frame that hangs turns the GPU off for everybody until a reboot (`s_nState = -1`; the compositing service then goes on with the CPU, `GPC_LOST`). The GPU's low memory (below 1 GB, `HEAP_LOW`) is shared with the kernel's other users.
 - **The compositing service** (`user/gpucomp/gpucomp.{h,c}`, a user library over v53 + v70; docs/03 *GPU compositing*): an app's layers — ARGB premultiplied textures of any size (past 2048 a side kept as tiles of 2048 with a one-texel border of their neighbours, so that bilinear filtering is seamless) — composited into a target (the window's canvas: the direct mode, no copy) with, per layer, a 2D affine matrix, a clip rectangle, an opacity, a blend mode (source-over; v72: multiply, screen, add, subtract, lighten, mask, cut out — multiply and subtract are two passes, `MULCOL` or `RSUB` then `UNDER`, so that they are exact over a transparent target too) and bilinear (or nearest) filtering; scrolling = the layer's source rectangle moved over its texture. A composite is one `gpu_render` frame (a target past 2048 a side: equal parts, one frame each): per layer and tile, the tile's inner rectangle (within the source rectangle) mapped into the target, clipped there (Sutherland-Hodgman against the clip rectangle and the target: the texture coordinates are affine, interpolated exactly) and drawn as a fan — the kernel's `FS_TEX` (texel × colour, the colour = the opacity), blend `PREMUL` (`ONE`, `ONE_MINUS_SRC_ALPHA`, or none for an opaque layer at full opacity), `Z_ALWAYS` + `NOZWRITE`, clamped, `NOMATRIX` clip-space vertices on the kernel's viewport (`x = x [w / 2] + [w / 2]`: the halves are integers in the binning list's scaling). Zero-copy textures are not possible: V3D 4.2's `TEXTURE_SHADER_STATE` has no raster layout for 2D images — the texture unit derives LT / UBLINEAR / UIF from the size (§ *Textures* above) —, so an upload always stores texels into the tiled layout (hence `gpu_texture_rect` for the damaged parts only); what is zero-copy is the target (the canvas, or a `gpu_vbuf` block). The same API is implemented on the CPU (the fallback: no GPU, an older kernel, the GPU stopped; a texture the GPU refused is drawn by the CPU between two GPU frames that keep the target), checked against the GPU path on the PC (`tools/tests/run_gpucomp_test.sh`: a software V3D runs this `FS_TEX` in the QPU simulator) and on the Pi (`/bin/gpcdemo test`).
 - **Not yet**: several frames in flight (the caller waits for its frame), mipmaps, shaders chosen by the app (v61 has them), antialiased polygon edges (the compositor's rotated layers have hard edges: MSAA 4x in the tile buffer would give them).
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16. The RAM: volume

Source: `kernel/sys/ramfs.cpp`, `kern/ramfs.h`; the stream: `CRamStream` (`sys/stream.cpp`); the kapi side: `sys/kapi.cpp`; host test tools: `tests/run_ramfs_test.sh`; Pi test user: `bin/ramtest.c`.

RAM: is a volume in memory, reached by the **same file calls** as the card: `open` / `read` / `fsize` / `fsize64` / `seek` / `close`, `save_file`, `file_in` / `file_out` (streams, appending too: `cp`, `cat`, the shell's redirections), `opendir` / `readdir` / `closedir`, `mkdir`, `remove`, `rename` (within `RAM:`; across volumes `-1`, the caller copies then removes, as between `SD:` and `SD1:`), `chdir` (`cd RAM:/x`, relative paths there), and `vol_info` (`v71`). `newlib's fopen` / `fread` / `fwrite` / `remove` and `newlib's opendir` sit on these, so a program needs nothing new to use it. Its files live **until the Pi restarts**: never written anywhere, not tied to any app (an app that ends leaves its files; its open handles are closed, `RamFsOnProcessGone` from `IpcOnProcessGone`). Programs cannot be started from it (`exec`, `spawn`: the card only). Browser keeps its disk cache and its JS code cache there (`RAM:/jet/cache`, `RAM:/jet/jscache`, `docs/08`): no browsing data on the card, and no SD write freezing core 0.

The tree. Folders and files are `TNode` records on the kernel heap — one size, so a freed one is reused as is (Circle's heap keeps freed blocks by size, it never merges them): a name of up to 255 characters (**case-insensitive, case kept**, as on the card's FAT), the parent, the children (a list in creation order: `readdir's` order), the open count (`nRefs`). A file removed while open leaves its folder at once and is freed at its last close (it can still be read to its end); a folder must be empty to be removed; a folder cannot be moved into itself; `rename` onto an existing name fails (as `FatFs`). Lookups walk the path's names (a folder's children one by one: fine for the few thousand files of a cache).

The bytes are in **64 KB pages of the page allocator** (`palloc_high`: the apps' pool, the high zone; without one — a 1 GB Pi — the low pager), never in the kernel's BSS (the 2 MB image limit) nor in the heap. A file is a list of **extents**: a whole page, or a **chunk** of 256 B ... 32 KB (8 sizes, powers of two) of a page cut into chunks of one size (`TSlab`: the page, a bitmap of its chunks; each size keeps the pages that have a free chunk). Every extent but the last is full. A file written whole (`save_file`) is cut exactly: whole pages, then the rest in falling powers of two (40 KB + 100 B → 32 KB, 8 KB, 256 B: under 256 B lost a file). A stream writes whole pages; at its close its **tail** (the chunks after its last whole page, and that page) is cut again exactly (`Trim`) — a log appended a line at a time stays compact. The extent list itself is in a chunk (a page holds ~2700 extents: 170 MB; a file is limited to `RAMFS_FILE_MAX`, 128 MB, and to the volume). A chunk freed gives its page back as soon as nothing else in it is used: removing a file returns its memory (`vol_info` shows it, `ramtest` checks it).

Limits (`kern/ramfs.h`):

The volume's size	<code>SD:/etc/system.ini ramfs=</code> : megabytes (<code>ramfs=64</code>) or a share of the page memory free at boot (<code>ramfs=10%</code> , at most 90 %); <code>ramfs=0</code> (or under 1 MB): no <code>RAM:</code> volume (<code>vol_info -1</code> , its paths fail). Default (no line): 128 MB, at most a quarter of the free page memory (a 1 GB Pi: ~60 MB). Never more than the free memory less the reserve. Read at boot (a restart applies a change).

The reserve	<code>RAMFS_RESERVE</code> = 32 MB of page memory always left to the apps: a write that would go below it fails (the volume is then "full" before its size).
Files and folders	<code>RAMFS_MAX_NODES</code> = 16384 (their records: ~250 bytes of heap each); a new record also needs 8 MB of heap never handed out (Circle panics when its heap runs out).
A file	<code>RAMFS_FILE_MAX</code> = 128 MB (and the volume's room).
Open handles	256 files, 64 folder listings (on the heap); streams are not counted.

A full volume: `save_file` fails (-1) and **leaves no half file** (a file that cannot fit is removed — the old content was already replaced); a stream's `write` answers the bytes it could store (-1 for none). The boot log says the size: `ramfs: RAM: volume, up to 128 MB (2900 MB of page memory free)`.

Locking. One re-entrant **sleeping lock** for the volume (an atomic owner, its waiters `Yield` — as the FatFs volume lock, §5); nothing yields while holding it. A big read or write is cut into **1 MB slices** (`RAMFS_SLICE`) with a `Yield` between them, the lock free meanwhile — a 16 MB copy does not hold the other tasks (the kernel is not preempted, §5). A `save_file` in progress is in a **no-kill** section (`EnterNoKill`): killed between two slices, the task ends once the file is complete or removed. The file calls run on core 0's tasks (and core 3's with `netcore=1`); an app core never calls the kernel (§14). Opening a file / listing a folder walks the tree under the lock.

v75 (POSIX open files, §8 v75: files and processes). `file_open` on a `RAM:` path holds the file's node (`RamFsNodeOpen`, its `nRefs`): `RamFsPRead` / `RamFsPWrite` anywhere (overwriting in place, a gap past the end filled with zeros, `O_APPEND` at the end under the lock), `RamFsTruncate` (shrinking gives the extents back, growing writes zeros), `RamFsNodeStat` / `RamFsStat`, `RamFsUnlink` / `RamFsRenameEx` (replaces the target; an open one is unlinked and freed at its last close), `RamFsMkdirEx`, `RamFsUtime`, `RamFsReadDir2` (`dir_read`). Every node keeps a write time (UTC) and a number (stat's `ino`, unique, kept by a rename); `dev` 64, `blksize` 64 KB. A node's last close cuts its tail exactly (`Trim`), as a stream's.

Speed (no card): a `save_file` or a `read` is a `memcpy` into / from the pages (GBs a second), plus the yields between the slices; `ramtest` prints the time of a 16 MB file on the Pi.

Tests. On the PC: `sh tools/tests/run_ramfs_test.sh` — `sys/ramfs.cpp` built with `RAMFS_HOST_TEST` (the pages from the PC's allocator, no scheduler: its few platform calls are at the top of the file) under AddressSanitizer: thousands of random operations (saves, streams written / appended / truncated, reads whole / in pieces / after a seek, listings, mkdir, remove, rename across folders) checked against a model, for 5 seeds; then the edges: the names (case, 127 / 128 characters), a folder not empty, a folder into itself, a file removed while open, a dead process's handles, small files taking their size rounded to 256 B, a streamed file's tail fitted, a log appended 300 times, the volume filled (a save that does not fit leaves no half file, the room back after the removes), and every page given back at the end. On the Pi: `/bin/ramtest` (`ramtest full` also fills the volume).

Annex — useful constants

Constant	Value	File
KPAGE_SIZE	64 KB	layout.h
L2_BLOCK_SIZE	512 MB	layout.h
USER_VA_BASE	8 GB	layout.h
USER_WINDOW_CANVAS	12 GB	layout.h
USER_WALLPAPER_CANVAS	13 GB	layout.h
KAPI_TABLE_VA	14 GB (the EL0 table)	kapi_abi.h
KAPI_STUBS_VA	14 GB + 64 KB (the EL0 code page: stubs, then the blob at +8 KB)	el0.h
USER_STACK_TOP	16 GB	layout.h
EL0_USTACK_MIN	8 MB (an app's user stack, lazy since v75; <code>app.txt stack</code> , 8–64 MB)	el0.h
USER_MMAP_BASE .. USER_MMAP_END	34 GB .. 60 GB (the mmap arena, v75)	layout.h
USER_THREAD_STACKS	32 GB (+ 32 MB a thread)	el0.h
KAPI_ABI_VERSION	77	kapi_abi.h
RAM: volume	128 MB by default ($\leq \frac{1}{4}$ of the free page memory; <code>system.ini ramfs=</code>), 32 MB reserve, 16384 files + folders, 128 MB a file	ramfs.h
USER_HEAP_BASE	10 GB	layout.h
Tasks	no limit (a linked list, §5; Circle's <code>MAX_TASKS</code> is not used)	scheduler.h
ASID	8 bits (1..255; 0 = kernel)	layout.h
Kernel stack of an app task (EL0_KSTACK_SIZE)	256 KB	el0.h
Screen resolution	1024×768 by default (<code>cmdline.txt width= / height=</code> ; changed while running: <code>screen_set</code> , v66)	window.h / cmdline.txt
GIMAGE_TRANSPARENT	0xFF00FF	gimage.h